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1 Linear Algebra

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using *Linear Algebra with Applications*, by Otto Bretsch

1.1 Linear Equations

Linear algebra begins as the study of solving linear equations. For example, consider a system of three equations given by

$$\begin{aligned} a_1x + a_2y + a_3z &= d_1 && (A) \\ b_1x + b_2y + b_2z &= d_2 && (B) \\ c_1x + c_2y + c_3z &= d_3 && (C) \end{aligned}$$

for $a_1, \dots d_3 \in \mathbb{R}$. Each equation defines some plane in $\mathbb{R}^3$, depending on the coefficients of x, y, and z. Geometrically, solutions to this system of equations are equivalent to the intersections of these planes; then, solutions may take one of three forms for a linear system in $\mathbb{R}^3$:

1. No solutions. In this case, there are no points on which all three planes intersect. The system of equations defining these planes is said to be **inconsistent**.
2. One solution. In this case, all three planes intersect at a single point in space. The system of equations defining these planes is said to be **consistent** with a **unique** solution.
3. Infinitely many solutions. In this case, all three planes intersect along a line (or plane) in space. The system of equations defining these planes is said to be **consistent**, but solutions are **not unique**.

In higher dimensions, a more general definition of the solution to a linear system involves the intersection of sets. In particular, let A, B, and C be sets containing all points defined by the equations (A), (B), and (C) respectively. Then, $A \cap B \cap C$ is the set of solutions. A system of equations is inconsistent only if the solution set is equal to the empty set, $\varnothing$; otherwise, the system is said to be consistent and at least one solution is contained in the intersection of the sets comprising the system.

For convenience, linear systems are often represented as a **matrix** with m rows and n columns (and therefore dimension $m \times n$). For a linear system in the form previously shown, the **augmented matrix** representing the system is given by

$$\left[\begin{array}{ccc|c} a_1 & a_2 & a_3 & d_1 \\ b_1 & b_2 & b_3 & d_2 \\ c_1 & c_2 & c_3 & d_3 \end{array}\right]$$

A series of row operations may be used to reduce any matrix to a simpler form. The following operations are the same as those allowed on a system of linear equations in standard form, and are also allowed when dealing with augmented matrices:

1. Multiply a row by a scalar.
2. Subtract a multiple of one row from another.
3. Swap the position of any two rows.

When a matrix A is in its simplest form, it is said to be in **reduced row echelon form**, denoted $\text{rref}(A)$. $\text{rref}(A)$ the above augmented matrix may be similar to the form

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & k_1 \\ 0 & 1 & 0 & k_2 \\ 0 & 0 & 1 & k_3 \end{array}\right]$$

Sometimes, it may be impossible to reduce a matrix directly to this form; in this case, there may be some constants located above the diagonal.

The **rank** of a matrix is defined loosely to be the number of leading 1s in the reduced row echelon form of the coefficient matrix. A square coefficient matrix which is of **full rank** (as many 1s as variables as equations) has one unique solution.

Sometimes, equations may "drop out" of a system. This is the case when two rows (equations) are **linearly dependant**; that is to say, they may be written as a **linear combination** of each other. For vectors (any object which may be multiplied by a scalar and added to another vector of equal size) to be **linearly independent**, $\vec{v_1} \neq k\vec{v_2},\ k \in \mathbb{R}$. Then, computing whether a vector $\vec{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix}$ is a linear combination of $\vec{v_1}, \dots \vec{v_n} = \begin{bmatrix} v_{11} \\ \vdots \\ v_{1n} \end{bmatrix}, \dots, \begin{bmatrix} v_{n1} \\ \vdots \\ v_{nn} \end{bmatrix}$, the question is simply whether $\vec{w} = x_1\vec{v_1} + \cdots + x_n\vec{v_n}$ for some $x_1, \dots, x_n$. But this is simply a linear system given by

$$\begin{cases} w_1 & = x_1 v_{11} + \cdots + x_n v_{n1} \\ & \vdots \\ w_n & = x_1 v_{1n} + \cdots + x_n v_{nn} \end{cases}$$

and therefore by the augmented matrix

$$\left[\begin{array}{ccc|c} v_{11} & \dots & v_{n1} & w_1 \\ \vdots & \ddots & \vdots & \vdots \\ v_{1n} & \dots & v_{nn} & w_n \end{array}\right]$$

Then, the columns of a matrix are simply vectors. In particular, a matrix A with m rows and n columns may be written with n vectors of length m:

$$A = \begin{bmatrix} | & & | \\ \vec{v_1} & \dots & \vec{v_n} \\ | & & | \end{bmatrix}$$

The product $A\vec{x}$ is given by the product of $\vec{x}$ with the rows of A:

$$A\vec{x} = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + \cdots + a_{1n}x_n \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n \end{bmatrix}$$

This is the special case $A\vec{x} = \vec{b}$, which is equivalent to the augmented matrix formulation which merely suppresses $\vec{x}$ and places $\vec{b}$ at the rightmost column of the matrix. The same operation is allowed on matrices under certain dimensional restrictions. Namely, for AB to be defined for matrices A and B, A must have the same number of columns as B has rows, and the product will have as many rows as A and as many columns as B. In this case,

$$AB = \begin{bmatrix} | & & | \\ A\vec{v_1} & \dots & A\vec{v_n} \\ | & & | \end{bmatrix}$$

where $A\vec{v_k}$ is the product of A with the kth column of B. Alternatively (and more simply for computations), AB may be computed by dotting the rows of A by the columns of B.

1.2 Linear Transformations

A **linear transformation**, T, is any mapping which satisfies certain properties of linearity:

A **linear transformation** $T : \mathbb{R}^m \to \mathbb{R}^n$ is a map (function) satisfying both of the following:

1. $T(\vec{x} + \vec{y}) = T(\vec{x}) + T(\vec{y})\ \forall\ \vec{x}, \vec{y} \in \mathbb{R}^m$.
2. $T(a\vec{x}) = aT(\vec{x})\ \forall\ a \in \mathbb{R},\ \vec{x} \in \mathbb{R}^m$.

A *matrix* is one such linear transformation. Concurrent linear transformations are also linear (i.e. for matrix A, B, if AB is defined, AB is also a linear transformation).

A matrix is **invertible** only if it is square (i.e. A is $n \times n$) and $\text{rref}(A) = I$ (or equivalently, if $\text{rank}(A) = n$). This is to say, if $A\vec{x} = \vec{b}$ has a unique solution, A is an invertible matrix. This inverse matrix A^{-1} satisfies the property $A^{-1}A = AA^{-1} = I$.

Finding the inverse of a matrix essentially involves computing $\vec{x} = A^{-1}\vec{b}$. This involves solving the $n \times 2n$ augmented matrix $[\ A \mid I\ ]$ until the left hand side is the identity matrix, resulting in $[\ I \mid A^{-1}\ ]$. If this process is not possible, A is not invertible. Since any matrix A is unique (i.e. no two matrices produce the exact same transformation; a matrix product or sum may be written as a new, unique matrix), every inverse matrix is also unique. There exists only one A^{-1} such that $AA^{-1} = A^{-1}A = I$.

Given unit vectors $\vec{e_1}, \dots \vec{e_n} = \begin{bmatrix} 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 \\ \vdots \\ 1 \end{bmatrix}$ and $A = \begin{bmatrix} \vec{v_1} \dots \vec{v_n} \end{bmatrix}$, multiplication shows that $A\vec{e_1} = \vec{v_1}, \dots, A\vec{e_n} = \vec{e_n}$ such that each unit vector simply returns its associated column of A. This gives an *unreasonably useful lemma*:

Unreasonably Useful Lemma. For a linear transformation matrix $A \in \mathbb{R}^{m \times n}$ and vector $\vec{e_j} \in \mathbb{R}^m$ being the jth canonical unit vector in $\mathbb{R}^m$, the jth column of A is given by $A\vec{e_j}$.

Then, it becomes clear that a linear transformation A essentially transforms each coordinate axis into each column of A. It is useful to geometrically interpret some common transformation matrices, using the knowledge that each coordinate axis takes on the form of the columns of the matrix. Common examples are given in $\mathbb{R}^2$:

Scaling. The matrix which scales a vector by c is given by cI where I is the identity matrix. In $\mathbb{R}^2$, this is simply

$$A = \begin{bmatrix} c & 0 \\ 0 & c \end{bmatrix}$$

Orthogonal Projection. Any vector $\vec{x}$ may be written in terms of its components, $\vec{x} = \vec{x_{||}} + \vec{x_{\perp}}$. Projection onto the x-axis is simply given by

$$P_x = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

and projection onto the y-axis is simply given by

$$P_y = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Considering the general projection onto a line L, given a unit vector $\vec{u}$ in the direction of L, the projection onto L is given by

$$\text{proj}_L(\vec{x}) = \boxed{(\vec{x} \cdot \vec{u})\vec{u}}$$

By multiplying the dot product through and then factoring out x_1, x_2 into a separate vector, this leaves the projection matrix:

$$P_L = \begin{bmatrix} u_1^2 & u_1u_2 \\ u_1u_2 & u_2^2 \end{bmatrix}$$

Reflection. For any vector $\vec{x}$ reflected over a line L, $\vec{x_{||}} - \vec{x_{\perp}}$. By some addition, the reflection matrix M is given by $2P_L - I$, or

$$M = \begin{bmatrix} 2u_1^2 - 1 & 2u_1u_2 \\ 2u_1u_2 & 2u_2^2 - 1 \end{bmatrix}$$

Rotation. The rotation matrix uses trigonometric functions to map any vector $\vec{x}$ to itself rotated by θ. This matrix is given by

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Shear. A shear matrix causes a shift on vector $\vec{x}$ in either the horizontal or vertical direction. The horizontal shear matrix is given by

$$S_h = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$$

and the vertical shear matrix is given by

$$S_v = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$$

1.3 Vector Spaces and Subspaces

The **image** of a function f refers to the set of all outputs which may be reached from a particular set of inputs. In particular,

$$\begin{aligned}\text{image}(f) &= \{f(x) \mid x \in X\} \\ &= \{y \in Y \mid y = f(x) \text{ for } x \in X\}\end{aligned}$$

The image of a matrix $A \in \mathbb{R}^{m \times n}$ is then given by

$$\text{image}(A) = \{\vec{y} \in \mathbb{R}^m \mid \vec{y} = A\vec{x} \text{ for } \vec{x} \in \mathbb{R}^n\}$$

Alternatively, the image of any linear transformation $T(\vec{x}) = A\vec{x}$ is the **span** of the column vectors of A, where *span* refers to the set of linear combinations which may be produced from A. For vectors $\vec{v_1}, \dots \vec{v_m} \in \mathbb{R}^n$, $\text{span}(\vec{v_1}, \dots, \vec{v_m}) = \{c_1\vec{v_1} + \cdots + c_m\vec{v_m} | c_1, \dots c_m \in \mathbb{R}\}$. Then, image$(A)$ is given by

$$\text{image}(A) = \text{span}(A\vec{x}) = \begin{bmatrix} | & & | \\ \vec{v_1} & \dots & \vec{v_m} \\ | & & | \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} = x_1\vec{v_1} + \cdots + x_m\vec{v_m}$$

Properties of the image. For a linear transformation $T : \mathbb{R}^m \to \mathbb{R}^n$:

1. The zero vector $\vec{0} \in \mathbb{R}^n$ is in the image of T.
2. If $\vec{v_1}, \vec{v_2}$ are in the image of T, $\vec{v_1} + \vec{v_2}$ is in the image of T (the image is **closed under addition**).
3. If $\vec{v}$ is in the image of T and $k \in \mathbb{R}$, then $k\vec{v}$ is in the image of T (the image is **closed under scalar multiplication**).

The **kernel** of a linear transformation $T : \mathbb{R}^m \to \mathbb{R}^n$ is defined to be the set of all vectors $\vec{x} \in \mathbb{R}^m$ satisfying $T(\vec{x}) = \vec{0}$. Alternatively, the kernel of a matrix equivalent to T, $\ker(A)$, is simply the set of $\vec{x}$ which satisfy the equation $A\vec{x} = \vec{0}$. $\ker(A)$ is also known as the **null space** of A.

Properties of the kernel. For a linear transformation $T : \mathbb{R}^m \to \mathbb{R}^n$:

1. The zero vector $\vec{0} \in \mathbb{R}^m$ is in the kernel of T.
2. The kernel is closed under addition.
3. The kernel is closed under scalar multiplication.

In general, $\ker(A) = \{\vec{0}\}$ under certain conditions:

1. If, for $m \times n$ matrix A, $\text{rank}(A) = m$ and $m \leq n$ (for $m > n$, there are nonzero vectors in the set $\ker(A)$)
2. If and only if, for $n \times n$ (square) matrix A, A is invertible.

The image and kernel of a linear transformations are each special **subspaces**. A subspace has the following properties:

A subset W of the vector space $\mathbb{R}^n$ is called a (linear) subspace of $\mathbb{R}^n$ if is satisfies the following properties:

1. W contains the zero vector in $\mathbb{R}^n$.
2. W is closed under addition (for all $w_1, w_2 \in W$, $w_1 + w_2 \in W$).
3. W is closed under scalar multiplication (for all $k \in \mathbb{R}, w \in W$, $kw \in W$)

From this, it follows that a subspace is closed under any linear combination of its elements.

The **basis** for a subspace is a set of vectors which spans the space. For example, $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ is a basis for $\mathbb{R}^3$, but any other set of three linearly independent vectors in $\mathbb{R}^3$ also forms a basis for $\mathbb{R}^3$. If a set of vectors is not linearly independent, at least one of the vectors is **redundant** and is not included in the basis. A basis for a space always contains the same amount of vectors. For example, any basis for $\mathbb{R}^n$ must contain n vectors. The number of vectors in the basis for a space is known as the **dimension** of the space. For example, $\dim(\mathbb{R}^n) = n$, because n linearly independent basis vectors are needed to span $\mathbb{R}^n$. A subspace V of $\mathbb{R}^n$ may have a dimension less than or equal to, but not greater than, n.

It is possible to check for linear independence of a set of vectors $\{\vec{v}_1, \ldots, \vec{v}_n\}$ by considering the linear combination

$$c_1\vec{v}_1 + \cdots + c_n\vec{v}_n = \vec{0}$$

If this relation holds only when $c_1 = \cdots = c_n = 0$, then all vectors in the set are linearly independent. Notice that this equation may be converted into an equivalent augmented matrix, given by

$$\left[\begin{array}{ccc|c} | & & | & 0 \\ \vec{v}_1 & \ldots & \vec{v}_n & \vdots \\ | & & | & 0 \end{array}\right]$$

If this matrix reduces to $[\ I \mid \vec{0}\ ]$, then all $c_1, \ldots, c_n$ must be 0, meaning the vectors are linearly independent. In general, for a matrix $A = \begin{bmatrix} | & & | \\ \vec{v}_1 & \ldots & \vec{v}_n \\ | & & | \end{bmatrix}$, the columns of A are linearly independent if and only if $\ker(A) = \{\vec{0}\}$, or equivalently, $\operatorname{rank}(A) = n$.

By reducing A to reduced row echelon form, it is possible to pick a basis for $\operatorname{im}(A)$. In particular, each column which may be reduced to have a pivot is a non-redundant column, and is therefore necessary to span the image. The set of these columns forms the basis for $\operatorname{im}(A)$.

The rank-nullity theorem relates the dimension of the kernel and image to the number of columns in a matrix:

Rank-Nullity Theorem. Generally, for a linear transformation $T : V \to W$,

$$\dim(\operatorname{im}(T)) + \dim(\ker(T)) = \dim(V)$$

and equivalently (but with different terminology), for an $m \times n$ matrix such that $T(\vec{x}) = A\vec{x}$,

$$\operatorname{rank}(A) + \operatorname{null}(A) = n$$

In other words, the number of columns with pivots (rank) and free variables (nullity) must sum to the total number of columns in a matrix.

It is now possible to think of a matrix as a function, with all the properties of a function applying to a matrix as well:

For a function $f : A \to B$,

f is **surjective** (onto) if, for every $y \in B$, there exists an $x \in A$ such that $f(x) = y$.

f is **injective** (one-to-one) if, for all $x, y \in A$, $f(x) = f(y)$ only when $x = y$.

f is **bijective** (invertible) if it is both injective and surjective.

For an $m \times n$ matrix A, surjectivity requires $m \leq n$ (at least as many columns as rows, at least as many many variables as equations) and injectivity requires $m \geq n$ (at least as many rows as columns, at least as many equations as variables), so only when $m = n$ (so A is square) can A be bijective.

These definitions and properties extend to all vector spaces, not merely $\mathbb{R}^n$. For example, the complex numbers, polynomials, differentiable functions, and any other set of object with well-defined zero which is closed under addition and scalar

multiplication may constitute a vector space. For any vector space, a linear transformation $T : V \to W$ is an **isomorphism** if T is bijective, and V is **isomorphic** to W is there exists an isomorphism T from V to W.

1.4 Change of Coordinates

Coordinates are essential for any geometric interpretation of a vector space. In particular, Cartesian coordinates are generally used to describe the x-y plane or the x-y-z space.

For an ordered basis $\mathcal{B} = (\vec{v}_1, \ldots, \vec{v}_n)$ for a vector space V, all elements $\vec{v} \in V$ may be written as a linear combination

$$\vec{v} = a_1\vec{v}_1 + \cdots + a_n\vec{v}_n$$

and the $\mathcal{B}$-coordinates are defined as

$$[\vec{v}]_{\mathcal{B}} = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$$

Notice that V is now related to $\mathbb{R}^n$ by $[\vec{v}]_{\mathcal{B}}$. This makes bases particularly useful when working with vector spaces besides $\mathbb{R}^n$ (i.e. differentiable functions, polynomials, etc).

For example, for the set $\mathcal{P}_2$ of polynomials of at most degree two, a basis $\mathcal{B} = (1, x, x^2)$ spans $\mathcal{P}_2$ such that for any $p \in \mathcal{P}_2$, $p(x) = a + bx + cx^2$, so $[p]_{\mathcal{B}} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ relates $p \in \mathcal{P}_2$ to $\mathbb{R}^2$.

Formally, this process is an isomorphic mapping from V to $\mathbb{R}^n$:

The mapping

$$V \to \mathbb{R}^n, \text{ given by } \vec{v} \mapsto [\vec{v}]_{\mathcal{B}}$$

is an isomorphism of V and $\mathbb{R}^n$, so

$$[\vec{v} + \vec{w}]_{\mathcal{B}} = [\vec{v}]_{\mathcal{B}} + [\vec{w}]_{\mathcal{B}}$$

and

$$[c\vec{v}]_{\mathcal{B}} = c[\vec{v}]_{\mathcal{B}}$$

For a linear transformation T given by a square matrix such that $T(\vec{x}) = A\vec{x}$, it is relatively simple to compute the basis for the transformation. In particular, for $\vec{x} = c_1\vec{v}_1 + \cdots + c_n\vec{v}_n$ where $\vec{v}_1, \ldots \vec{v}_n$ form a basis,

$$[T(\vec{x})]_{\mathcal{B}} = \begin{bmatrix} | & & | \\ [T(\vec{v}_1]_{\mathcal{B}} & \ldots & [T(\vec{v}_n)]_{\mathcal{B}} \\ | & & | \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = B[\vec{x}]_{\mathcal{B}}$$

where B is the $\mathcal{B}$-coordinate matrix given by

$$B = \begin{bmatrix} | & & | \\ [T(\vec{v}_1]_{\mathcal{B}} & \ldots & [T(\vec{v}_n)]_{\mathcal{B}} \\ | & & | \end{bmatrix}$$

Alternatively, B may be computed by

$$B = S^{-1}AS$$

where S is the matrix of the bases of A, given by

$$S = \begin{bmatrix} | & & | \\ \vec{v}_1 & \ldots & \vec{v}_n \\ | & & | \end{bmatrix}$$

Then, the $\mathcal{B}$-matrix for a linear transformation may be constructed two ways:

For a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^n$ such that $T(\vec{x}) = A\vec{x}$ with basis $\vec{v}_1, \ldots, \vec{v}_n$ for $\mathbb{R}^n$, the $\mathcal{B}$-matrix $B = [T]_{\mathcal{B}}$ may be computed either column-wise, as

$$B = \begin{bmatrix} | & & | \\ [T(\vec{v}_1]_{\mathcal{B}} & \ldots & [T(\vec{v}_n)]_{\mathcal{B}} \\ | & & | \end{bmatrix}$$

or with the basis, as

$$B = S^{-1}AS, \qquad S = \begin{bmatrix} | & & | \\ \vec{v}_1 & \ldots & \vec{v}_n \\ | & & | \end{bmatrix}$$

And since S is invertible, the relationship

$$SB = AS$$

holds generally.

In general, *any isomorphic transformation $T : V \to V$ may be represented with a B-matrix*:

For any isomorphic transformation $T : V \to V$ with basis $\mathcal{B}$ for V given by $\mathcal{B} = \{f_1, \ldots, f_n\}$, the change of basis matrix B is given by

$$B = \begin{bmatrix} | & & | \\ [T(f_1)]_{\mathcal{B}} & \ldots & [T(f_n)]_{\mathcal{B}} \\ | & & | \end{bmatrix}$$

The entire change of basis process may be best understood with a commutative diagram:

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "L_{\mathcal{B}}"] & V \arrow[d, "L_{\mathcal{B}}"] \\ \mathbb{R}^n \arrow[r, "{[T]_{\mathcal{B}}}"] & \mathbb{R}^n \end{tikzcd}$$

$L_{\mathcal{B}}$ is the coordinate isomorphism associating an arbitrary vector space V with some basis $\mathcal{B}$ in $\mathbb{R}^n$. Direct paths (i.e. by T or $[T]_{\mathcal{B}}$) are equivalent to the change-of-coordinate path (i.e. by $L_{\mathcal{B}}[T]_{\mathcal{B}}L_{\mathcal{B}}^{-1}$ or $L_{\mathcal{B}}^{-1} \circ T \circ L_{\mathcal{B}}$ respectively).

1.5 Orthogonality in $\mathbb{R}^n$, Inner Product Spaces, and Least Squares

In higher dimensions, the concept of orthogonality (perpendicularity) and length of vectors becomes difficult to visualize. Instead, a rigorous definition replaces intuition:

Two vectors $\vec{v}$ and $\vec{w}$ in $\mathbb{R}^n$ are **orthogonal** if their dot product $\vec{v} \cdot \vec{w} = v_1w_1 + \cdots + v_nw_n = 0$.

The **length** of a vector is given by $||\vec{v}|| = \sqrt{\vec{v} \cdot \vec{v}} = \sqrt{v_1^2 + \cdots + v_n^2}$. A vector is a **unit vector** if its length is equal to 1. For a vector $\vec{v}$ which is not a unit vector, the corresponding unit vector $\vec{u}$ in the direction of $\vec{v}$ is given by $\vec{u} = \frac{1}{||\vec{v}||}\vec{v}$.

The concept of the dot product generalizes for any vector space as an **inner product**. In fact, the dot product is one example of an inner product in the *inner product space* $\mathbb{R}^n$. An inner product is defined as any transformation $\langle \cdot, \cdot \rangle : V \to \mathbb{R}$ satisfying the following properties:

For all $f, g, h \in V$ and all $c \in \mathbb{R}$, an inner product $\langle \cdot, \cdot \rangle$ satisfies the following:

1. $\langle f, g \rangle = \langle g, f \rangle$
2. $\langle f + g, h \rangle = \langle f, h \rangle + \langle g, h \rangle$
3. $\langle cf, g \rangle = c\langle f, g \rangle$
4. $\langle f, f \rangle > 0$ for all nonzero $f \in V$

An inner product space is a vector space which has an assigned inner product. In a particular inner product space, the magnitude of an element f is defined as $\sqrt{\langle f, f \rangle}$, and elements f, g are orthogonal if $\langle f, g \rangle = 0$.

Vectors are **orthonormal** if they are of unit length and orthogonal. Orthonormal vectors must be linearly independent. Then, a set of n orthonormal vectors in $\mathbb{R}^n$ have special properties. In particular, for orthonormal vectors $\vec{u}_1, \ldots, \vec{u}_n \in \mathbb{R}^n$, $\vec{u}_1, \ldots, \vec{u}_n$ must form a basis for $\mathbb{R}^n$.

The projection may be generalized in $\mathbb{R}^n$. In particular:

For a subspace V of $\mathbb{R}^n$ with orthonormal basis $\vec{u}_1, \ldots, \vec{u}_n$, the projection of any $\vec{x} \in \mathbb{R}^n$ onto the subspace V is given by

$$\text{proj}_V \vec{x} = \vec{x}_{||} = (\vec{u}_1 \cdot \vec{x}_1)\vec{u}_1 + \cdots + (\vec{u}_n \cdot \vec{x}_n)\vec{x}_n$$

The **orthogonal compliment** of a vector space V refers to the set of all vectors $\vec{x}$ which are perpendicular to every vector $\vec{v}$ in V. In set notation, $V_\perp$ is given by

$$V_\perp = \{\vec{x} \in \mathbb{R}^n \mid \vec{x} \cdot \vec{x} = 0 \ \forall \ \vec{v} \in V\}$$

$V_\perp$ has several unique properties:

Properties of the orthogonal complement of a subspace V of $\mathbb{R}^n$:

1. $V_\perp$ is a subspace of V.
2. $V \cap V_\perp = \{\vec{0}\}$
3. $\dim(V) + \dim(V_\perp) = n$
4. $(V_\perp)_\perp = V$

Using this information, any basis may be converted into an **orthonormal basis** by the **Gram-Schmidt process**, a recursive process outlined as follows:

Gram-Schmidt Process. For a basis $\mathcal{B} = \{v_1, \ldots, v_n\}$, take

$$\begin{aligned} u_1 &= v_1 \\ u_i &= v_i - v_i^\perp, \text{ where } v_i^\perp = \text{proj}_{u_1}(v_i) + \cdots + \text{proj}_{u_{i-1}} v_i, \qquad \text{for } i > 1 \end{aligned}$$

which yields the new orthonormal basis

$$\mathcal{B}' = \left\{ \frac{u_1}{||u_1||}, \ldots, \frac{u_n}{||u_n||} \right\}$$

A linear transformation is called an **orthogonal transformation** if it preserves length and orthogonality. In particular, a linear transformation has the following properties:

Properties of an Orthogonal Transformation. A linear transformation $T : V \to V$ is orthogonal if and only if any of the following hold:

1. $||T(\vec{x})|| = ||\vec{x}||$ for all $\vec{x} \in \mathbb{R}^n$
2. for orthogonal $\vec{v}, \vec{w} \in V$, $T(\vec{v})$ and $T(\vec{w})$ are orthogonal
3. The $n \times n$ standard matrix A of T is an **orthogonal matrix**, such that the columns of A form an orthonormal basis for $\mathbb{R}^n$
4. $A^T A = AA^T = I_n$, where A^T is the transpose of A such that entry a_{ij} of A becomes entry a_{ji} of A^T. Equivalently, A is orthogonal if $A^T = A^{-1}$.

The transpose has several other useful properties:

Properties of the Transpose. For any $A, B \in \mathbb{R}^{m \times n}$, the following properties of the transpose apply:

1. $(A + B)^T = A^T + B^T$
2. $(AB)^T = B^T A^T$
3. $(A^T)^{-1} = (A^{-1})^T$
4. $\text{rank}(A^T) = \text{rank}(A)$
5. $\ker(A) = \ker(A^T A)$
6. $(\text{im} A)^\perp = \ker(A^T)$

The transpose has several applications, particularly in data fitting. In particular, for any *inconsistent system* $A\vec{x} = \vec{b}$ (for which there is no solution), the **least-squares solution** approximates a solution.

Least Squares Solutions. Let $A \in \mathbb{R}^{m \times n}$. The least-squares solution of a linear system $A\vec{x} = \vec{b}$ is the vector $\vec{x}^*$ such that $||\vec{b} - A\vec{x}^*|| \leq ||\vec{b} - A\vec{x}||$ for all $\vec{x} \in \mathbb{R}^n$. In other words, $\vec{x}^*$ minimizes the error between $\vec{b}$ and $A\vec{x}$.

The least-squares solutions of a system $A\vec{x} = \vec{b}$ are the exact solutions of the consistent system

$$A^T A\vec{x} = A^T \vec{b}$$

If A is invertible, then $\vec{x}^* = (A^T A)^{-1} A^T \vec{b}$. For a consistent linear system, $\vec{X}^* = \vec{x}$ (the least squares solution is the solution to $A\vec{x} = \vec{b}$).

Least-squares solutions are frequently applied in data fitting to fit vectors (generally polynomials or some other differentiable function) to a set of data–for example, in fitting a trend line, a quadratic function, or an exponential curve to an over-constrained dataset (which cannot fit exactly to the function in question).

1.6 Determinants

The **determinant** of a matrix, $\det A$, is a real number associated with a square matrix. In general, the determinant is computed recursively by *Laplace expansion*. Several properties of the determinant allow its computation to be simplified at times:

For any square $n \times n$ matrix, the following properties hold:

1. $\det(AB) = (\det A)(\det B)$
2. $\det(A^n) = (\det A)^n$
3. $\det A^T = \det A$
4. $\det(A^{-1}) = \frac{1}{\det A}$

For special matrices, the following properties hold:

1. If a matrix has linearly dependent rows or columns, $\det A = 0$. Alternatively, a matrix is invertible if and only if $\det A \neq 0$.
2. The determinant of a triangular matrix is given by $\det A = \mathrm{tr} A$, where $\mathrm{tr} A$ is the sum of the diagonal entries of A.
3. If A and B are similar matrices, then $\det A = \det B$

1.7 Eigenvalues and Eigenvectors

For a linear transformation $T : V \to V$, an **eigenvalue** of T is a real number λ such that $T(\vec{v}) = \lambda\vec{v}$, where (nonzero) $\vec{v} \in V$ is an **eigenvector** corresponding to λ.

If there exist linearly independent vectors $\vec{v}_1, \ldots, \vec{v}_n \in \mathbb{R}^n$, with corresponding $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$ such that $A\vec{v}_i = \lambda_i\vec{v}_i$, then $\mathcal{B} = \{\vec{v}_1, \ldots, \vec{v}_n\}$ forms an **eigenbasis** for $\mathbb{R}^n$.

If an eigenbasis $\mathcal{B} = (\vec{v}_1, \ldots, \vec{v}_n)$ exists for some matrix A, then A is **diagonalizable** (similar to a diagonal matrix) by the diagonalization

$$D = S^{-1}AS, \qquad S = \begin{bmatrix} | & & | \\ \vec{v}_1 & \ldots & \vec{v}_n \\ | & & | \end{bmatrix}, \quad D = \begin{bmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \lambda_n \end{bmatrix}$$

A method to calculate eigenvalues emerges from the definition of an eigenvalue-eigenvector pair. In particular, if λ is an arbitrary real number and $\vec{v}$ is any nonzero vector, then the condition for an eigenvalue is:

$$A\vec{v} = \lambda\vec{v} \iff A\vec{v} - \lambda\vec{v} = \vec{0} \iff A\vec{v} - \lambda I\vec{v} = \vec{0} \iff (A - \lambda I)\vec{v} = \vec{0}$$

Thus, all vectors in $\ker(A - \lambda I)$ are eigenvectors to corresponding eigenvalues λ. If there are eigenvectors (in the kernel), then $\det(A - \lambda I) = 0$ as $A - \lambda I$ is not invertible in this case. Thus, (real or complex) eigenvalues and eigenvectors may be computed as follows:

Eigenvalues and eigenvectors satisfy the **characteristic equation**:

$$\det(A - \lambda I) = 0$$

For each λ_i found from the roots of the characteristic polynomial, eigenvectors consist of all vectors in $\ker(A - \lambda_i I)$.

Each eigenvalue has an **algebraic multiplicity** $\text{almu}(\lambda_i)$ and a **geometric multiplicity** $\text{gemo}(\lambda_i)$. Algebraic multiplicity refers to the number of times λ_i appears as a root in the characteristic equation, while geometric multiplicity refers to $\dim(\ker(A - \lambda_i I)$. If $\text{almu}(\lambda_i) \neq \text{gemu}(\lambda_i)$ for any eigenvector, then A is not diagonalizable.

In some cases, there exists an *orthonormal eigenbasis* for a transformation. This is only the case when S is orthogonal such that $S^{-1}AS = S^TAS = D$ for some diagonalization D. The **spectral theorem** outlines the condition for an orthonormal eigenbasis:

Spectal Theorem. A matrix A is orthogonally diagonalizable if and only if A is symmetric, where a symmetric matrix satisfies $A^T = A$.

2 Real Analysis

Notes from **Math 351: Principles of Analysis**

Taken at University of Michigan, Winter 2025

using *Elementary Analysis: The Theory of Calculus*, by Kenneth Ross

2.1 Real Numbers

This material is essentially a development of the two Fundamental Theorems of Calculus, beginning from sequences. To begin, certain axioms about real numbers and sets are taken for granted. These axioms are outlined:

Archimedean Property. If $a > 0$ and $b > 0$, then there exists a natural number $n \in \mathbb{N}$ such that $na > b$, or $n > b/a$.

Density of Rational Numbers. If $a, b \in \mathbb{R}$ and $a < b$, then there exists a rational number $r \in \mathbb{Q}$ such that $a < r < b$.

Density of Irrational Numbers. If $a, b \in \mathbb{R}$ and $a < b$, then there exists an irrational number $x \in \mathbb{R} \backslash \mathbb{Q}$ such that $a < x < b$.

For the final axiom about the real numbers, a definition is necessary:

Definition. Let $S \subseteq \mathbb{R}$. If there exists $M \in \mathbb{R}$ such that $s \leq M$ for all $s \in S$, then M is an **upper bound** for S. Similarly, if there exists $m \in \mathbb{R}$ such that $m \leq s$ for all $s \in S$, then m is a **lower bound** for S. S is said to be **bounded** if it has both an upper and lower bound. Equivalently, S is bounded if and only if there exists $M_0 \in \mathbb{R}$ such that if $s \in S$, then $|s| \leq M_0$.

Completeness Axiom: If S is a non-empty subset of $\mathbb{R}$ which has an upper bound, then S has a least upper bound $\sup S$. Similarly, if S has a lower bound, then S has a greatest lower bound $\inf S$.

2.2 Sequences

A sequence $s : \mathbb{N} \to \mathbb{R}$ is a mapping of natural numbers to real numbers. Generally, the nth term of a sequence $s(n)$ is denoted s_n.

Definition. A sequence $\{s_n\}$ is said to **converge** to $s \in \mathbb{R}$ if for every $\epsilon > 0$, there exists $N \in \mathbb{R}$ such that if $n > N$, then $|s_n - s| < \epsilon$. Here, s is equivalently denoted as $s = \lim s_n = \lim\limits_{n \to \infty} s_n$.

Essentially, if the values of a sequence become arbitrarily close to some number after a given point, then the sequence is said to converge to that number. It can be shown that this limit s is *unique*.

Example. Prove that $\lim \frac{1}{n} = 0$.

Proof. Fix $\epsilon > 0$. Let $N = \frac{1}{\epsilon}$. Then, if $n > N$,

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} < \frac{1}{N} = \frac{1}{1/\epsilon} = \epsilon$$

Thus, $\left| \frac{1}{n} - 0 \right| < \epsilon$, and $\lim \frac{1}{n} = 0$. ■

From the definition of the limit, various facts may be proved which aid in computing limits. Let $\lim s_n = s$ and $\lim t_n = t$. Then, the following limits hold:

Scalar Multiplication. If $c \in \mathbb{R}$, then $\lim cs_n = cs$

Addition. $\lim(s_n + t_n) = \lim s_n + \lim t_n = s + t$

Subtraction. $\lim(s_n - t_n) = \lim s_n - \lim t_n = s - t$

Multiplication. $\lim(s_n t_n) = st$

Division. If $t \neq 0$ and $t_n \neq 0$ for all $n \in \mathbb{N}$, then $\lim(s_n/t_n) = \lim(s_n)/\lim(t_n) = s/t$

Note that these limit laws *only apply to sequences known to converge.* A particularly important limit law, the *squeeze limit law*, is given:

Theorem. (Squeeze Principle) Let $a_n \leq s_n \leq b_n$ for all but finitely many values of n and $\lim a_n = \lim b_n = s$. Then, $\lim a_n = \lim s_n = \lim b_n = s$.

Proof. Fix $\epsilon > 0$. Since $a_n \leq s_n \leq b_n$ for all but finitely many values of n, there exists N_0 such that if $n > N_0$, then $a_n \leq s_n \leq b_n$. Similarly, since $\lim a_n = \lim b_n = s$, there exists N_1, N_2 such that if $n > N_1$, then $|a_n - s| < \epsilon$ and if $n > N_2$, then $|b_n - s| < \epsilon$. Let $N = \max(N_0, N_1, N_2)$. Then, if $n > N$,

$$|a_n - s| < \epsilon, \text{ so } s - \epsilon < a_n < s + \epsilon$$

and

$$|b_n - s| < \epsilon, \text{ so } s - \epsilon < b_n < s + \epsilon$$

Thus,

$$s - \epsilon < a_n \leq s_n \leq b_n < s + \epsilon$$

and thus $|s_n - s| < \epsilon$, so $\lim s_n = s$. ■

It is possible to derive other *comparison principles*, i.e. if $s_n \geq a$ for all but finitely many n, then $\lim s_n \geq a$, and if $s_n \leq b$ for all but finitely many n, then $\lim s_n \leq b$.

Definition. A sequence $\{s_n\}$ is said to **diverge to** ∞ if, for all $M \in \mathbb{R}$, there exists a real number N such that if $n > N$, then $s_n > M$. This is denoted $\lim s_n = +\infty$. A sequence $\{s_n\}$ is said to **diverge to** $-\infty$ if, for all $M \in \mathbb{R}$, there exists a real number N such that if $n > N$, then $s_n < M$. This is denoted $\lim s_n = -\infty$.

A sequence will either *converge* to a finite limit, *diverge* to $+\infty$ or $-\infty$, or *not converge* if neither convergence nor divergence are met. For example, the sequence $s_n = (-1)^n$ does not converge.

Definition. A sequence $\{s_n\}$ is **monotone increasing** if, for all $n \in \mathbb{N}$, $s_{n+1} \geq s_n$. $\{s_n\}$ is **monotone decreasing** if, for all $n \in \mathbb{N}$, $s_n \geq s_{n+1}$. $\{s_n\}$ is **monotone** if it is monotone increasing and/or monotone decreasing.

Monotone sequences are special sequences for two reasons: (1) a bounded monotone sequence is guaranteed to converge, and it converges to its infimum if it is monotone decreasing or its supremum if it is monotone increasing; and (2) a *monotone subsequence* can be found in any sequence. The meaning of subsequence is an infinite sampling of terms of an original sequence, ordered in the same manner:

Definition. A sequence $\{s_{n_k}\}$ is a **subsequence** of $\{s_n\}$ if each $\{n_k\}$ is a strictly increasing sequence of natural numbers (i.e. $n_{k-1} < n_k$), indexed by $k \in \mathbb{N}$.

Using these two facts, it is possible to prove a key result in analysis:

Bolzano-Weierstrass Theorem. Every bounded sequence has a convergent subsequence.

Proof. Let $\{s_n\}$ be a bounded sequence. Then, there exists a subsequence $\{s_{n_k}\}$ which is monotone. Since $\{s_n\}$ is bounded, $\{s_{n_k}\}$ must also be bounded. Since $\{s_{n_k}\}$ is monotone and bounded, $\{s_{n_k}\}$ is convergent. Thus, $\{s_n\}$ has a convergent subsequence. ■

Subsequences of convergent or divergent sequences have special properties. If a sequence $\{s_n\}$ is convergent to s, then every subsequence s_{n_k} converges to s. Similarly, if $\{s_n\}$ diverges to $+\infty$, then every $\{s_{n_k}\}$ also diverges to $+\infty$, and if $\{s_n\}$ diverges to $-\infty$, then every $\{s_{n_k}\}$ also diverges to $-\infty$. Therefore, if the convergence (or divergence) of any two subsequences disagree, then the sequence is not convergent.

Example. Prove that $\{s_n\} = \begin{cases} 1, & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$ does not converge by using subsequences.

Proof. Observe that

$$\lim s_{2k} = \lim 1 = 1$$

and

$$\lim s_{2k+1} = \lim 0 = 0$$

and thus $\lim s_{2k} \neq \lim s_{2k+1}$, so $\{s_n\}$ does not converge. ■

A final method to evaluate the convergence of a sequence is to determine whether the sequence is *Cauchy*.

Definition. A sequence $\{s_n\}$ is a **Cauchy sequence** if, for any $\epsilon > 0$, there exists a real number N such that if $m, n > N$, then $|s_m - s_n| < \epsilon$.

In other words, if the terms in a sequence become arbitrarily close together for sufficiently large index values, the sequence is Cauchy. An important equivalence between convergent sequences and Cauchy sequences can be shown:

Theorem. A sequence $\{s_n\}$ is a Cauchy sequence if and only if it is convergent.

This theorem provides a criteria for convergence even when the value to which a sequence converges is not obvious; it is sufficient to shown that the difference between consecutive terms becomes arbitrarily small for large enough indexing terms.

A special type of sequence is the **infinite series**. A series $\sum_{n=m}^{\infty} a_n$ is said to converge if its **partial sums**, a sequence with nth term $s_n = a_m + a_{m+1} + \cdots + a_n$, is convergent.

Theorem. If $\sum_{n=m}^{\infty} a_n$ is an infinite series, then $\lim a_n = 0$.

Discussion of closedness and compactness.

Definition. A subset $S \subseteq \mathbb{R}$ is **closed** if every convergent sequence with values in S has limit in S. A subset $S \subseteq \mathbb{R}$ is **compact** if every sequence with values in S has a convergent subsequence with limit in S.

An important theorem in analysis, the Heine-Borel theorem, relates compactness and closedness of a set:

Heine-Borel Theorem. A set $S \subseteq \mathbb{R}$ is compact if and only if S is closed and bounded.

The Heine-Borel theorem is readily applied to an interval $[a, b]$ of $\mathbb{R}$. The set $[a, b]$ clearly is bounded below by a and above by b, and this set is closed. Therefore, any closed interval of $\mathbb{R}$ is compact.

2.3 Continuity

Continuity of a function may be defined immediately from sequences.

Definition. A function $f : D \to \mathbb{R}$ is **continuous** at $x \in D$ if for every sequence $\{x_n\}$ converging to x with values in D, $\lim f(x_n) = f(x)$.

This is the *sequence definition of continuity.* The criteria here is essentially that the limit distributes inside the function, as this definition requires $\lim f(x_n) = f(\lim x_n)$.

Example. Prove that $f(x) = x^2$ is continuous on $\mathbb{R}$.

Proof. Let $x \in \mathbb{R}$ be arbitrary. Let $\{x_n\}$ converge to x. Then,

$$\lim f(x_n) = \lim(x_n^2) = \lim(x_n) \cdot \lim(x_n) = (\lim x_n)^2 = f(\lim x_n) = f(x)$$

Our choice of x is arbitrary, so for any $x \in \mathbb{R}$, $\lim f(x_n) = f(x)$, so x^2 is continuous on $\mathbb{R}$. ■

$f(x_n)$ is simply a sequence, and if f is continuous, this is a convergent sequence. Thus, limit laws for sequences essentially carry through to continuous functions, with these operations preserving continuity. In particular, for functions f, g, h continuous at x, $(f+g)(x)$ is continuous, $cf(x)$ is continuous for any $c \in \mathbb{R}$, and $\left(\frac{f}{g}\right)(x)$ is continuous if g is nonzero on the domain D. Additionally, a composition of continuous functions $f(g(x))$ is continuous at x. Finally, the squeeze principle carries through; if $f(x) \leq g(x) \leq h(x)$ for all $x \in D$ and $f(x_0) = h(x_0)$, then g must be continuous at x_0.

Two key theorems may be proven immediately from these facts:

Extreme Value Theorem. If $f : D \to \mathbb{R}$ is continuous on $[a, b]$, then f is bounded and it achieves its maximum and its minimum on $[a, b]$. Equivalently, there exists $x_1, x_2 \in [a, b]$ such that $f(x_1) \leq f(x) \leq f(x_2)$ for all $x \in [a, b]$.

Using the extreme value theorem, another key theorem may be proven:

Intermediate Value Theorem. If f is continuous on $[a, b]$ and y is between $f(a)$ and $f(b)$, then there exists $x_0 \in [a, b]$ such that $f(x_0) = y$.

Continuity may also be defined without reference to sequences. This definition sometimes simplifies the process of proving or disproving continuity. This definition, which is equivalent to the sequence definition of continuity, is given below:

Definition. A function $f : D \to \mathbb{R}$ is continuous at $x_0 \in D$ if, for all $\epsilon > 0$ there exists $\delta > 0$ such that if $x \in D$ and $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \epsilon$.

In the epsilon-delta definition of continuity, a function f is continuous at x_0 if for any inputs δ-close to x_0, the outputs are ϵ-close to $f(x_0)$.

Example. Prove that $f(x) = x^3$ is continuous at $x = 1$.

Scratch Work. We often make the assumption that δ is small, i.e. $\delta < 1$. Making this assumption, we have

$$|x - 1| < 1 \implies -1 < x - 1 < 1 \implies 0 < x < 2$$

And we also have

$$|f(x) - f(1)| = |x^3 - 1| = |x - 1||x^2 + x + 1| < \delta \underbrace{|2^2 + 2 + 1|}_{\text{given } x<2} < 7\delta$$

If $\epsilon \geq 7\delta$, we are done. Thus, we choose δ to be the minimum $\epsilon/7$ (as found by assumption) and 1 (as an upper bound).

Proof. Fix $\epsilon > 0$. Choose $\delta = \min\left(\frac{\epsilon}{7}, 1\right)$. If $|x - 1| < \delta$, then

$$|f(x) - f(1)| = |x^3 - 1| = |x - 1||x^2 + x + 1| < 7\delta \leq 7\frac{\epsilon}{7} = \epsilon$$

For our chosen δ, we are guaranteed that if $|x - 1| < \delta$, then $|f(x) - f(1)| < \epsilon$. Thus, f is continuous at $x = 1$. ■

The $\epsilon - \delta$ definition lends itself to the definition of a stronger form of continuity, *uniform continuity*:

Definition. A function $f : D \to \mathbb{R}$ is **uniformly continuous** on $S \subseteq D$ if for all $\epsilon > 0$, there exists $\delta > 0$ such that if $x, y \in S$ and $|x - y| < \delta$, then $|f(x) - f(y)| < \epsilon$.

Functions which grow at increasing rate are generally *not uniformly continuous* on an unbounded interval. For example, $f(x) = x^2$ can be shown to be continuous for any value in $\mathbb{R}$, but x^2 is *not uniformly continuous* on $\mathbb{R}$ because δ cannot take on a form to account for arbitrary ϵ. By contrast, $f(x) = x$ is both continuous for every point in $\mathbb{R}$ and uniformly continuous on $\mathbb{R}$, with $\delta = \epsilon$. The most important result of uniform continuity is the following theorem:

Theorem. If f is continuous on a closed interval $[a, b]$, then it is uniformly continuous on $[a, b]$.

2.4 Differentiation

Before defining the derivative, it is necessary to define the limit at a point.

Definition. Suppose $f : D \to \mathbb{R}$ is a function, $x_0 \in \mathbb{R}$, and that there exists $\beta > 0$ such that D contains $(x_0 - \beta, x_0) \cup (x_0, x_0 + \beta)$. The *limit* as $x \to x_0$, denoted

$$\lim_{x \to x_0} f(x) = L$$

if for any sequence $\{x_n\}$ with values in $D \backslash \{x_0\}$ such that $\lim x_n = x_0$, $\lim f(x_n) = L$.

Notice that x_0 is not considered in evaluating the limit. Furthermore, the limit is only defined if the function can be evaluated within β of x_0 (in either direction). Since $\lim\limits_{x \to x_0} f(x)$ is defined in terms of sequences, the limit laws derived for sequences still apply. Furthermore, the limit as $x \to x_0$ may be equivalently defined in terms of the $\epsilon - \delta$ definition:

Theorem. $\lim\limits_{x \to x_0} f(x) = L$ if and only if for all $\epsilon > 0$, there exists $\delta > 0$ such that if $|x - x_0| < \delta$ and $x \in D$, then $|f(x) - L| < \epsilon$.

An important result relates limits to continuity:

Theorem. A function f is continuous at x_0 if and only if $\lim\limits_{x \to x_0} f(x) = f(x_0)$.

With the limit definition, it is possible to define the derivative:

Definition. Suppose $f : D \to \mathbb{R}$, $x_0 \in \mathbb{R}$ and D contains an open interval around x_0. f is called *differentiable* at x_0 with derivative $f'(x_0)$ if

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0)$$

If this limit does not exist, then f is said to be *not differentiable* at x_0 and $f'(x_0)$ does not exist. Notice that x_0 is *not* in the domain of $\frac{f(x) - f(x_0)}{x - x_0}$ to avoid division by zero. Differentiability and continuity are related by the following theorem:

Theorem. Suppose f is differentiable at x_0. Then, f is continuous at x_0.

Notice that the opposite statement is clearly *false*. As a counterexample, the function $|x|$ may be shown to be continuous but not differentiable at $x = 0$.

The common differentiation rules may be derived:

Linearity of derivatives. If f, g are differentiable at x_0 and $c \in \mathbb{R}$, then $(f+g)'(x_0) = f'(x_0) + g'(x_0)$ and $(cf)'(x_0) = cf'(x_0)$

Product Rule. If f, g are differentiable at x_0, then $(fg)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0)$

Quotient Rule. If f, g are differentiable at x_0 and $g(x_0) \neq 0$, then $(f/g)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{[g(x_0)]^2}$

Chain Rule. If g is differentiable at x_0 and f is differentiable at $g(x_0)$, then $(f(g(x_0))' = f'(g(x_0))g'(x_0)$

Two important theorems involving the existence of derivatives may be derived directly from the definition:

Rolle's Theorem. If f is continuous on $[a, b]$, differentiable on (a, b) and $f(a) = f(b)$, then there exists $x_0 \in (a, b)$ such that $f'(x_0) = 0$.

Rolle's Theorem is used in the proof of a stronger existence theorem, the *mean value theorem*:

Mean Value Theorem. If f is continuous on $[a, b]$ and differentiable on (a, b), then there exists $x_0 \in (a, b)$ such that $f'(x_0) = \frac{f(b)-f(a)}{b-a}$

In other words, for any interval where f is continuous and differentiable, the average slope over the interval must equal the instantaneous slope, the derivative, for at least one point on that interval. Observe that Rolle's Theorem is the Mean Value Theorem under the special case $f(b) - f(a) = 0$.

Following immediately from the Mean Value Theorem, derivatives can be shown to relate to the behavior of their original function.

Function behavior and derivatives. Suppose f is differentiable on (a, b).

If $f'(x) = 0$ for all $x \in (a, b)$, then $f(x) = c$ on (a, b) for some constant $c \in \mathbb{R}$.

If $f'(x) > 0$ for all $x \in (a, b)$, then f is strictly increasing on (a, b).

If $f'(x) < 0$ for all $x \in (a, b)$, then f is strictly decreasing on (a, b).

Each result follows from Mean Value Theorem. The case of strictly decreasing is illustrated as example:

Proof. Suppose $f'(x) < 0$ for all $x \in (a, b)$. Let $x_1, x_2 \in (a, b)$ such that $x_2 > x_1$. Then, f is differentiable on (x_1, x_2), and there exists some value x_0 where $x_1 < x_0 < x_2$ and

$$f'(x_0) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} < 0$$

Thus, $f(x_2) < f(x_1)$, so f is strictly increasing on (a, b). ■

As a final result, the intermediate value theorem may be extended to derivatives.

Intermediate Value Theorem for Derivatives. If f is differentiable on (a, b) and $x_1, x_2 \in (a, b)$ where $x_1 < x_2$ and c is between $f'(x_1)$ and $f'(x_2)$, then there exists $x_0 \in (x_1, x_2)$ such that $f'(x_0) = c$.

If a function is at least twice differentiable, then this result is trivial as f' is continuous. However, this theorem applies more generally, including to functions which are only once differentiable, making this result particularly useful.

2.5 Integration

Definition. A **partition** P of $[a, b]$ is a finite collection of points $P = \{a = t_0 < t_1 < \cdots < t_{n-1} < t_n = b\}$

Notably, each point $t_k \in [a, b]$ is distinct, and the endpoints are *always included* in the partition. Thus, the smallest partition contains two points, i.e. the two endpoints.

Integration is then defined by summation over a partition:

Let f be a bounded function on $[a, b]$. Let P be a partition of $[a, b]$.

The **upper Darboux sum** of f with partition P is defined by

$$U(f, P) = \sum_{k=1}^{n} \sup \{f[t_{k-1}, t_k]\} (t_k - t_{k-1})$$

and the **upper Darboux integral** of f on $[a, b]$ is defined as

$$U(f) = \inf\{U(f, P) | P \text{ partition of } [a, b]\}$$

Similarly, the **lower Darboux sum** of f with partition P is defined by

$$L(f, P) = \sum_{k=1}^{n} \inf \{f[t_{k-1}, t_k]\} (t_k - t_{k-1})$$

and the **lower Darboux integral** of f on $[a, b]$ is defined as

$$L(f) = \sup\{L(f, P) | P \text{ partition of } [a, b]\}$$

f is said to be **integrable** on $[a, b]$ if $U(f) = L(f)$, and thus

$$L(f) = \int_a^b f(x)dx = U(f)$$

In other words, integration is defined by a special type of summation, where a *partition* splits up an interval $[a, b]$ into subintervals. On each subinterval defined by the partition, the greatest or least value is multiplied by the length of that subinterval, thus computing the area of a rectangle defined by f. When the greatest possible lower Darboux sums and the smallest possible upper Darboux sums are equal, this value is said to be the *integral* of f on $[a, b]$.

The following inequality can be shown:

$$\inf\{f[a, b]\}(b - a) \leq L(f, P) \leq L(f) \leq U(f) \leq U(f, P) \leq \sup\{f[a, b]\}(b - a)$$

Informally, this means that for any partition P, the lower and upper Darboux sums are bounded respectively by rectangles with width $b - a$ and the least and greatest value of f on the interval respectively. Additionally, the greatest lower Darboux sum is naturally bounded from below by the lower Darboux sum on any partition, and the least upper Darboux sum is bounded from above by the upper Darboux sum on any partition.

Theorem (Cauchy Integrability Criterion). Let f be bounded on $[a, b]$. f is integrable on $[a, b]$ if and only if for all $\epsilon > 0$, there exists a partition P over $[a, b]$ such that $U(f, P) - L(f, P) < \epsilon$.

Basic facts of integrability can be shown from the definition:

If f is continuous on $[a, b]$, then f is integrable on $[a, b]$.

Let f be integrable on $[a, b]$. If $c, d \in (a, b)$ and $c < d$, then f is integrable on $[c, d]$.

Integration is linear. In particular, if f, g are integrable on $[a, b]$ and $c \in \mathbb{R}$, then $\int_a^b (f+g)(x)dx = \int_a^b f(x)dx + \int_a^b g(x)dx$ and $\int_a^b cf(x)dx = c\int_a^b f(x)dx$.

It is now possible to prove the fundamental theorems of calculus. These theorems essentially relate differentiation and integration as, in essence, "inverse" operations. The 2nd Fundamental Theorem is simpler to prove, and is therefore outlined first:

2nd Fundamental Theorem of Calculus. Suppose f is continuous on $[a, b]$ and differentiable on (a, b). If f' is integrable on $[a, b]$, then

$$\int_a^b f'(x)dx = f(b) - f(a)$$

Proof. Fix $\epsilon > 0$. Since f' is integrable, f' meets the Cauchy criterion, so there exists a partition P such that $U(f', P) - L(f', P) < \epsilon$. By the Mean Value Theorem, for each interval $[t_k, t_{k+1}]$ there exists some x_k such that

$$f'(x_k) = \frac{(f(t_{k+1}) - f(t_k))}{(t_{k+1} - t_k)}$$

and thus $f(t_{k+1}) - f(t_k) = f'(x_k)(t_{k+1} - t_k)$. Thus,

$$f(b) - f(a) = \sum_{k=1}^{n} [f(t_{k+1}) - f(t_k)] = \sum_{k=1}^{n} f'(x_k)(t_{k+1} - t_k)$$

Therefore,

$$L(f', P) \leq f(b) - f(a) \leq U(f', P)$$

Since $U(f', P) - L(f', P) < \epsilon$, and by definition of integration $L(f', P) \leq \int_a^b f'(x)dx \leq U(f', P)$, then

$$0 \leq \int_a^b f'(x)dx - L(f', P) \leq U(f', P) - L(f', P) < \epsilon$$

and $L(f', P) \leq f(b) - f(a)$, so

$$\int_a^b f'(x)dx - L(f', P) \leq \left| \int_a^b f'(x)dx - [f(b) - f(a)] \right| < \epsilon$$

■

In other words, if a function f' is known to be the derivative of another function f, computing $\int_a^b f'(x)dx$ simply involves computing the difference between f at the endpoints of the interval.

1st Fundamental Theorem of Calculus. Let f be integrable on $[a, b]$. Define $F(x) = \int_a^x f(t)dt$, then F is uniformly continuous on $[a, b]$. If f is continuous at $x_0 \in (a, b)$, then F is differentiable at x_0 with $F'(x_0) = f(x_0)$.

Proof. Uniform Continuity. Fix $\epsilon > 0$. f is integrable and therefore bounded by the Cauchy criterion for integration, so there exists some $M > 0$ such that $|f(x)| \leq M$ for all $x \in [a, b]$. If $x, y \in [a, b]$, and $|x - y| < \epsilon/M$, then (assuming without loss of generality that $x < y$):

$$|F(y) - F(x)| = \left| \int_x^y f(t)dt \right| \leq \int_x^y |f(t)|dt \leq \int_x^y M dt = M(y - x) = M|x - y| < \epsilon$$

Thus, $|F(y) - F(x)| < \epsilon$ and F is uniformly continuous on $[a, b]$.

Differentiability. Suppose f is continuous at $x_0 \in (a, b)$. For any $x \neq x_0$,

$$\frac{F(x) - F(x_0)}{x - x_0} = \frac{1}{x - x_0} \int_{x_0}^x f(t)dt$$

and

$$f(x_0) = \frac{1}{x - x_0} \int_{x_0}^x f(x_0)dt$$

by simple evaluation of the constant integral. Thus,

$$\frac{F(x) - F(x_0)}{x - x_0} - f(x_0) = \int_{x_0}^x [f(t) - f(x_0)]dt$$

Fix $\epsilon > 0$. Because f is continuous at x_0, there exists $\delta > 0$ such that if $t \in (a, b)$ and $|t - x_0| < \delta$, then $|f(t) - f(x_0)| < \epsilon$. Thus,

$$\left| \frac{F(x) - F(x_0)}{x - x_0} - f(x_0) \right| \leq \frac{1}{x - x_0} \int_{x_0}^x \epsilon dt = \epsilon$$

Therefore,

$$\lim_{x \to x_0} \frac{F(x) - F(x_0)}{x - x_0} = f(x_0)$$

and thus $F'(x_0) = f(x_0)$. ■

The fact that $\left| \int_x^y f(t)dt \right| \leq \int_x^y |f(t)|dt$ can be shown by evaluating the Darboux sums of each respectively. Intuitively, $|f(t)|$ is always nonnegative, so summing multiples of $|f(t)|$ must be at least as large as the absolute value of the original integral (which may come out positive, negative, or zero before taking the absolute value, i.e. a different number and one necessarily smaller than $\int_x^y |f(t)|dt$ if f is negative for any value on $[x, y]$).

This fundamental theorem gives two facts: first, if f is integrable, its integral function F is uniformly continuous and therefore continuous; second, if the original function f is continuous, then F is differentiable and its derivative at a point is the value of f at that point.

3 Point-Set Topology

Notes from **Math 490: Introduction to Topology**

Taken at University of Michigan, Fall 2025

3.1 Metric Spaces

Definition. Let X be a set. A **metric** is a function

$$d : X \times X \to \mathbb{R}$$

satisfying the following conditions for all $x, y, z \in X$:

Positivity. $d(x, y) \geq 0$ and $d(x, y) = 0$ if and only if $x = y$

Symmetry. $d(x, y) = d(y, x)$

Triangle Inequality. $d(x, y) + d(y, z) \geq d(x, z)$

A set X with a metric d is called a **metric space**, denoted (X, d).

d is sometimes called the *distance* from x to y. The most familiar metric, the **Euclidean metric**, mapping from $\mathbb{R}^n \times \mathbb{R}^n$, is given by

$$d(\vec{x}, \vec{y}) = ||\vec{x} - \vec{y}|| = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2} \qquad \text{(Euclidean Metric)}$$

Note that the Euclidean metric is indeed a metric, and thus $(\mathbb{R}^n, d)$ is a metric space:

Proof. Let $d : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ such that $d(\vec{x}, \vec{y}) = ||\vec{x} - \vec{y}||$. Let $\vec{x}, \vec{y}, \vec{z} \in \mathbb{R}^n$.

Positivity. Note that $d(\vec{x}, \vec{y}) = ||\vec{x} - \vec{y}|| = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}$. For any $a, b \in \mathbb{R}$, $(a - b)^2 \geq 0$, so $\sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2} \geq \sqrt{0} = 0$, so $d(\vec{x}, \vec{y}) \geq 0$.

Suppose $\vec{x} = \vec{y}$. Then $x_1 - y_1 = \cdots = x_n - y_n = 0$, so $d(\vec{x}, \vec{y}) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2} = \sqrt{0} = 0$.

Now suppose instead that $d(\vec{x}, \vec{y}) = 0$. Then, $\sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2} = 0$. Since $(a - b)^2$ is nonnegative for all $a, b \in \mathbb{R}$, then $(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2 = 0$ implies $x_1 - y_1 = \cdots = x_n - y_n = 0$. Thus, $\vec{x} - \vec{y} = 0$, or $\vec{x} = \vec{y}$.

Thus, d satisfies positivity.

Symmetry. Note that $(a - b)^2 = (b - a)^2$ for all $a, b \in \mathbb{R}$. Thus, $d(\vec{x}, \vec{y}) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2} = \sqrt{(y_1 - x_1)^2 + \cdots + (y_n - x_n)^2} = d(\vec{y}, \vec{x})$.

Triangle Inequality. Note that:

$$\begin{aligned}
d(\vec{x}, \vec{z})^2 &= (x_1 - z_1)^2 + \cdots + (x_n - z_n)^2 \\
&= (x_1 - y_1 + y_1 - z_1)^2 + \cdots + (x_n - y_n + y_n - z_n)^2 \\
&= (x_1 - y_1)^2 + (y_1 - z_1)^2 + 2(x_1 - y_1)(y_1 - z_1) + \cdots + (x_n - y_n)^2 + (y_n - z_n)^2 + 2(x_n - y_n)(y_n - z_n) \\
&= (x_1 - y_1)^2 + \cdots + (x_n - y_n)^2 + 2\left[(x_1 - y_1)(y_1 - z_1) + \cdots + (x_n - y_n)(y_n - z_n)\right] \\
&\leq d(\vec{x}, \vec{y})^2 + d(\vec{y}, \vec{z})^2 + 2d(\vec{x}, \vec{y})d(\vec{y}, \vec{z}) \\
&= (d(\vec{x}, \vec{y}) + d(\vec{y}, \vec{z}))^2
\end{aligned}$$

Because square roots are increasing and squares are strictly nonnegative, by taking square roots we have $d(\vec{x}, \vec{z}) \leq d(\vec{x}, \vec{y}) + d(\vec{y}, \vec{z})$.

■

The following definitions define what it means for a subset of a metric space to be open or closed (or neither, or both). These definitions are particularly useful because they generalize the properties of open and closed intervals of Euclidean space to any metric space.

Definition. (Open Ball). Let (X, d) be a metric space. Let $x_0 \in X$. Let $r > 0$. Then, the **open ball** of radius r about x_0 is the subset of X defined by

$$B_r(x_0) = \{x \in X \mid d(x_0, x) < r\} \subseteq X$$

Definition. (Interior Point and Open Sets in a Metric Space). Let (X, d) be a metric space. Let $U \subseteq X$. A point $u \in U$ is called an **interior point** of U if there exists some radius $r_u > 0$ such that $B_r(u) \subseteq U$. The set $U \subseteq X$ is called **open** if every point $u \in U$ is an interior point of U.

Definition. (Closed Sets in a Metric Space). A subset $U \subseteq X$ is **closed** if its complement $X \backslash U$ is open.

Note that it is possible for a set to be both open and closed, or neither open nor closed. In other words, openness and closure are *not* mutually exclusive.

Definition. (Accumulation and Isolation Points of a Set). Let (X, d) be a metric space. Let $S \subseteq X$ be a set. A point $x \in X$ is an **accumulation point** of S if for all $r > 0$, the open ball $B_r(x)$ contains a point of S distinct from x. An element $s \in S$ which is not an accumulation point of S is called an **isolated point** of S if there exists $r > 0$ such that $B_r(s)$ contains no points of S except for s.

Note that, by definition, an isolation point of S must be an element of S, but an accumulation point of S need not be an element of S. For example, the interval $S = (0, 1) \subseteq \mathbb{R}$ has accumulation points $[0, 1]$, even thought $0, 1 \notin S$.

Definition. ("The" Product Metric). Let (X, d_X) and (Y, d_Y) be metric spaces. Then, the Cartesian product $X \times Y$ may be endowed with a metric $d_{X \times Y} : (X \times Y) \times (X \times Y) \to \mathbb{R}$, most commonly defined by

$$d_{X \times Y}((x_1, y_1), (x_2, y_2)) = \sqrt{d_X(x_1, x_2)^2 + d_Y(y_1, y_2)^2}$$

With these definitions, it is possible to discuss **continuity** on a metric space. Note that there are several equivalent definitions of continuity, but the most simple definition is essentially a generalization of the definition of continuity on the real number line.

Continuity of functions. Let (X, d_X) and (Y, d_Y) be metric spaces. Let $x, x' \in X$.

Definition. ($\varepsilon\delta$ Continuity). A function $f : X \to Y$ is called *continuous at* x if, for all $\varepsilon > 0$, there exists $\delta > 0$ such that, if $d_X(x, x') < \delta$, then $d_Y(f(x), f(x')) < \varepsilon$. f is called *continuous* if it is continuous at every $x \in X$.

A function which is continuous also satisfies the following equivalent definitions:

1. f is continuous if for all $\varepsilon > 0$, there exists $\delta > 0$ such that $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$.
2. f is continuous if, given any open set $U \subseteq Y$, $f^{-1}(U) \subseteq X$ is open.
3. f is continuous if, given any open ball $B = B_r(y) \subseteq Y$, its preimage $f^{-1}(B)$ is an open subset of X.

Note that, considering the real numbers under the Euclidean metric, the $\varepsilon\delta$ definition of continuity reduces to the familiar definition of continuity. Using this analogy, alternative definition (1) becomes fairly obvious: if a point is within the δ-ball, its image maps within the ε-ball.

Alternative definition (2) is particularly useful, as it will generalize to spaces which do not have a notion of distance (i.e. generalized topological spaces). Because of its importance, a proof of alternative definition (2) is provided.

Proof. Let (X, d_X) and (Y, d_Y) be metric spaces.

Suppose f is continuous. Let $U \subseteq Y$ be open. If $x \in f^{-1}(U)$, then $f(x) \in U$. Since U is open, there exists $\varepsilon > 0$ such that $B_\varepsilon(f(x)) \subseteq U$. Since f is continuous, for all $\varepsilon > 0$ there exists $\delta > 0$ such that $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$. Thus, $f(B_\delta(x)) \subseteq U$. Therefore, $B_\delta(x) \subseteq f^{-1}(U)$. Therefore, x is an interior point of $f^{-1}(U)$, so $f^{-1}(U)$ is open.

Suppose f is a function such that, for any open set $U \subseteq Y$, $f^{-1}(U) \subseteq X$ is open. Let $x \in X$. Fix $\varepsilon > 0$. Because the preimage of an open set is open, by assumption we have that $f^{-1}(B_\varepsilon(f(x)))$ is open because open balls are open. Since $x \in f^{-1}(B_\varepsilon(f(x)))$, x is an interior point of $f^{-1}(B_\varepsilon(f(x)))$. Therefore, there exists $\delta > 0$ such that $B_\delta(x) \subseteq f^{-1}(B_\varepsilon(f(x)))$. Therefore, $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$.

Therefore, the $\varepsilon\delta$ definition of continuity holds if and only if for all open sets $U \subseteq Y$, the preimage $f^{-1}(U)$ is open. Therefore, the two definitions are equivalent. ■

Similarly to continuity, a convergent sequence in a metric space can be defined by extending the familiar definition from analysis.

Definition. (Convergent Sequences). Let (X, d) be a metric space. Let (x_n) be a sequence indexed by the natural numbers. The sequence *converges* to x_∞, denoted $\lim_{n\to\infty} x_n = x_\infty$, if, for all $\varepsilon > 0$, there exists $N > 0$ such that, if $n > N$, then $d(x_n, x_\infty) < \varepsilon$.

Note that, for the metric space $\mathbb{R}$ endowed with the Euclidean metric, the definition of convergence reduces to the definition from analysis. As with continuity, convergence may be equivalently defined with open sets.

Definition. (Equivalent Definition of Convergence). Let (X, d) be a metric space. A sequence (x_n) is convergent if, for any open set $U \subseteq X$ which contains x_∞, there exists some $N > 0$ such that, if $n > N$, then $x_n \in U$.

The definition of convergent sequences yields yet another (equivalent) definition for continuous functions:

Sequence Definition of Continuity. f is continuous if, given any sequence (x_n) of points in X with limit $x_\infty \in X$, $\lim\limits_{n\to\infty} f(x_n) = f(\lim\limits_{n\to\infty} x_n)$.

The following is a useful criteria which can often be used to construct a convergent sequence in a metric space:

Convergence Criteria for Sequences in Metric Spaces. Suppose (x_n) is a sequence of points in X with the property that $x_n \in B_{\frac{1}{n}}(x)$. Then, (x_n) converges to x.

Proof. Fix $\varepsilon > 0$. Choose $N > \frac{1}{\varepsilon}$. Then, if $n > N$, $x_n \in B_{\frac{1}{N}}(x)$ and $d(x_n, x) < \frac{1}{N} < \varepsilon$. Therefore, (x_n) converges to x. ■

A similarly useful criteria characterizes *closed* sets and is provided without proof. The criteria essentially provides a way to consider whether a set is closed without considering the complement set:

Alternate Characterization of Closed Subsets. A set $A \subseteq X$ is closed if any only if for any convergent sequence of points (a_n) in A converging to a point $a_\infty \in X$, the limit $a_\infty \in A$.

Note that many results which hold on $\mathbb{R}$ with the standard Euclidean metric do *not* hold in a general metric space. For example, Cauchy sequences are *not in general* equivalent to convergent sequences. In fact, $\mathbb{R}$ with the Euclidean metric is an example of a *complete* metric space, a special type of metric space where Cauchy sequences are equivalent to convergent sequences.

Complete Metric Spaces. Let (X, d) be a metric space. Let (x_n) be a sequence in X.

Definition. (Cauchy Sequence). (x_n) is a Cauchy sequence if, for all $\varepsilon > 0$, there exists $N > 0$ such that, if $n, m > N$, then $d(x_n, x_m) < \varepsilon$.

Definition. (Complete Metric Spaces). (X, d) is a **complete** metric space if every Cauchy sequence in X converges.

Fact. If (x_n) is a convergent sequence, (x_n) is a Cauchy sequence.

A metric space fails to be complete if a sequence in X is able to become arbitrarily close to a point not in X. In precisely this case, there exist Cauchy sequences which are not convergent. Specifically, in this case, there exist Cauchy sequences which limit to a point not contained in X, therefore failing to converge.

Definition. Let (X, d) be a metric space. Let (x_n) be a sequence in X. Let (n_i) be a strictly increasing sequence of natural numbers (note that $n_i \geq i$). The sequence (x_{n_i}) is called a **subsequence** of (x_n).

Familiarly, if (x_n) converges to x_∞, then any subsequence x_{n_i} must also converge to x_∞.

Sequentially Compact Subspaces. A metric space (X, d) is called **sequentially compact** if every sequence in X has a convergent subsequence. Similarly, a set $S \subseteq X$ is sequentially compact if every sequence of points in S has a subsequence that converges to a point in S.

A sequentially compact subspace $S \subseteq X$ has the following properties:

1. S **is closed**. That is, $X \backslash S$ is open.
2. S **is bounded**. That is, there exists $R > 0$ such that, for every $x, y \in S$, $d(x, y) < R$.

Theorem. (Extreme Value Theorem for Metric Spaces). Let C be a nonempty, sequentially compact subset of X. Let $f : X \to \mathbb{R}$ be a continuous function. Then, there exists a point $c \in C$ such that

$$f(c) = \sup_{x \in C} f(x)$$

Theorem. Closed Subsets of Sequentially Compact Sets are Sequentially Compact. Suppose that (X, d) is a sequentially compact metric space, and $C \subseteq X$ is closed. Then, C is sequentially compact.

Theorem. Sequentially Compact Implies Complete. Suppose (X, d) is a sequentially compact metric space. Then, it is complete.

On a subspace of $\mathbb{R}^n$, sequential compactness is equivalent to closure and boundedness (by the Heine-Borel theorem from analysis). Again, this result does *not hold* for metric spaces in general.

Theorem. Sequentially Compact Continuous Images are Sequentially Compact. Suppose $f : X \to Y$ is a continuous function. If $C \subseteq X$ is sequentially compact, then $f(C) \subseteq Y$ is sequentially compact.

Proof. Suppose C is sequentially compact. Let (c_n) be a sequence in C. Because C is sequentially compact, there exists a subsequence (c_{n_k}) converging to $c_{n_\infty} \in C$. Since f is continuous, $\lim_{n \to \infty} f(c_{n_k}) = f(c_{n_\infty}) \in f(C)$. The choice of (c_n) is arbitrary, so for any sequence $(f(c_n))$ in $f(C)$, there exists a subsequence $(f(c_{n_k}))$ in $f(C)$ converging in $f(C)$. Thus, $f(C)$ is sequentially compact. ■

3.2 Topological Spaces

Definition. Topological Space. Let X be a set. A *topology* $\mathcal{T}$ on X is a collection of subsets of X that satisfies the following properties:

The ambient space and empty set are open. $\varnothing \in \mathcal{T}$ and $X \in \mathcal{T}$.

Arbitrary unions are open. If $\{U_i\}_{i \in I}$ is any collection of elements of $\mathcal{T}$, then $\bigcup_{i \in I} U_i \in \mathcal{T}$

Finite intersections are open. If $U_1, U_2 \in \mathcal{T}$, then $U_1 \cap U_2 \in \mathcal{T}$.

A set X endowed with a topology $\mathcal{T}$ is called a **topological space** and is denoted $(X, \mathcal{T})$. A set $C \subseteq X$ is called **closed** if its complement $X \backslash C \in \mathcal{T}$.

Essentially, a topology specifies a space X and the collection of subsets of X, i.e. those contained in $\mathcal{T}$, which are open. Note that $\mathcal{T}$ is a subset of the power set of X. Some examples of topologies on a set X are listed:

The **indiscrete topology**, where the only open sets are $\varnothing$ and X. This topology is given by $\mathcal{T} = \{\varnothing, X\}$. This is the "coarsest" topology on X, containing the fewest allowable elements.

The **discrete topology**, where *every* subset of X is open. This topology is given by the power set of $\mathcal{P}(X)$; i.e. $\mathcal{T} = \mathcal{P}(X) = \{U \mid U \subseteq X\}$. This is the "finest" topology on X, containing maximum elements (every subset of X, including singletons, is open).

The **topology induced by a metric space**. Let (X, d) be a metric space. Define $\mathcal{T}_d$ as the collection of all open sets in X under the metric d. Then, $(X, \mathcal{T}_d)$ forms a topological space. A topology $\mathcal{T}$ on X is called *metrizable* if there exists a metric d on X such that $\mathcal{T}$ is the set $\mathcal{T}_d$ of open sets in the metric space (X, d).

Example of Metrizability. Let X be a set. Consider the metrizability of the discrete and indiscrete topology:

The discrete topology *is* metrizable. Let d be the discrete metric, given by

$$d(x, y) = \begin{cases} 0, & x = y \\ 1, & x \neq y \end{cases}$$

Note that if $x \in X$, then $x \in B_{\frac{1}{2}}(x) = \{x\}$, so singletons are open. Because singletons are open, the union of singletons is open, so every set in X is open with the discrete metric. Thus, (X, d) induces the discrete topology.

The indiscrete topology is *not* metrizable if X contains at least two distinct elements. Let d be any metric on X. If $x_1, x_2 \in X$ are distinct, then for $r = d(x_1, x_2)$, $x_2 \notin B_r(x_1)$ and $x_1 \in B_r(x_1)$. Therefore, $B_r(x_1) \neq \varnothing$ and $B_r(x_1) \neq X$, so d does not induce the indiscrete topology. The choice of metric is arbitrary, so the indiscrete topology is not metrizable.

Note that the following properties hold in general topological spaces:

1. X and $\varnothing$ are closed.

   Proof. Note that $X \backslash X = \varnothing$ which is open. Therefore, X is closed. Similarly, $X \backslash \varnothing = X$ which is open. Therefore, $\varnothing$ is closed. ■

2. If B and C are closed subsets of X, then $B \cup C$ is closed.

   Proof. Note that $X \backslash (B \cup C) = (X \backslash B) \cap (X \backslash C)$ which is a finite intersection of open sets and is therefore open. Therefore, $B \cup C$ is closed. ■

3. For a collection $\{C_i\}_{i\in I}$ of closed subsets of X, the arbitrary intersection $\bigcap_{i\in I} C_i$ is closed.

Proof. Note that $X\backslash\bigcap_{i\in I} C_i = \bigcup_{i\in I}(X\backslash C_i)$ by DeMorgan's laws, which is open. Therefore, $\bigcap_{i\in I} C_i$ is closed. ■

Subspace Topology. Let $(X, \mathcal{T}_X)$ be a topological space. Let $S \subseteq X$. Then, S inherits the structure of a topological space called the *subspace topology*, $\mathcal{T}_S$, defined by:

$$\mathcal{T}_S = \{U \cap S \mid U \in \mathcal{T}_X\}$$

It is often convenient to work with some collection of sets which *generates* a topology, rather than with a topology itself. Such a generating set is called a *basis* for a topology.

Definition. (Basis of a topology). Let $(X, \mathcal{T})$ be a topological space. A collection $\mathcal{B}$ of subsets of X is called a **basis** for the topology $\mathcal{T}$ if:

- $\mathcal{B} \subseteq \mathcal{T}$, so every basis element in open
- Every element of $\mathcal{T}$ can be expressed as a union of elements of $\mathcal{B}$.

The basis $\mathcal{B}$ "generates" the topology $\mathcal{T}$.

Note that the term "basis" is analogous to a "spanning set", as a basis for a topology has no requirement of minimality. In fact, the basis $\mathcal{B} = \mathcal{T}$ is a (trivial) example of a basis, and many topologies do not have a finite basis (and therefore a minimal basis does not necessarily exist). For example, a basis for the Euclidean topology on the real numbers contains infinitely many open intervals.

The following set of definitions generalize several concepts from metric spaces; for this reason, these definitions align with intuition on certain familiar metric spaces as Euclidean space, but in fact generalize concepts without a notion of distance (i.e. to non-metrizable topological spaces).

Let $(X, \mathcal{T})$ be a topological space, and $A \subseteq X$.

Definition. (Interior of a Set). The interior of A, denoted $\text{Int}(A)$, is defined as the set

$$\text{Int}(A) = \{a \in A \mid a \text{ is an interior point of } A\}$$

$\text{Int}(A)$ is an open set, and is essentially the "largest" open subset of A.

Definition. (Closure of a Set). The closure of A, denoted $\overline{A}$, is defined as the set

$$\overline{A} = \{x \in X \mid \text{every open neighborhood } U \text{ of } x \text{ contains a point of } A\}$$

The closure $\overline{A}$ is a closed set, and is essentially the "smallest" closed set containing A.

Definition. (Accumulation Points of a Set). The accumulation points of A, contained in the set denoted A', are defined by

$$A' = \{x \in X \mid \text{every open neighborhood } U \text{ of } x \text{ contains a point of } A \text{ distinct from } x\}$$

Definition. (Boundary of a Set). The boundary of A, denoted ∂A, is defined as the set

$$\partial A = \{x \in X \mid \text{every open neighborhood } U \text{ of } x \text{ contains a point in } A \text{ and a point in } X\backslash A\}$$

Note that $\partial A = \partial(X\backslash A)$, i.e. ∂A is a "shared" boundary between A and the ambient space.

Note that every point of X is contained in precisely one of: $\text{Int}(A)$, $\text{Int}(X\backslash A)$, or $\partial A = \partial(X\backslash A)$.

By considering the equivalent open-set definition of continuity, the notion of continuous functions can immediately be generalized in a topological space.

Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces. Let $f : X \to Y$. f is called *continuous* if, for every set $U \subseteq Y$ open in Y, $f^{-1}(U) \subseteq X$ is open in X.

It can be shown that, equivalently, f is continuous if for every closed set $C \subseteq Y$, $f^{-1}(C) \subseteq X$ is closed.

By considering the open-set definition of sequential convergence, this notion is also immediately generalized.

Let $(X, \mathcal{T}_X)$ be a topological space. A sequence (x_n) converges to x_∞ if, for any open neighorhood $U \subseteq X$ of x_∞, there exists some $N > 0$ such that, if $n > N$ then $x_n \in U$.

Note that, in a general topological space, *limits are not necessarily unique.* That is, a sequence may converge to more than one limit.

Example (Sierpinski Topology). Consider the set $\mathbb{S} = \{0, 1\}$ endowed with the topology $\mathcal{T}_\mathbb{S} = \{\varnothing, \{0\}, \{0, 1\}\}$. Consider the following sequences:

$$\{a_n\} = \{0,\ 0,\ 0,\ 0,\ \dots\}, \qquad \{b_n\} = \{1,\ 1,\ 1,\ 1,\ \dots\}, \qquad \{c_n\} = \{1,\ 0,\ 1,\ 0,\ \dots\}$$

Note that the open neighborhoods of 0 are $\{0\}$ and $\{0, 1\}$. Then, we see that (a_n) converges to 0 because $a_n \in \{0\}$ and $a_n \in \{0, 1\}$, so (a_n) is contained in every open neighborhood of 0 for all n. Similarly, the only open neighborhood of 1 is $\{0, 1\}$, and $a_n \in \{0, 1\}$ for all n, so (a_n) converges to 1 as well.

By similar logic, we can see that (b_n) converges to 1, and we also have that (c_n) converges to 1, which follows from the fact that the only open neighborhood containing 1 is $\{0, 1\}$, which is the whole space.

The Product Topology. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces. Then, the *product topology* on $X \times Y$, denoted $\mathcal{T}_{X \times Y}$, is generated by the basis

$$\mathcal{B} = \{U \times V \mid U \subseteq X \text{ is open, and } V \subseteq Y \text{ is open}\}$$

Some special features of the product topology will be discussed.

Property 1. Suppose (X, d_X) and (Y, d_Y) are metric spaces. Then, the product metric $d_{X \times Y}$ induces the product topology on $X \times Y$. That is to say, the product metric has the same structure as the product topology. For this reason, the following properties also apply to the product metric, but are shown in more generality here.

Property 2. The projection maps $\pi_X : X \times Y \to X$, which maps $(x, y) \mapsto x$, and $\pi_Y : X \times Y \to Y$, which maps $(x, y) \mapsto y$, are continuous.

Proof. Without loss of generality, we show that π_X is continuous. Let $U \subseteq X$ be an open subset of X. Then, $\pi_X^{-1}(U) = U \times Y$. Note that $U \subseteq X$ is open in X and $Y \subseteq Y$ is open in Y, so $U \times Y \in \mathcal{B}$. Basis elements are open, so $U \times Y = \pi_X^{-1}(U)$ is open in the product topology on $X \times Y$. Our choice of U is arbitrary, so π_X is continuous. ■

The projection maps are often useful when something is known about X or Y, but not their product; because they are continuous, the project maps allow one to work with subsets of either X or Y while preserving the structure of the topology.

Property 3. A function $f : X \to Y_1 \times Y_2$, given by $f(x) = (f_1(x), f_2(x))$ is continuous if and only if $f_1(x)$ and $f_2(x)$ are continuous.

Proof. Suppose f is continuous. Then $\pi_{Y_1}(f)$ and $\pi_{Y_2}(f)$ are continuous, because the composition of continuous functions is continuous. Note that $\pi_{Y_1}(f) = f_1$ and $\pi_{Y_2}(f) = f_2$, so both f_1 and f_2 are continuous.

Suppose f_1 and f_2 are continuous. Let $W \subseteq Y_1 \times Y_2$ be any open subset. Note that W can be expressed as a union of basis elements, so $W = \bigcup_{i \in I}(U_i \times V_i)$ for some index I, where each $U_i \subseteq Y_1$ is open and each $V_i \subseteq Y_2$ is open. Then,

$$f^{-1}(W) = f^{-1}\left(\bigcup_{i \in I}(U_i \times V_i)\right) = \bigcup_{i \in I} f^{-1}(U_i \times V_i)$$

Note that this is an arbitrary union of basis elements, so it suffices to show that $f^{-1}(U_i \times V_i)$ is open for an arbitrary basis element. Note that, if $x \in f^{-1}(U_i \times V_i)$, then $x \in f_1^{-1}(U_i)$ and $x \in f_2^{-1}(V_i)$. Then, $x \in f_1^{-1}(U_i) \cap f_2^{-1}(V_i)$. Note that $f_1^{-1}(U_i)$ is open and $f_2^{-1}(V_i)$ is open, so their finite intersection is open. Therefore, $f^{-1}(W)$ is open, so f is continuous.

Therefore, f is continuous if and only if f_1 and f_2 are continuous. ■

3.3 Structured Topological Spaces

When the discussion of topological spaces is restricted to the less general case of topologies which have some additional structure, several useful results can be shown.

Separation Axioms. Let $(X, \mathcal{T}_X)$ be a topological space. Let $x, y \in X$ be any distinct points.

T_1 **Space**. X is called T_1 if there exists an open neighborhood U_x of x that does not contain y, and an open neighborhood U_y of y that does not contain x.

T_2 **Space**. X is called T_2 if there exist disjoint open neighborhoods U_x of x and U_y of y. This is called the **Hausdorff property**, and T_2 spaces are equivalently called Hausdorff spaces.

T_3 **Space**. X is called T_3 if it is a *regular* T_0 space. X is called *regular* if, for every closed subset C not containing x, there exist disjoint open subsets U_x and U_C containing x and C, respectively.

T_4 **Space**. X is called T_4 if it is a normal T_2 space. X is called *normal* if, for any pair of disjoint closed subsets C and D, there exist disjoint open sets U_C and U_D containing C and D, respectively.

There are more separation axioms, but these ones are sufficient for many results. In particular, T_1 and T_2 spaces have several nice properties which facilitate useful results. The separation axioms give strictly stronger conditions on disjoint open neighborhoods around distinct points. As a consequence, it can be shown that $T_4 \implies T_3 \implies T_2 \implies T_1$, and so on for other separation axioms. The visual meaning of the separation axioms is shown below, where dotted balls represent open sets and solid boxes represent closed sets:

A topology having the T_1 separation axiom is equivalent to the statement that "points are closed," i.e. every singleton is a closed set.

Proof. Suppose $(X, \mathcal{T}_X)$ is T_1. Consider the singleton $\{x\}$. We wish to show that $X \backslash \{x\}$ is open, meaning any point $y \in X \backslash \{x\}$ is an interior point. Because X is T_1, there exists an open neighborhood U_y of y which does not contain x. Then, $U_y \subseteq X \backslash \{x\}$ is an open neighborhood of y contained in $X \backslash \{x\}$ and therefore y is an interior point. Then, $X \backslash \{x\}$ is open, so $\{x\}$ is closed.

Suppose $(X, \mathcal{T}_X)$ has the property that every singleton $\{x\}$ is closed. Then, $X \backslash \{x\}$ is open. Then, for every $y \in X$ distinct from x, y is an interior point of $X \backslash \{x\}$. Then, there exists some open neighborhood U_y of y contained in $X \backslash \{x\}$. Then, $x \notin U_y$. Then, X is T_1.

Therefore, X is T_1 if and only if points (i.e. singleton sets) are closed. ■

Another important result is that metric spaces are T_2 topological spaces.

Proof. Suppose (X, d) is a metric space. Let $x, y \in X$ be distinct points. Let $r = d(x, y)/2$. We now show that $B_r(x)$ and $B_r(y)$ are disjoint open neighborhoods of x and y respectively.

Suppose for sake of contradiction that $B_r(x)$ and $B_r(y)$ are not disjoint, so there exists some $z \in B_r(x) \cap B_r(y)$. Then, $d(z, x) < r$ and $d(z, y) < r$. By the triangle inequality, $d(x, z) + d(y, z) \geq d(x, y)$. But then,

$$r + r > d(x, z) + d(y, z) \geq d(x, y) = 2r$$

Then, we have that $2r > 2r$, which is not true. Therefore, no such z can exist. Therefore, $B_r(x) \cap B_r(y)$ are disjoint. Clearly $B_r(x)$ is an open neighborhood of x, and clearly $B_r(y)$ is an open neighborhood of y, so there exist disjoint open neighborhoods of x and y, specifically $B_r(x)$ and $B_r(y)$. Therefore, (X, d) is T_2. ■

To summarize, this results in the following theorems:

Theorem. Metric spaces are Hausdorff. Notably, if a topological space is not Hausdorff, it is not metrizable.

Theorem. A topological space is T_1 if and only if points are closed.

Note that, because $T_2 \implies T_1$, points are closed in T_2 spaces as well.

Hausdorff spaces tend to generalize many properties of metric spaces; in particular, Hausdorff spaces generalize the notion of "disjoint open balls" without the notion of distance. One such "nice" property is that sequences in Hausdorff spaces have unique limits.

Theorem. If X is Hausdorff and (x_n) is a convergent sequence, then its limit is unique.

Proof. Let X be a Hausdorff space. Suppose (x_n) converges to a and (x_n) converges to b. We seek to show that $a = b$. Assume for sake of contradiction that $a \neq b$. Since X is Hausdorff, there exist disjoint open neighborhoods U_a of a and U_b of b. Since (x_n) converges to a, there exists $N_a > 0$ such that if $n > N_a$, then $x_n \in U_a$. Similarly, there exists $N_b > 0$ such that if $n > N_b$, then $x_n \in U_b$. Let $N = \max(N_a, N_b)$. Then, if $n > N$, we have that $x_n \in U_a$ and $x_n \in U_b$. However, we have that U_a and U_b are disjoint, which is a contradiction. Then, it must be the case that $a = b$, so limits are unique. ■

Connected and Path-Connected Subspaces. Let $(X, \mathcal{T})$ be a topological space. We say that X is **disconnected** if there exist disjoint, nonempty, open subsets A and B such that $X = A \cup B$. The sets A and B are a *separation* of X. If no such separation exists, X is called *connected*. A *connected component* of X is the maximal connected subset of X (and a disconnected set will have more than one connecting component).

Two important results are given without proof:

- X is disconnected if and only if there exists a nontrivial subset A which is both open and closed (i.e. there exists clopen A where $A \neq \varnothing$ and $A \neq X$).
- **Generalized Intermediate Value Theorem**. Let X be connected and $f : X \to (\mathbb{R}, \text{Euclidean})$ be continuous. If $x, y \in X$ and $f(x) < c < f(y)$, then there exists some $z \in X$ such that $f(z) = c$.

Let X be a topological space, with $x, y \in X$. A *path* from x to y is any continuous function $\gamma : ([0, 1], \text{Euclidean}) \to (X, \mathcal{T})$ such that $\gamma(0) = x$ and $\gamma(1) = y$. The space $(X, \mathcal{T})$ is **path-connected** if given any x and y, there exists a path from x to y.

Three important results of path-connectedness are given without proof:

- If X is path-connected, then X is connected. Note that the opposite is not true, i.e. there exists connected spaces which are *not* path connected.
- Suppose $f : X \to Y$ is a continuous map between topological spaces. If X is path-connected, then $f(X)$ is path-connected.
- The product $X \times Y$ with the product topology is connected if and only if both X and Y are connected.

Definition (Open Cover). Let $(X, \mathcal{T})$ be a topological space. A collection $\{U_i\}_{i\in I}$ such that $\bigcup_{i\in I} U_i = X$ is called an *open cover* of X. A sub-collection $\{U_i\}_{i\in I'}$, with $I' \subseteq I$, is called an *open subcover* of X if $\{U_i\}_{i\in I'}$ is also a cover of X.

Note that one can also talk about an open cover of some subset $A \subseteq X$. There are two equivalent ways to describe covers of A: either $A = \bigcup_{i\in I} U_i$ where each U_i is open in the subspace topology $\mathcal{T}_A$, or $A \subseteq \bigcup_{i\in I} U_i$ where each U_i is open in X. Note that every set has at least one cover, with the entire space X being a trivial example.

Compactness. A topological space $(X, \mathcal{T})$ is **compact** if **every** open cover of X has a finite subcover.

Several important results for compact spaces are outlined:

Theorem. Closed subsets of compact spaces are compact.

Proof. Let $(X, \mathcal{T})$ be a compact topological space. Suppose $A \subseteq X$ is closed. Let $\{U_i\}_{i\in I}$ be any open cover of A. We wish to show that there exists a finite subcover of A. A is closed, so $X\backslash A$ is open. Since $\{U_i\}_{i\in I}$ is an open cover of A, we have that $(X\backslash A) \cup \left(\bigcup_{i\in I} U_i\right) = X$. Since this collection covers X, there exists a finite collection $(X\backslash A) \cup \left(\bigcup_{i\in I'} U_i\right)$ of sets which also covers X, where $I' \subseteq I$ is finite. Then, it must be the case that $A \subseteq \bigcup_{i\in I'} U_i$. Then, there exists some finite collection $\{U_i\}_{i\in I'}$ which is a finite subcover of A. Then, A is compact. ■

Theorem. A subset of a Hausdorff topological space is closed if and only if it is compact.

Proof. Let $(X, \mathcal{T})$ be a Hausdorff topological space. We have already shown that, if X is compact, then closed subsets are compact. We must only show that, if subsets are compact, then they are closed.

Let $A \subseteq X$ be a compact subset of X. We seek to show that A is closed, meaning $X\backslash A$ is open. Fix $x \in X\backslash A$. Because X is Hausdorff, for every $a \in A$, there exist disjoint open neighborhoods U_a of a and V_a of x. Note that $\{U_a\}_{a\in A}$ is an open cover of A, because every point of A is included in some U_a. Because A is compact, there exists some finite collection of points $a_1, \ldots, a_n$ where $\{U_{a_i}\}_{i\leq n}$ is a finite subcover of A. Define $V = \bigcap_{i=1}^{n} V_{a_i}$. Note that $x \in V_{a_i}$ for each (finitely many) i, so V is an open neighborhood of x contained in $X\backslash A$. Then, x is an interior point of $X\backslash A$. Our choice of x is arbitrary, so $X\backslash A$ is open. Therefore, A is closed.

Therefore, we have that a subset of a compact Hausdorff space is compact if and only if it is closed. ■

Theorem. Suppose $f : X \to Y$ is a continuous map between topological spaces. If X is compact, then $f(X) \subseteq Y$ is compact.

Proof. Let X be compact, and let $f : X \to Y$ be a continuous function. Suppose $\{U_i\}_{i\in I}$ is an open cover of $f(X)$. Then, $f(X) \subseteq \bigcup_{i\in I} U_i$. Then, $X = f^{-1}(f(X)) \subseteq f^{-1}\left(\bigcup_{i\in I} U_i\right) = \bigcup_{i\in I} f^{-1}(U_i)$ Note that this follows because $X \subseteq f^{-1}(f(X))$ is a property of preimages, and f is defined on X so $f^{-1}(U_i) \subseteq X$ for each $i \in I$. Because f is continuous, $f^{-1}(U_i)$ is open for each $i \in I$, and we have that $\bigcup_{i\in I} f^{-1}(U_i) = X$, so $\{f^{-1}(U_i)\}_{i\in I}$ is an open cover of X. Because X is compact, there exists some finite $I' \subseteq I$ where $\{f^{-1}(U_i)\}_{i\in I'}$ is a finite subcover of X. Because $X \subseteq \bigcup_{i\in I'} f^{-1}(U_i)$, we have that $f(X) \subseteq \bigcup_{i\in I'} U_i$. Therefore, $\{U_i\}_{i\in I'}$ is a finite subcover of $f(X)$. Therefore, $f(X)$ is compact. ■

A final result is presented without proof.

Generalized Extreme Value Theorem. Let X be a nonempty compact topological space, and $f : X \to (\mathbb{R}, \text{Euclidean})$ be continuous. Then, $\sup(f(X))$ is finite, and there exists some $z \in X$ such that $f(z) = \sup(f(X))$.

Recall that a *sequentially compact* space has the property that every sequence has a convergent subsequence. It should be noted that, despite the similarity in naming convention, compactness is *not*, in general, related to sequential compactness. In fact, neither implies the other in general. However, it *can* be shown that these two concepts are actually *equivalent* in metric spaces. The proof of this fact is quite involved, and it is presented here without proof:

Theorem. Let (X, d) be a **metric space**. Then, X is compact if and only if X is sequentially compact.

Some other useful results about compactness on metric spaces are provided without proof:

- Compact subsets of a metric space are *closed and bounded.*
- A subspace of Euclidean $\mathbb{R}^n$ space is compact if and only if it is closed and bounded.
- $X \times Y$ with the product topology is compact if and only if both X and Y are compact.

Homeomorphisms of Topological Spaces. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces. A map $f : X \to Y$ is called a *homeomorphism* if f is continuous, f has an inverse $f^{-1} : Y \to X$, and f^{-1} is continuous. X is called *homeomorphic* to Y if there exists a homeomorphism from X to Y.

Note that for a continuous bijective map f, its inverse need not be continuous; for example, consider the function $f : (\mathbb{R}, \text{discrete}) \to (\mathbb{R}, \text{indiscrete})$, given by $f : x \mapsto x$. Note that only $\mathbb{R}$ and $\varnothing$ are open in the indiscrete topology, and both $f^{-1}(\mathbb{R}) = \mathbb{R}$ and $f^{-1}(\varnothing) = \varnothing$ are also open in the discrete topology, so f is continuous. f is simply the identity map, so f is clearly invertible; however, the inverse map f^{-1} is *not* continuous (i.e. $\{0\}$ is open in the discrete topology, but $(f^{-1})^{-1}(\{0\}) = \{0\}$ is not open in the indiscrete topology).

If two topological spaces are homeomorphic, they are essentially "the same," up to a relabeling of the elements. This means the overall structure is preserved by homeomorphisms. This is summarized in the statement that homeomorphisms are continuous, bijective, open maps (and continuous, bijective, closed maps), where an open map takes open sets in the input space to open sets in the output space (and similarly, a closed map takes closed sets to closed sets).

Theorem. A map $f : (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$ is a homeomorphism if and only if f is a continuous, invertible, open and closed map.

This theorem follows quickly from the definition of continuity. This means that, for any two homeomorphic spaces, any property which is defined in terms of the topology alone which is possessed by one space must necessarily be possessed by the other.

Using this fact, a useful theorem can be proved.

Theorem. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism.

Proof. Suppose $(X, \mathcal{T}_X)$ is compact and $(Y, \mathcal{T}_Y)$ is Hausdorff. Let $f : X \to Y$ be a continuous bijection. By assumption, f is continuous and f^{-1} exists, so to show that f is a homeomorphism it suffices to show that f^{-1} is continuous.

Suppose $C \subseteq X$ is closed. To show continuity, we seek to show that $(f^{-1})^{-1}(C)$ is also closed. Note that, by definition, $(f^{-1})^{-1}(C) = f(C)$. Because X is compact and C is closed, C is compact. Therefore, $f(C)$ is compact. Compact subsets of Hausdorff space are closed, so $f(C)$ is closed. This means that $(f^{-1})^{-1}(C)$ is closed. Therefore, the preimage of f^{-1} on closed sets is closed, so f^{-1} is continuous. Therefore, we have that f is a homeomorphism. ■

4 Complex Analysis

Notes from **Math 555: Introduction to Complex Variables**

Taken at University of Michigan, Winter 2026

using *Complex Analysis*, by Lars Ahlfors

4.1 Complex Numbers

An initial motivation for complex numbers is in solution to equations such as $x^2 + 1 = 0$. No real number satisfies this equation, so it seems natural to introduce a number i such that $i^2 = -1$, and therefore $\pm i$ satisfies this equation. It also seems natural to extend the typical rules of arithmetic from the real numbers to also apply to the complex numbers. A *complex number* z has the form $z = a + ib$, where $a, b \in \mathbb{R}$. Note that z has a real part, with $\mathrm{Re}(z) = a$, and an imaginary part, with $\mathrm{Im}(z) = b$.

The complex numbers can be viewed as $\mathbb{R}^2$ with some additional structure. In particular, a complex number $z \in \mathbb{C}$ is a vector $(a, b) \in \mathbb{R}^2$ with component-wise vector addition and the standard multiplication for real numbers. That is, for $z = a+ib$ and $w = c+id$, we have $z+w = (a+c)+i(b+d) = (a+c, b+d)$ and $zw = (a+ib)(c+id) = (ac-bd)+i(ad+bc) = (ac-bd, ad+bc)$.

Note that $\mathbb{C}$ is a *field*, and in particular $\mathbb{C}$ is a *field extension* of $\mathbb{R}$. That is, the real line can be viewed as a subset of $\mathbb{C}$. Namely, the set $\{z \in \mathbb{C} \mid \mathrm{Im}(z) = 0\}$ is precisely the real number line. Because $\mathbb{C}$ is a field extension of $\mathbb{R}$, it inherits the usual rules for addition, subtraction, multiplication, and division. That is, we have the associative, commutative, and distributive laws, and we have the neutral elements 0 and 1 under addition and multiplication respectively.

Some notation for complex numbers will now be introduced. Let $z = a + ib$ be a complex number. The *complex conjugate* of z, denoted $\overline{z}$, is given by

$$\overline{z} := a - ib$$

Suppose w is another complex number. Then, the following properties hold:

$$\overline{w + z} = \overline{w} + \overline{z} \tag{1}$$

$$\overline{wz} = \overline{w} \cdot \overline{z} \tag{2}$$

$$\overline{(w/z)} = \overline{w}/\overline{z} \tag{3}$$

Property (3) follows naturally from property (2). A proof is provided for (1) and (2).

Proof. Let $w = a + ib$, $z = x + iy$. Then, we have that

$$\begin{aligned}
\overline{w + z} &= \overline{a + ib + x + iy} \\
&= \overline{(a + x) + i(b + y)} \\
&= (a + x) - i(b + y) \\
&= (a - ib) + (x - iy) \\
&= \overline{w} + \overline{z}
\end{aligned}$$

Similarly, we have that

$$\begin{aligned}
\overline{wz} &= \overline{(a + ib)(x + iy)} \\
&= \overline{(ax - by) + i(ay + bx)} \\
&= (ax - by) - i(ay + bx) \\
&= ax - iay - ibx - by \\
&= (a - ib)(x - iy) \\
&= \overline{w} \cdot \overline{z}
\end{aligned}$$

■

The *modulus* of z, denoted $|z|$ and effectively representing the norm of z, is computed by the familiar Euclidean metric on $\mathbb{R}^2$. The modulus can be equivalently written in terms of the complex conjugate of z:

$$|z| := \sqrt{\mathrm{Re}(z)^2 + \mathrm{Im}(z)^2} = \sqrt{\overline{z}z} \geq 0$$

Note that this implies that $\overline{z}z \geq 0$, which is indeed the case. It can easily be shown that $|zw| = |z||w|$. In particular, this follows directly from the following relationship:

$$|zw| = \sqrt{\overline{z}z\overline{w}w} = \sqrt{\overline{z}z}\sqrt{\overline{w}w} = |z||w|$$

This result can extend to arbitrarily many complex numbers by induction.

Geometry of the Complex Numbers. As mentioned, a complex number $z \in \mathbb{C}$ is essentially a two-dimensional vector with some additional structure. It is natural to imagine z to exist on the *complex plane*, with the x-axis corresponding to the real part of z and the y-axis corresponding to the imaginary part of z.

In rectangular coordinates, $z = a + ib$ is simply the vector (a, b). z may also naturally be expressed in terms of polar coordinates; that is, there exists $r \geq 0$ and an angle $\theta \geq 0$ such that

$$z = r\,\mathrm{cis}(\theta) := (r\cos\theta, r\sin\theta) = r\cos\theta + ir\sin\theta$$

Note that $r = |z|$ uniquely, but θ, which is called the *argument* of z, is not uniquely defined; some notation for argument will now be introduced.

The set of possible values for θ is denoted $\arg(z)$, and is precisely the set

$$\arg(z) = \{\theta \in \mathbb{R} \mid z = (r\cos\theta, r\sin\theta)\}$$

The *principal branch* of the argument, denoted $\mathrm{Arg}(z)$, is a function $\mathrm{Arg} : \mathbb{C} \setminus \{0\} \to (-\pi, \pi]$ defined by $\mathrm{Arg}(z) = \theta_0$, where $\theta_0 \in \arg(z)$ also satisfies the condition $-\pi < \theta_0 \leq \pi$. Sometimes this function is defined instead on $[0, 2\pi)$, in which case it will be denoted HArg in this context, with $\mathrm{HArg}(z) = \theta_0$ similarly producing a $\theta_0 \in \arg(z)$ with the additional property that $0 \leq \theta_0 < 2\pi$. Note that $\arg(z)$ may be expressed in terms of the principal argument, with $\arg(z) = \{\mathrm{Arg}(z) + 2\pi n \mid n \in \mathbb{Z}\}$.

The argument has the following properties:

$$\arg(wz) = \arg(w) + \arg(z) \tag{1}$$

$$\arg(w/z) = \arg(w) - \arg(z) \tag{2}$$

These properties can be shown directly by the definition of cis(z) and applying trigonometric identities.

4.2 Analysis on $\mathbb{R}^n$

Because $\mathbb{C}$ essentially has the structure of $\mathbb{R}^2$, it is beneficial to summarize some general results from analysis on $\mathbb{R}^n$, which can then be restricted to the case of $\mathbb{R}^2$ and therefore to $\mathbb{C}$. Most of these results are provided without proof because the result (or a more general form) has been shown in real analysis or topology, or the result readily generalizes from $\mathbb{R}$ or general metric spaces.

Sequences in $\mathbb{R}^n$. Let S be any set. A sequence in S is a function $(x_n)_{n\in\mathbb{N}} : \mathbb{N} \to S$. In other words, $(x_n)_{n\in\mathbb{N}}$ assigns a value in S to every natural number. Some relevant definitions are provided: **Definition**. Let $x \in \mathbb{R}^n$, and let $(x_n)_{n\in\mathbb{N}}$ be a sequence in $\mathbb{R}^n$.

1. x is a **limit point** of $(x_n)_{n\in\mathbb{N}}$ if for all $\varepsilon > 0$, the set $\{n \in \mathbb{N} \mid |x_n - x| < \varepsilon\}$ is infinite.
2. $(x_n)_{n\in\mathbb{N}}$ **converges to** x if $\lim_{n\to\infty} |x_n - x| = 0$; in other words, x is a **limit** of $(x_n)_{n\in\mathbb{N}}$ if for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $|x_n - x| < \varepsilon$.
3. $(x_n)_{n\in\mathbb{N}}$ is **bounded** if there exists $M \in \mathbb{R}$ such that $|x_n| \leq M$ for all $n \in \mathbb{N}$.
4. $(x_n)_{n\in\mathbb{N}}$ is **Cauchy** if for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$, $|x_m - x_n| < \varepsilon$.

Another way to view this is as follows: x is a limit point of $(x_n)_{n\in\mathbb{N}}$ if $x_n \in B_\varepsilon(x)$ for infinitely many values of n, and x is a limit if there is some $N \in \mathbb{N}$ where $x_n \in B_\varepsilon(x)$ for all $n > N$, which is the topological definition of sequential convergence. Note that a sequence can have multiple limit points, but a sequence can have at most one (unique) in $\mathbb{R}^n$.

The two main theorems concerning sequences in $\mathbb{R}^n$ are provided without proof:

Bolzano-Weierstrass Theorem (in $\mathbb{R}^n$). Suppose $(x_n)_{n\in\mathbb{N}}$ is a bounded sequence in $\mathbb{R}^n$. Then, $(x_n)_{n\in\mathbb{N}}$ has a limit point.
Completeness of $\mathbb{R}^n$. A sequence $(x_n)_{n\in\mathbb{N}}$ is convergent if and only if it is Cauchy.

An alternate characterization relating limit points and limits is given. Let (x_n) be a sequence in $\mathbb{R}^n$ and $x \in \mathbb{R}^n$.

(1) (x_n) converges to x if and only if for every open neighborhood U of x, there exists an $N \in \mathbb{N}$ such that $x_n \in U$ for all $n \geq N$.

(2) x is a limit point of (x_n) if and only if there exists a subsequence (x_{n_k}) converging to x.

(3) If $S \subseteq \mathbb{R}^n$, then $\overline{S}$ is precisely the set of points $y \in \mathbb{R}^n$ such that there exists a sequence in S converging to y. That is, a closed set contains the limits of all its sequences.

Series in $\mathbb{R}^n$. Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in $\mathbb{R}^n$. Then, the sequence of *partial sums* is denoted by:

$$(s_n)_{n\in\mathbb{N}} := \sum_n x_n := \left(\sum_{n=1}^{N} x_n\right)_{n\in\mathbb{N}}$$

If $(s_n)_{n\in\mathbb{N}}$ converges to some $s \in \mathbb{R}^n$, then s is called the **sum** of the series, and we write $\displaystyle\sum_{n=1}^{\infty} x_n := \lim_{N\to\infty} \sum_{n=1}^{N} x_n = s$.

$\sum_n x_n$ **converges absolutely** if $\displaystyle\sum_{n=1}^{\infty} |x_n| < \infty$, and $\sum_n x_n$ **converges conditionally** if $\displaystyle\sum_{n=1}^{\infty} |x_n| = \infty$.

An example of a conditionally convergent series is given by $\sum_n (-1)^n/n$. This series converges, but $\sum_n |(-1)^n/n| = \sum_n 1/n$ which does not converge.

Theorem. Suppose (x_n) is a sequence in $\mathbb{R}^n$ with the property that $\sum_n x_n$ is absolutely convergent. Then, the following properties hold:

$$\sum_n x_n \text{ is convergent} \tag{1}$$

$$\left|\sum_{n=1}^{\infty} x_n\right| \leq \sum_{n=1}^{\infty} |x_n| \tag{2}$$

Property (2) is called the "infinite triangle inequality."

Let $\rho : \mathbb{N} \to \mathbb{N}$ be a bijection. Then, the series $\sum_n x_{\rho(n)}$ is a *rearrangement* of $\sum_n x_n$. The rearrangement of a series is not in general equivalent to the original series, except under the condition of absolute convergence. If (x_n) is not absolutely convergent, this statement fails (in spectacular fashion):

Riemann's Rearrangement Theorem. Suppose $\sum_n x_n$ is a conditionally convergent series of real numbers.

Let $s \in \mathbb{R} \cup \{-\infty, \infty\}$. Then, there exists a bijection $\rho : \mathbb{N} \to \mathbb{N}$ such that $\displaystyle\sum_{n=1}^{\infty} x_{\rho(n)} = s$. There also exists a bijection ρ such that $\displaystyle\sum_{n=1}^{\infty} x_{\rho(n)}$ does not converge to a real limit or $\pm\infty$.

Cauchy Product of Infinite Series. Let (w_n) and (z_n) be sequences of complex numbers (or, therefore, real numbers). From analysis, we have that sequences can sum:

$$\sum_{n=1}^{\infty}(w_n + z_n) = \sum_{n=1}^{\infty} w_n + \sum_{n=1}^{\infty} z_n$$

which converges (or converges absolutely) if the individual series converge (or converge absolutely). The **Cauchy product** of $\sum_n z_n$ and $\sum_n w_n$ is a similar procedure, which is defined as follows:

$$\sum_n \left(\sum_{i=0}^{n} z_i w_{n-1} \right) = \sum_n \left(\sum_{i+j=n} z_i w_j \right)$$

Theorem (Convergence of Cauchy Products). If $\sum_n z_n$ converges *absolutely* and $\sum_n w_n$ converges, then their Cauchy product converges, and:

$$\sum_{n=0}^{\infty} \left(\sum_{i=0}^{\infty} z_i w_{n-i} \right) = \sum_{n=0}^{\infty} z_n \sum_{n=0}^{\infty} w_n$$

There are several useful tests for whether a series converges or diverges; these are familiar from standard analysis, and the results extend generally from $\mathbb{R}$ to $\mathbb{R}^n$:

Series Tests. The following tests can be used to conclude whether a series converges or diverges. Let (x_n) be a sequence in $\mathbb{R}^n$.

Divergence Test. If $\sum_n x_n$ converges, then $\lim_{n\to\infty} x_n = 0$. Importantly, the negation is that if $\lim_{n\to\infty} x_n \neq 0$, then $\sum_n x_n$ does not converge.

Comparison Test. If (y_n) is a sequence in $\mathbb{R}^n$ such that $\sum_n y_n$ converges absolutely and $|x_n| \leq |y_n|$ for all $n \in \mathbb{N}$, then $\sum_n x_n$ converges absolutely.

Root Test. If $\limsup_{n\to\infty} |x_n|^{1/n} < 1$, then $\sum_n x_n$ converges absolutely. If $\limsup_{n\to\infty} |x_n|^{1/n} > 1$, then $\sum_n x_n$ diverges.

Ratio Test. Suppose $x_n \neq 0$ for all $n \in \mathbb{N}$. If $\limsup_{n\to\infty} \frac{|x_{n+1}|}{|x_n|} < 1$, then $\sum_n x_n$ converges absolutely. If $\liminf_{n\to\infty} \frac{|x_{n+1}|}{|x_n|} > 1$, then $\sum_n x_n$ diverges.

Compact Subsets of $\mathbb{R}^n$. Recall from topology that a set $U \subseteq \mathbb{R}^n$ is called *compact* if every cover of U has a finite subcover. In $\mathbb{R}^n$, there is a more useful characterization from analysis:

Heine-Borel Theorem. A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.

Because sequential compactness is equivalent to compactness in Euclidean space, we also have a sequential characterization of compactness:

Sequential Characterization of Compactness. A subset $U \subseteq \mathbb{R}^n$ is compact if and only if every sequence in U has a convergent subsequence with limit in U.

The familiar $\varepsilon - \delta$ definition of convergence and continuity generalize to $\mathbb{R}^n$:

Convergence and Continuity. Let $S \subseteq \mathbb{R}^m$, $T \subseteq \mathbb{R}^n$, and $f : S \to T$.

Definition of Convergence. Let $x_0 \in \overline{S}$. $f(x)$ **converges** to $L \in \mathbb{R}^n$ if for all $\varepsilon > 0$, there exists $\delta > 0$ such that if $x \in S$ and $0 < |x - x_0| < \delta$, then $|f(x) - L| < \varepsilon$. In this L is the *limit* as $x \to x_0$, denoted:

$$\lim_{x \to x_0, x \in S} f(x) = L$$

Definition of Continuity. Let $x_0 \in S$. f is **continuous** at x_0 if for all $\varepsilon > 0$ and all $y \in S$, there exists $\delta > 0$ such that if $x \in S$ and $|x - y| < \delta$, then $|f(x) - f(y)| < \varepsilon$. This is equivalent to the statement that:

$$\lim_{x \to x_0, x \in S} f(x) = f(x_0)$$

Definition of Uniform Continuity. f is **uniformly continuous** on S if for all $\varepsilon > 0$, there exists $\delta > 0$ such that if $x, y \in S$ and $|x - y| < \delta$, then $|f(x) - f(y)| < \varepsilon$.

Note that in the definition of continuity, δ *can* depend on y, i.e. $\delta = \delta(\varepsilon, y)$, but in the uniform continuity definition δ may *not* depend on y, i.e. $\delta = \delta(\varepsilon)$ alone.

Using the fact that for compact sets C, $f(C)$ is compact, the extreme value theorem can be shown:

Extreme Value Theorem. Suppose $C \subseteq \mathbb{R}^n$ is a nonempty, compact set and $f : C \to \mathbb{R}$ is continuous. Then, there exist $x_m, x_M \in C$ such that $f(x_m) = \min\{f(x) \mid x \in C\}$ and $f(x_M) = \max\{f(x) \mid x \in C\}$.

In one dimension, a function $f : (a, b) \to \mathbb{R}$ is *real-differentiable* at $x_0 \in (a, b)$ if:

$$f'(x_0) := \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} \quad \text{exists}$$

This definition does not immediately generalize to higher dimensions; however, an equivalent characterization is that f is differentiable if there is a line passing through $(x_0, f(x_0))$ which accurately approximates the function. Then, f is *real-differentiable* at x_0 if

$$\lim_{x \to x_0} \frac{|f(x) - f(x_0) - m(x - x_0)|}{|x - x_0|} = 0$$

Note that the slope $m = f'(x_0)$. This definition can be readily generalized to the $\mathbb{R}^n$ case.

Definition of Real-Differentiability. Let $U \subseteq \mathbb{R}^m$ be open. Let $x_0 \in U$. A function $f : U \to \mathbb{R}^n$ is **real-differentiable** at x_0 if there exists an $m \times n$ matrix A such that

$$\lim_{x \to x_0} \frac{|f(x) - f(x_0) - A(x - x_0)|}{|x - x_0|} = 0$$

Alternatively, we may define the error $\varepsilon(x)$ of the linear approximation, and rewrite in planar form:

$$f(x) = f(x_0) + A(x - x_0) + \varepsilon(x), \text{ where } \lim_{x \to x_0} \frac{|\varepsilon(x)|}{|x - x_0|} = 0$$

Note that this can equivalently be rewritten as:

$$\varepsilon(x) = f(x) - f(x_0) - A(x - x_0) \implies \lim_{x \to x_0} \frac{|f(x) - f(x_0) - A(x - x_0)|}{|x - x_0|} = 0 \text{ if } f \text{ is real-differentiable}$$

The matrix A is unique, and A is called the **Jacobian matrix of f at x_0**, or the **total derivative**. A essentially contains the "slopes" (the derivatives) of f. That is, if f is represented as the vector $f = (f_1, \ldots, f_m)$ where $f_i : U \to \mathbb{R}$, then $a_{ij} = \left.\frac{\partial f_i}{\partial x_j}\right|_{x=x_0}$, i.e. the ijth entries are the partial derivatives of f_i with respect to x_j.

Note that if f is real-differentiable at x_0, then the partial derivatives $\partial_j f_i(x_0)$ exist. The opposite is **not** true. Additionally, the existence of partial derivatives does not guarantee a function of several variables is real-differentiable, and a function which is real-differentiable can have discontinuous partial derivatives.

Sequences of Functions. Let $S \subseteq \mathbb{R}^m$, $f : \mathbb{R}^m \to \mathbb{R}^n$, and $(f_n)_{n\in\mathbb{N}} = (f_n : S \to \mathbb{R}^n)_{n\in\mathbb{N}}$.

Definition (Pointwise and Uniform Convergence). $(f_n)_{n\in\mathbb{N}}$ is **pointwise convergent** to f if for all $x \in S$, $(f_n(x))_{n\in\mathbb{N}}$ converges to $f(x) \in \mathbb{R}^n$. f is **uniformly convergent** to f if $\lim_{n\to\infty} \left(\sup_{x\in S} |f_n(x) - f(x)| \right) = 0$.

It is useful to translate these definitions into quantifiers:

Pointwise Convergence	**Uniform Convergence**
$\forall x \in S$ and $\forall \varepsilon > 0$, $\exists N(x, \varepsilon) \in \mathbb{N}$ s.t. $\forall n \geq N$, $\lvert f_n(x) - f(x)\rvert < \varepsilon$	$\forall \varepsilon > 0$, $\exists N(\varepsilon) \in \mathbb{N}$ s.t. $\forall n \geq N$ and $\forall x \in S$, $\lvert f_n(x) - f(x)\rvert < \varepsilon$

General Properties of Uniform Convergence. Suppose $(f_n)_{n\in\mathbb{N}}$ converges uniformly to f. Then:

1. If f_n is uniformly continuous for all $n \in \mathbb{N}$, then f is uniformly continuous.
2. If f_n is continuous for all $n \in \mathbb{N}$, then f is continuous.
3. If f_n is bounded for all $n \in \mathbb{N}$, then f is bounded.

An important equivalence exists between uniformly convergent and *uniformly Cauchy* sequences of functions:

Definition (Uniform Cauchy). $(f_n)_{n\in\mathbb{N}}$ is **uniformly Cauchy** if, for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$, we have $\sup_{x\in S} |f_n(x) - f_m(x)| < \varepsilon$.

Theorem (Equivalence between Uniform Cauchy and Uniform Convergence). $(f_n)_{n\in\mathbb{N}}$ is uniformly Cauchy if and only if $(f_n)_{n\in\mathbb{N}}$ is uniformly convergent to some function f.

As with standard sequences, one can consider the partial sums of a sequence of functions, i.e. the *series* of functions. Let $(f_n)_{n\in\mathbb{N}}$ be a sequence of functions from S to $\mathbb{R}^n$, and $f : S \to \mathbb{R}^n$ be another function.

Definition (Series of Functions). The sequence $\sum_n f_n := (\sum_{n=1}^{\infty} f_n)_N \in \mathbb{N}$ is called the **sequence of partial sums** of $(f_n)_n \in \mathbb{N}$. If $\sum_n f_n$ converges pointwise to f, then f is called the *sum* of the series $\sum_n f_n$, and $\sum_{n=1}^{\infty} f_n := f$. $\sum_n f_n$ **converges absolutely uniformly** if the series $\sum_n |f_n|$ of functions $|f_n| : S \to \mathbb{R}$ converges uniformly.

An equivalent way to state the definition of absolute uniform convergence is that there is some function $s : S \to \mathbb{R}$ such that, for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, then

$$\sup_{x\in S} \left| \sum_{i=1}^{n} |f_i(x)| - s(x) \right| < \varepsilon$$

That is, $s(x)$ is the sum of f_n, and $s(x) = \sum_{n=1}^{\infty} |f_n(x)|$ for all $x \in S$.

Theorem (Weierstrass's M-Test). Suppose

$$\sum_{n=1}^{\infty} \sup_{x\in S} |f_n(x)| < \infty$$

Then, $\sum_n f_n$ converges absolutely uniformly.

Weierstrass's M-test can be used to show that a series of functions converges uniformly (or absolutely uniformly). Note that the pre-condition $\sum_{n=1}^{\infty} \sup_{x\in S} |f_n(x)| < \infty$ can be verified by the comparison test with a convergent (real-valued) series $\sum_n M_n$, i.e. where each M_n bounds $f_n(x)$ for fixed n and varying over all x in the domain of f.

4.3 Complex Functions

Complex functions are essentially functions on $\mathbb{R}^2$, so the discussion of $\mathbb{R}^n$ can generally be restricted to $\mathbb{R}^2$ (i.e. for sequences, series, limits, and continuity). However, $\mathbb{C}$ has a multiplication (and division) structure, so the definition of complex-differentiability immediately generalizes from the definition on $\mathbb{R}$.

Complex Differentiable Functions. Let $U \subseteq \mathbb{C}$ be an open set. Let $f : U \to \mathbb{C}$.

f is **complex differentiable** at $z_0 \in U$ if the following complex limit exists:

$$f'(z_0) := \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} := \left.\frac{df}{dz}\right|_{z=z_0}$$

If f is differentiable at all $z_0 \in U$, then f is called *complex differentiable* on U.

f is **holomorphic at** z_0 if f is complex-differentiable on a *neighborhood of* z_0, i.e. there exists $\varepsilon > 0$ such that $\mathbb{D}_\varepsilon(z_0) \subseteq U$ and f is complex differentiable on $\mathbb{D}_\varepsilon(z_0)$ (where $\mathbb{D}_\varepsilon(z_0)$ is the disk of radius ε centered at z_0). f is holomorphic on U if f is holomorphic at every point of U.

If f is holomorphic on all of $\mathbb{C}$, then f is called **entire**.

The standard properties from $\mathbb{R}$ hold: complex differentiation is a linear operator, f is continuous at every point at which it is differentiable, and the product and quotient rule remain the same. The chain rule also remains the same, and in fact extends to curves in $\mathbb{C}$:

Definition of a Curve. Let $I \subseteq \mathbb{R}$ be an interval, and let $S \subseteq \mathbb{R}^n$. Let $\gamma : I \to S$ be a curve in S (i.e. a continuous map from I so S). γ is *differentiable at* t_0 if the following limit exists:

$$\gamma'(t_0) = \lim_{t \to t_0} \frac{\gamma(t) - \gamma(t_0)}{t - t_0}$$

The chain rule for curves on $\mathbb{C}$ is given as follows:

Theorem. Let $I \subseteq \mathbb{R}$ be an open interval and $U \subseteq \mathbb{C}$ be open. If the curve $\gamma : I \to U$ is differentiable at $t_0 \in I$ and $f : U \to C$ is complex differentiable at $\gamma(t_0)$, then $f \circ \gamma$ is differentiable at t_0, and:

$$(f \circ g)'(t_0) = f'(\gamma(t_0))\gamma'(t_0)$$

A curve γ is called **regular** if $\gamma'(t) \neq 0$ for all $t \in I$. Regular curves allow for a special characterization of the properties of complex derivatives.

Geometric Interpretation of Complex Differentiation. Let $I \subseteq \mathbb{R}$ be an interval and $U \subseteq \mathbb{C}$ be open. Let $\gamma : I \to \mathbb{C}$ be a regular curve, and let $f : U \to \mathbb{C}$. Suppose $\gamma(t_0) = z_0$. Then, by the chain rule:

$$(f \circ \gamma)'(t_0) = f'(\gamma(t_0))\gamma'(t_0) = f'(z_0)\gamma'(t_0)$$

Two properties follow from the chain rule:

$$|(f \circ \gamma)'(t_0)| = |f'(z_0)||\gamma'(t_0)| \qquad \text{(Stretching Property)}$$

$$\arg[(f \circ \gamma)'(t_0)] = \arg[f'(z_0)] + \arg[\gamma'(t_0)] \qquad \text{(Rotation Property)}$$

In other words, complex differentiation on a curve results *only* in a (uniform) stretch or shrink of the tangent vector by $|f'(z_0)|$, and a rotation by $0 \leq \theta < 2\pi$, where $\theta \in \arg[f'(z_0)]$. Importantly, for two curves, f preserves the *signed angle* between tangent vectors. A function with this property is called a **conformal map.**

Definition of Conformal Map (in $\mathbb{C}$). Let $U \subseteq \mathbb{C}$ be open. Let $f : U \to \mathbb{C}$. f is called **conformal** at $z_0 \in U$ if, for any two regular curves $\gamma_1 : (a_1, b_1) \to U$ and $\gamma_2 : (a_2, b_2) \to U$ with $\gamma_1(t_1) = \gamma_2(t_2) = z_0$, $f \circ \gamma_1$ and $f \circ \gamma_2$ are regular at t_1 and t_2 respectively, i.e. $(f \circ \gamma_1)'(t_1)$ and $(f \circ \gamma_2)'(t_2)$ exists, is nonzero, and:

$$\arg[(f \circ \gamma_1)'(t_1)] - \arg[(f \circ \gamma_2)'(t_2)] = \arg[\gamma_1'(t_1)] - \arg[\gamma_2'(t_2)]$$

Note that the additive property of the arguments is precisely the additive property already established for complex derivatives. This gives the following theorem:

Theorem. Suppose $f : U \to \mathbb{C}$ is complex differentiable at $z_0 \in U$ and $f'(z_0) \neq 0$. Then, f is conformal at z_0.

This condition can be stated more strictly if U is viewed as a subset of $\mathbb{R}$.

Theorem. Suppose $f : U \to \mathbb{C}$ is *real-differentiable* when $U, \mathbb{C}$ are viewed as subsets of $\mathbb{R}^2$. Then, f is conformal at z_0 if and only if f is complex differentiable at z_0 with $f'(z_0) \neq 0$.

It is possible to completely characterize the complex differentiability of a function. Let $U \subseteq \mathbb{C}$ be open, and let $f : U \to \mathbb{C}$. Note that f can be viewed as a function $f(z) = u(z) + iv(z)$, where $u(z) := \mathrm{Re}(z)$ and $v(z) := \mathrm{Im}(z)$. The following gives a complete characterization of complex differentiability.

Theorem (Cauchy-Riemann Equations). f is complex-differentiable at z_0 if and only if:

1. f is real differentiable when viewed as a function from $U \to \mathbb{R}$; that is, $f(z) = [u(z), v(z)]^\top$ is real-differentiable.
2. f satisfies the Cauchy-Riemann Equations:
$$\begin{cases} \partial_1 u = & \partial_2 v \\ \partial_2 u = & -\partial_1 v \end{cases}$$

In Cartesian form, the Cauchy Riemann equations are equivalently written as:

$$\begin{cases} \dfrac{\partial}{\partial x} u(x,y) & = \dfrac{\partial}{\partial y} v(x,y) \\ \dfrac{\partial}{\partial y} u(x,y) & = -\dfrac{\partial}{\partial x} v(x,y) \end{cases}$$

It is important to note that satisfying the Cauchy-Riemann equations are **not a sufficient condition** to satisfy complex differentiability; f must also be real differentiable when viewed as a function between subsets of $\mathbb{R}^2$.

Definition (Power Series). Let $a \in \mathbb{C}$. The power series centered at a is a series of functions $\sum_n f_n$ where $f_n(z) = c_n(z-a)^n$ for each $n \in \mathbb{N}_0$, where $c_n \in \mathbb{C}$ are the coefficients of the power series.

Cauchy-Hadamard Theorem. Let $\sum_n c_n(z-a)^n$ be a power series. Define

$$R := \frac{1}{\limsup\limits_{n\to\infty} |c_n|^{\frac{1}{n}}}$$

Then, the following conditions completely characterize the convergence and divergence of the power series:

1. If $z \in \mathbb{D}_R(a)$, then $\sum_n c_n(z-a)^n$ converges pointwise.
2. If $z \notin \mathbb{D}_R(a)$, then $\sum_n c_n(z-a)^n$ diverges.
3. If $z \in \partial\mathbb{D}_R(a)$, then $\sum_n c_n(z-a)^n$ may either converge or diverge.

A few important notes about the Cauchy-Hadamard theorem:

- R is called the **radius of convergence**; note that R may be either 0 (i.e. the power series does not converge) or $+\infty$ (i.e. the power series converges on the entire complex plane).
- Because $\sum_n c_n(z-a)^n$ converges pointwise inside $\mathbb{D}_R(a)$, for any $0 \leq r < R$, the power series converges *absolutely uniformly* on the closed sub-disk $\overline{\mathbb{D}}_r(a)$.

- $\partial\mathbb{D}_R(a)$ is called the **circle of convergence**; a particular power series may converge at all points on the circle of convergence, at finitely many points on the circle of convergence, or at no points on the circle of convergence.
- It is often easier to compute R with the following equivalent process:
  1. Find $z_0 \in \mathbb{C}$ such that $\sum_n c_n(z-a)^n$ converges and $R_0 := |z_0 - a|$. Then, this guarantees that $R_0 \leq R$.
  2. Find $z_0 \in \mathbb{C}$ such that $\sum_n c_n(z-a)^n$ diverges and $R_0 := |z_0 - a|$. Then, this guarantees that $R_0 \geq R$.
  3. Combining (1) and (2) gives that $R = R_0$. This avoids the explicit computation of $\limsup\limits_{n\to\infty} |c_n|^{1/n}$.

A proof of the Cauchy-Hadamard theoerm is provided:

Proof. Let $\sum_n c_n(z-a)^n$ be a power series. Define $R := \dfrac{1}{\limsup\limits_{n\to\infty} |c_n|^{\frac{1}{n}}}$. We show each part in turn.

(1) Suppose $0 \leq r < R$ and let $z \in \overline{\mathbb{D}}_r(a)$. We apply Weierstrass's M-Test:

$$\sum_{n=1}^{\infty} \sup_{z\in\overline{\mathbb{D}}_R(a)} |c_n(z-a)^n| = \sum_{n=1}^{\infty} \sup_{z\in\overline{\mathbb{D}}_r(a)} |c_n||z-a|^n \leq \sum_{n=1}^{\infty} |c_n| r^n$$

We apply the root test. Note that:

$$\limsup_{n\to\infty} \left(|c_n|\, r^n\right)^{\frac{1}{n}} = \limsup_{n\to\infty} |c_n|^{\frac{1}{n}} r = \frac{r}{R} < 1$$

Therefore, $\sum_n^{\infty} |c_n| r^n < \infty$. Then, by the Weierstrass M-test, $\sum_n c_n(z-a)^n$ converges absolutely uniformly on $\overline{\mathbb{D}}_r(a)$. Note that this also means that the power series converges pointwise on $\mathbb{D}_r(a)$.

(2) Suppose $z \in \mathbb{C}$ and $|z-a| > R$. We apply the root test:

$$\limsup_{n\to\infty} |c_n(z-a)^n|^{\frac{1}{n}} = \limsup_{n\to\infty} |c_n|^{\frac{1}{n}} |z-a| = \frac{|z-a|}{R} > 1$$

Therefore, $\sum_n c_n(z-a)^n$ diverges. ■

Power series are especially important in complex analysis because they lead to the concept of *analyticity*, where the *analytic* functions are those which are locally expressible as a power series.

Analytic Functions. Let $U \subseteq \mathbb{C}$ be open. Let $f : U \to \mathbb{C}$, and let $z_0 \in U$.

Definition. f is **analytic** at z_0 if there exists $r > 0$ such that $\mathbb{D}_r(z_0) \subseteq U$ and there exists a power series $\sum_n c_n(z-z_0)^n$ with radius of convergence $R \geq r$ with that property that $f(z) = \sum_{n=1}^{\infty} c_n(z-z_0)^n$ for all $z \in \mathbb{D}_r(z_0)$.

Fact. If $\sum_n c_n(z-a)^n$ is a power series with radius of convergence $0 < r \leq \infty$, and $f(z) := \sum_{n=0}^{\infty} c_n(z-a)^n$ for all $z \in \mathbb{D}_r(a)$, then $f : \mathbb{D}_r(a) \to \mathbb{C}$ is analytic on all of $\mathbb{D}_r(a)$.

The concept of analyticity is especially powerful because it can immediately be shown that the analytic functions are holomorphic. The other direction is true, but less immediately obvious. As a consequence, any results which apply to analytic functions will also apply to holomorphic functions. The holomorphicity of an analytic function follows from the fact that derivatives distribute over power series, a fact which is true but which involves a tedious proof (which is omitted).

Differentiating Power Series. Let $\sum_n c_n(z-a)^n$ be a power series with radius of convergence $0 \leq R \leq +\infty$.

Theorem (analytic $\implies$ holomorphic) Define $f(z) := \sum_{n=0}^{\infty} c_n(z-a)^n$. Then, the following properties hold:

(i) $\sum_n (n+1)c_{n+1}(z-a)^n$ has radius of convergence R.

(ii) f is holomorphic on $\mathbb{D}_R(a)$, and $f'(z) = \sum_{n=0}^{\infty} \frac{d}{dz}\left[c_n(z-a)^n\right]$

As a result, if f is analytic at z_0, then f is also holomorphic at z_0. Another possibly unexpected consequence is that f must be infinitely complex-differentiable.

Theorem. Suppose f is analytic on $\mathbb{D}_R(a)$. Then, f is infinitely complex-differentiable, and its kth derivative is given by:

$$\frac{d^k f(z)}{dz^k} = \sum_{n=0}^{\infty} \frac{d^k}{dz^k}\left[c_n(z-a)^n\right] = \sum_{n=k}^{\infty} n(n-1)\cdots(n-k+1)c_n(z-a)^{n-k}$$

This result can be shown by induction.

A special application is that several special functions can be exactly written as a power series. In fact, power series allows one to define the exponential, as well as the trigonometric and hyperbolic trigonometric functions.

Exponential and Trigonometric Functions with Power Series. The following functions are defined in terms of power series:

$$\begin{aligned}
\exp(z) &:= e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} \\
\cosh(z) &:= \frac{e^z + e^{-z}}{2} = \sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!} \\
\sinh(z) &:= \frac{e^z - e^{-z}}{2} = \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!} \\
\cos(z) &:= \frac{e^{iz} + e^{-iz}}{2} = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n)!} \\
\sin(z) &:= \frac{e^{iz} - e^{-iz}}{2} = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}
\end{aligned}$$

Using these definitions, the following properties can be shown with relative ease:

$$e^z = \cosh(z) + \sinh(z)$$

$$\cosh(iz) = \cos(z) \quad \text{and} \quad \sinh(iz) = i\sin(z)$$

$$e^{iz} = \cos z + i \sin z \qquad \text{(Euler's Formula)}$$

Analyticity can be equivalently characterized by local Taylor expandability.

Equivalency between Analyticity and Taylor Expandability. Let $U \subseteq \mathbb{C}$ be an open set. Let $z_0 \in U$. A function $f : U \to \mathbb{C}$ is analytic at z_0 if and only if there exists $r > 0$ such that $\mathbb{D}_r(z_0) \subseteq U$ and the following hold:

(a) f is infinitely complex differentiable on $\mathbb{D}_r(z_0)$

(b) The radius of convergence of the Taylor series $\displaystyle\sum_n \frac{f^{(n)}(z_0)}{n!}(z-z_0)^n$ is at least r

(c) $f(z) = \displaystyle\sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z-z_0)^n$ for all $z \in \mathbb{D}_r(z_0)$

Proof. Suppose (a), (b), and (c). Note that f is expressible as

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z-z_0)^n$$

on $\mathbb{D}_r(z_0)$, which is a power series. Thus, f is analytic on $\mathbb{D}_r(z_0)$.

Suppose f is analytic. Then,

$$f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$$

We have already shown that (a) follows. We also have that the kth derivative is given by:

$$f^{(k)}(z) = \sum_{n=k}^{\infty} n(n-1)\cdots(n-k+1)c_n(z-z_0)^{n-k}$$

Evaluating at $z = z_0$, we have:

$$f^{(k)}(z_0) = \sum_{n=k}^{\infty} n(n-1)\cdots(n-k+1)c_n(0)^{n-k} = k(k-1)\cdots(k-k+1)c_k \cdot 1 = k!$$

Dividing by $k!$, we have:

$$c_k = \frac{f^{(k)}(z_0)}{k!}$$

This holds for all k, including $k = 0$ which is simply the original function f. Since c_n is precisely the coefficient of the Taylor expansion for each $n \in \mathbb{N}_0$, the Taylor expansion is identical to the power series, and therefore has the same radius of convergence. Therefore, (b) and (c) follow as well. ■

The fact that analytic functions are precisely those functions which are locally Taylor-expandable is a very rigid condition, which gives rise to some special, equivalent properties.

Identity Theorem. Let $U \subseteq \mathbb{C}$ be a nonempty *domain*, where a domain is an open and connected set. If $f, g : U \to \mathbb{C}$ are analytic, then the following are equivalent:

(a) $f = g$ on all of U.

(b) $\{z \in U \mid f(z) = g(z)\}$ has an accumulation point in U.

(c) There exists $z_0 \in U$ such that $f^{(n)}(z_0) = g^{(n)}(z_0)$ for all $n \in \mathbb{N}_0$.

An immediate consequence of (b), perhaps more important than (b) itself, is that zeroes are isolated; that is, if $f(z) \neq 0$ for some $z \in U$, then $\{z \in U \mid f(z) = 0\}$ *cannot* have an accumulation point. In a sense, this is due to the fact that analytic function locally behave "like a polynomial," rather than any arbitrary smooth function. On top of this, analytic functions uniquely behave like their Taylor expansions, which can only happen if all the derivatives agree.

4.4 Complex Integration

Riemann integrability in $\mathbb{C}$ is essentially identical to Riemann integrability in $\mathbb{R}$. In particular, if $g : [a, b] \to \mathbb{C}$ is a (piecewise) continuous function expressed as $g(t) = u(t) + iv(t)$, then its Riemann integral is simply:

$$\int_a^b g(t)dt = \int_a^b u(t)dt + i\int_a^b v(t)dt$$

where both $u(t)$ are $v(t)$ are functions from $[a, b] \to \mathbb{R}$, and therefore standard integration rules and properties simply carry over from the real case.

If one instead considers functions of the form $f : U \to \mathbb{C}$, where $U \subseteq \mathbb{C}$, such an approach is *not* sufficient. In particular, one must specify a path to integrate over. A particularly useful path would have some minimal "smoothness," or at least could be broken into several "smooth" pieces. This motivates the definition of a *contour*.

Definition (Contour). A curve $\gamma : [a, b] \to \mathbb{C}$ is **continuously differentiable**, or C^1, if γ is differentiable on (a, b), γ' is continuous on (a, b), and $\gamma'(a)$ and $\gamma'(b)$ both exist and agree with the limits of $\gamma'(t)$ approaching from the right and left respectively. γ is **piecewise** C^1 if there exists $a = t_0 < t_1 < \cdots < t_n = b$ such that $\gamma|_{[t_{i-1},t_i]} \to \mathbb{C}$ is C^1 for all $i = 1, \ldots, n$. A curve in the complex plane is called a **contour** if it is piecewise continuously differentiable (i.e. piecewise C^1).

Definition (Contour Integral). Let $\gamma : [a, b] \to \mathbb{C}$ be a contour. Define γ^* as the forward image of γ, i.e. $\gamma^* := \gamma[a, b]$. If $f : \gamma^* \to \mathbb{C}$ is continuous, then the **contour integral** of f along γ is defined as:

$$\int_\gamma f(z)dz := \int_a^b f(\gamma(t))\gamma'(t)dt$$

Note: if γ is only piecewise C^1, this definition is still applicable by simply splitting the integral into a sum of integrals over each differentiable piece.

Contour integrals are sometimes called *complex line integrals*, which

Properties of Contour Integrals. Let $\gamma : [a, b] \to \mathbb{C}$ be a contour, and $f : \gamma^* \to \mathbb{C}$ a continuous function. The following properties hold for contour integration.

(i) Contour integrals are linear.

(ii) Time-reversals are negative; i.e. integrating over the contour $\overleftarrow{\gamma} := \gamma(a + b - t)$, which is γ "in reverse," gives:

$$\int_{\overleftarrow{\gamma}} f(z)dz = -\int_\gamma f(z)dz$$

(iii) Concatenated paths add. If $\eta : [c, d] \to \mathbb{C}$ is a path, then integration over the concatenation

$$\gamma \cdot \eta := \begin{cases} \gamma(t), & a \le t \le b \\ \eta(t + c - b), & b \le t \le b + d - c \end{cases}$$

gives the following:

$$\int_{\gamma\cdot\eta} f(z)dz = \int_\gamma f(z)dz + \int_\eta f(z)dz$$

(iv) ML ("max times length") Estimate. If $f(z) \le M$ for all $z \in \gamma^*$ and length$(\gamma) \le L$, then:

$$\left|\int_\gamma f(z)dz\right| \le \int_\gamma |f(z)||dz| := \int_a^b |f(\gamma(t))||\gamma'(t)|dt \le ML$$

Definition (Antiderivative). Let $U \subseteq \mathbb{C}$. Let $f : U \to \mathbb{C}$. A function $F : U \to \mathbb{C}$ is called an **antiderivative** of f if F is holomorphic on U and $F'(z) = f(z)$ for all $z \in U$.

Note that for any $c \in \mathbb{C}$, $F + c$ is *also* an antiderivative of f. However, **not every antiderivative of f is necessarily expressible in this form**. In particular, if F is piecewise, different constants may be added on each piece of F, producing a different antiderivative. However, for connected U, all antiderivatives *can* be expressed as $F + c$.

Fact. If U is open *and connected*, all antiderivatives F of f are of the form $F + c$ for some $c \in \mathbb{C}$.

Proof. Suppose U is an open, *connected* subset of $\mathbb{C}$. Suppose F_1, F_2 are antiderivatives of f. Then, for all $z \in U$, we have:

$$(F_1 - F_2)'(z) = F_1'(z) - F_2'(z) = f(z) - f(z) = 0$$

$F_1 - F_2$ has derivative 0, so $F_1 - F_2$ is constant. That is, $F_1(z) = F_2(z) + c$ for some $c \in \mathbb{C}$. ■

Complex Fundamental Theorem of Calculus. Suppose $f : U \to \mathbb{C}$ is continuous, and $\gamma : [a, b] \to \mathbb{C}$ is a contour in U. If G is an antiderivative of g, then:

$$\int_\gamma g(z)dz = G(\gamma(b)) - G(\gamma(a))$$

Proof. Note that:

$$\int_\gamma g(z)dz = \int_\gamma G'(z)dz = \int_a^b G'(\gamma(t))\gamma'(t)dt = \int_a^b (G \circ \gamma)'(t)dt = \underbrace{(G \circ \gamma)(b) - (G \circ \gamma)(a)}_{\text{by real FTC}}$$

■

Antiderivative Theorem. Let $f : U \to \mathbb{C}$ be a continuous function. The following statements are equivalent:

(a) f has an antiderivative on U.

(b) $\int_\gamma f(z)dz = 0$ for *all closed contours* in U, i.e. contours for which $\gamma(b) = \gamma(a)$.

(c) $\int_\gamma f(z)dz = \int_\eta f(z)dz$ for all contours $\gamma : [a, b] \to \mathbb{C}$ and $\eta : [c, d] \to \mathbb{C}$ where $\gamma(a) = \eta(c)$ and $\gamma(b) = \eta(d)$. That is, integration is path independent.

(d) If U is convex, then $\int_\tau f(z)dz = 0$ for all triangular contours in U.

The proof of this characterization is omitted, but (b) follows quickly from (a) and (c) quickly from (b). (a) can be shown from (c) by first considering convex domains (i.e. only nonempty, open, connected sets which contain all line segments defined by points $z, w \in U$), and then generalizing by writing an arbitrary open set as a (possibly infinite) union of convex domains and constructing the antiderivative piecewise.

The antiderivative theorem hints at a particularly powerful check for whether a function has an antiderivative; in particular, if one finds a single closed contour over which the contour integral of f is nonzero, f does not have an antiderivative. This is precisely the statement $\neg(b) \implies \neg(a)$.

Example. Consider the function:

$$f(z) := \frac{1}{z}$$

defined on $\mathbb{D}$. Consider the following integral:

$$\int_{\partial\mathbb{D}} f(z)dz = \int_{e^{it}} \frac{1}{z}dz = \int_0^{2\pi} \frac{1}{e^{it}} ie^{it}dt = i\int_0^{2\pi} e^{it}e^{-it}dt = 2\pi i \neq 0$$

Therefore, f does not have an antiderivative.

A related result, which is possibly the most important result in complex analysis, is the *Cauchy-Goursat Theorem*, which is sometimes also called the *Cauchy Integral Theorem*. At this point, a weaker (although still quite useful) version of this theorem can be shown.

Cauchy-Goursat Theorem for Convex Domains. Suppose U is a convex domain, and $f : U \to \mathbb{C}$ is holomorphic. Then,

$$\int_\gamma f(z)dz = 0 \qquad \forall \text{ closed contours } \gamma \text{ in } U \tag{Cauchy-Goursat}$$

Proof. First, recall a result from topology:

Fact. Suppose C_n is a nonempty compact subset of $\mathbb{R}^n$ for all $n \in \mathbb{N}$ where $C_{n+1} \subseteq C_n$. If $\operatorname{diam}(C_n) := \sup_{x,y\in C_n} |x - y|$ has the property that $\operatorname{diam}(C_n) \to 0$ as $n \to \infty$, then there exists $x_0 \in \mathbb{R}^n$ such that $\bigcap_{n\in\mathbb{N}} C_n = \{x_0\}$.

Now, let τ be a triangular contour in U. Define $M := \left|\int_\tau f(z)dz\right|$. Define T as the set of points enclosed by τ, i.e. $\tau^* = \partial T$. We wish to show that $M = 0$.

Without loss of generality, assume τ is oriented counterclockwise. We construct four triangular contours from τ as follows: connect the midpoints of each line segment comprising τ, which gives four new triangles; define new triangular contours $\sigma_i,\ i = 1, 2, 3, 4$ following these contours clockwise. Notice that, due to the orientation,

$$M = \left|\int_\tau f(z)dz)\right| = \left|\sum_{i=1}^4 \int_{\sigma_i} f(z)dz\right| \leq \sum_{i=1}^4 \left|\int_{\sigma_i} f(z)dz\right|$$

Note that, for this to be true, for at least one i, we have that $\left|\int_{\sigma_i} f(z)dz\right| \geq M/4$. Define $\tau_0 = \tau$ and $\tau_1 = \sigma_i$ for i which satisfies this condition, and note that the perimeter of τ_1 is precisely half the perimeter of τ_0.

Continue this devision inductively to obtain a sequence $(\tau_n)_{n\in\mathbb{N}}$ of triangular contours such that $T_{n+1} \subseteq T_n$ and $\text{perimeter}(\tau_n) = \frac{\text{perimeter}(\tau_0)}{2^n}$ and $\left|\int_{\tau_n} f(z)dz\right| \geq M/4^n$. By the topology fact, T_n is closed and bounded and therefore compact. Since $\text{perimeter}(\tau_n) \to 0$ as $n \to \infty$, we have that $\text{diam}(T_n) \to 0$ as $n \to \infty$. Therefore,

$$\bigcap_{n\in\mathbb{N}_0} T_n = \{z_0\}$$

for some $z_0 \in U$. Now, we use the holomorphicity of f to show that $M = 0$.

Fix $\varepsilon > 0$. Since f is complex-differentiable on U, it is complex differentiable at z_0, so there exists $\delta > 0$ such that $\mathbb{D}_\delta(z_0) \subseteq U$ and for all $z \in \mathbb{D}_\delta(z_0) \setminus \{z_0\}$,

$$\left|\frac{f(z) - f(z_0)}{z - z_0} - f'(z_0)\right| < \varepsilon$$

By rearrangement,

$$|f(z) - f(z_0) - f'(z_0)(z - z_0)| < |z - z_0|\varepsilon$$

Since $\text{diam}(T_n) \to 0$ as $n \to \infty$, there exists $N \in \mathbb{N}_0$ such that if $n > N$, then $T_n \subseteq \mathbb{D}_\delta(z_0)$. Since $f(z_0)$ is a constant and $f'(z_0)(z - z_0)$ is a polynomial in z, it has an antiderivative, so by part (d) of the antiderivative theorem, we have:

$$\int_{\tau_n} [f(z_0) + f'(z_0)(z - z_0)]dz = 0$$

because τ_n is a closed contour. Then, we have:

$$\frac{M}{4^n} \leq \left|\int_{\tau_n} f(z)dz\right| = \left|\int_{\tau_n} [f(z) \underbrace{- f(z_0) - f'(z_0)(z - z_0)}_{0}]dz\right| \leq \underbrace{\text{perimeter}(\tau_n^*) \sup_{z\in\tau_n^*} |f(z) - f(z_0) - f'(z_0)(z - z_0)|}_{\text{ML Estimate}}$$

Note that $|f(z) - f(z_0) - f'(z_0)(z - z_0)| < \varepsilon|z - z_0| \leq \varepsilon\,\text{perimeter}(\tau_n^*)$. Substituting that $\text{perimeter}(\tau_n^*) = \text{perimeter}(\tau_0^*)/2^{\text{n}}$, and this gives:

$$0 \leq \frac{M}{\cancel{4^n}} \leq \frac{\text{perimeter}(\tau_0^{*2})\varepsilon}{\cancel{2^{2n}}}$$

Since $\text{perimeter}(\tau_0^*)$ becomes arbitrarily small as $n \to \infty$, we have that $0 < M < \varepsilon$ for all $\varepsilon > 0$ and therefore $M = 0$. ■

The requirement of convexity in Cauchy's Integral Theorem can be reduced to simply connected domains, but it cannot be weakened further. Several important consequences immediately follow from the stated result, with a special case of the more general *Cauchy's Integral Formula* following immediately for the special case of circular contours.

Cauchy's Integral Formula for Disks and Circles. Let $z_0 \in \mathbb{C}$ and $R > 0$. If $f : \mathbb{D}_R(z_0) \to \mathbb{C}$ is holomorphic and $0 < r < R$, then

$$f(\zeta) = \frac{1}{2\pi i}\int_{\partial\mathbb{D}_r(z_0)} \frac{f(z)}{z - \zeta}dz$$

That is, the value $f(\zeta)$ is fully determined by the set of points a distance r away.

Another interesting result is Liouville's theorem, which clearly contrasts the behavior of complex-differentiable and real-differentiable functions:

Liouville's Theorem. Suppose $f : \mathbb{C} \to \mathbb{C}$ is entire and bounded. Then, f is constant.

Proof. Let $z_0 \in \mathbb{C}$, $r > |z_0|$. First, note that by Cauchy's Integral Formula, we have:

$$\begin{aligned} f(z_0) - f(0) &= \frac{1}{2\pi i}\int_{\partial \mathbb{D}_r} \frac{f(z)}{z - z_0} dz - \frac{1}{2\pi i}\int_{\partial \mathbb{D}_r} \frac{f(z)}{z} dz \\ &= \frac{1}{2\pi i}\int_{\partial \mathbb{D}_r} f(z)\left(\frac{1}{z - z_0} - \frac{1}{z}\right) dz \\ &= \frac{z_0}{2\pi i}\int_{\partial \mathbb{D}_r} \frac{f(z) z_0}{(z - z_0) z} dz \end{aligned}$$

From this, we can apply the ML estimate. Let M bound f, i.e. $|f(z)| < M$ for all $z \in \mathbb{C}$. Then:

$$\begin{aligned} |f(z_0) - f(0)| &= \left|\frac{z_0}{2\pi i}\int_{\partial \mathbb{D}_r} \frac{f(z)}{z(z - z_0)} dz\right| \\ &\leq \frac{|z_0|}{2\pi} \sup_{z \in \partial \mathbb{D}_r} \left|\frac{f(z)}{z(z - z_0)}\right| \cdot 2\pi r \\ &\leq M|z_0| \sup_{z \in \partial \mathbb{D}_r} \frac{1}{|z - z_0|} \\ &\leq \frac{M|z_0|}{r - |z_0|} \end{aligned}$$

The quantity $\frac{M|z_0|}{r-|z_0|}$ can be made arbitrarily small by sufficiently large choice of r; therefore, $f(z_0) = f(0)$. z_0 is arbitrary, so f is identically equal to $f(0)$, i.e. a constant. ■

Liouville's Theorem can be applied to show that every complex polynomial can be factored completely.

Fundamental Theorem of Algebra. Let $p(z) := \sum_{k=0}^{n} c_k z^k$ be a complex polynomial of degree $n \geq 1$, i.e. p is a non-constant complex polynomial. Then, there exists $z_0 \in \mathbb{C}$ such that $p(z_0) = 0$.

Proof. Suppose for sake of contradiction that $p(z) \neq 0$ for all $z \in \mathbb{C}$. Then, since p is entire and nonzero, $f := \frac{1}{p}$ is entire. Note that, if $|z| \geq 1$, then:

$$\begin{aligned} |f(z)| &= \left|\frac{1}{c_n z^n + \sum_{k=0}^{n-1} c_k z^k}\right| \\ &\leq \frac{1}{|c_n||z|^n - |\sum_{k=0}^{n-1} c_k z^k} \\ &\leq \frac{1}{|c_n||z|^n - |z|^{n-1}\sum_{k=0}^{n-1} |c_k|} \\ &= \frac{1}{|z|^{n-1}\left(|c_n||z| - \sum_{k=0}^{n-1} |c_k|\right)} \end{aligned}$$

Then, $|f(z)|$ can be made arbitrarily close to 0 for sufficiently large $|z|$, so there exists $R > 0$ such that $|f(z)| \leq 1$ for $|z| > R$. Choose $M := \max\left(1, \sup_{z \in \overline{\mathbb{D}}_R} |f(z)|\right)$, which is finite. Then, $|f(z)| \leq M$ for all $z \in \mathbb{C}$.

We now have that f is entire and bounded. Then, by Liouville's Theorem, f is constant. Then, p must be constant, which is a contradiction to our assumption. Therefore, there must exist $z_0 \in \mathbb{C}$ such that $p(z_0) = 0$. ■

A corollary of this result is that every complex polynomial can be reduced to linear factors; that is, if $p(z)$ is a complex polynomial of degree $n \geq 1$, then there exists (constant) $c, z_1, \ldots, z_n \in \mathbb{C}$ such that

$$p(z) = c(z - z_1)(z - z_2)\cdots(z - z_n)$$

which can be shown by algebra and induction from the case $n = 1$ (where p is already fully factored).

Equivalency between Analyticity and Holomorphicity. Several key results can be shown It has already been shown that the analytic functions are holomorphic. An equivalency can be shown, and to do so it suffices to show that the holomorphic functions are analytic.

Theorem (Holomorphic $\implies$ Analytic). Suppose $z_0 \in \mathbb{C}$, $R > 0$, and $f : \mathbb{D}_R(z_0) \to \mathbb{C}$ is holomorphic. If $0 < r < R$, $\zeta \in \mathbb{D}_r(z_0)$, and $\rho := r - |\zeta - z_0| > 0$, then

$$f(z) = \sum_{n=0}^{\infty} \underbrace{\left(\frac{1}{2\pi i} \int_{\partial \mathbb{D}_r(z_0)} \frac{f(w)}{(w-\zeta)^{n+1}} dw \right)}_{c_n} (z - \zeta^n)$$

for all $z \in \mathbb{D}_\rho(\zeta)$. In particular, f has a power series expansion and is therefore analytic.

Proof. Fix $z \in \mathbb{D}_\rho(\zeta)$. The general idea is to form a power-series expression for f, which will be done in particular by forming a geometric series formulation. If $w \in \partial \mathbb{D}_r(z_0)$, then:

$$\begin{aligned}
\left| \frac{f(w)}{w-z} \right| &= \left| \frac{f(w)}{(w-\zeta)-(z-\zeta)} \right| \\
&= \left| \frac{f(w)}{w-\zeta} \cdot \frac{1}{1 - \frac{z-\zeta}{w-\zeta}} \right| \\
&= \left| \frac{f(w)}{w-\zeta} \right| \cdot \frac{|z-\zeta|}{|w - z_0 - (\zeta - z_0)|} \\
\text{(reverse } \Delta \text{ inequality)} \quad &\leq \frac{|z-\zeta|}{|w-z_0| - |\zeta - z_0|} \\
&\leq \frac{|z-\zeta|}{\rho} \\
&< 1
\end{aligned}$$

Now, applying the geometric series formula:

$$\begin{aligned}
\frac{f(w)}{w-\zeta} \cdot \frac{1}{1 - \frac{z-\zeta}{w-\zeta}} &= \frac{f(w)}{w-\zeta} \sum_{n=0}^{\infty} \left(\frac{z-\zeta}{w-\zeta} \right)^n \\
&= \sum_{n=0}^{\infty} \frac{f(w)}{(w-\zeta)^{n+1}} (z-\zeta)^n
\end{aligned}$$

The Weierstrass M-test can be used to show that this series converges uniformly. For this reason, the order of limits can be interchanged. Then, applying the Cauchy integral formula:

$$\begin{aligned}
f(z) &= \frac{1}{2\pi i} \int_{\partial \mathbb{D}_r(z_0)} \frac{f(w)}{w-z} dz \\
&= \frac{1}{2\pi i} \int_{\partial \mathbb{D}_r(z_0)} \sum_{n=0}^{\infty} \frac{f(w)}{(w-\zeta)^{n+1}} (z-\zeta)^n dw \\
&= \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_{\partial \mathbb{D}_r(z_0)} \frac{f(w)}{(w-\zeta)^{n+1}} dw \right) (z-\zeta)^n
\end{aligned}$$

which is precisely the power series desired. ■

As a result of this discussion, it has been shown that analyticity is equivalent to holomorphicity, and the two concepts are interchangeable.

There are several interesting results from the equivalency between holomorphicity and analyticity. One such result is *Morera's Theorem*:

Theorem (Morera's Theorem). Let $U \subseteq \mathbb{C}$ be an open set and $f : U \to \mathbb{C}$ continuous. If $\int_\tau f(z)dz = 0$ for every triangular contour τ in U such that the set T enclosed by τ is contained in U, then $f : U \to \mathbb{C}$ is holomorphic.

Proof. Let $z_0 \in U$. z_0 is an interior point, so there exists $r > 0$ such that $\mathbb{D}_r(z_0) \subseteq U$. Since $\mathbb{D}_r(z_0)$ is convex, $\int_\tau f(z)dz = 0$ for all triangular contours τ in $\mathbb{D}_r(z_0)$. Since $\mathbb{D}_r(z_0)$ is convex, f has an antiderivative F. Since F is holomorphic, F is infinitely complex differentiable. Therefore, $F' = f|_{\mathbb{D}_r(z_0)}$ is holomorphic. Since $z_0 \in U$ is arbitrary, f is holomorphic on U. ■

Several other interesting results also follow, particularly the *differentiated Cauchy integral formula*, and also Cauchy's estimates, which follows from the differentiated Cauchy integral formula.

Corollary (Differentiated Cauchy Integral Formula for Disks and Circles). Suppose $z_0 \in \mathbb{C}$, $R > 0$, and $f : \mathbb{D}_R(z_0) \to \mathbb{C}$ is holomorphic. If $0 < r < R$, $\zeta \in \mathbb{D}_r(z_0)$, and $\rho := r - |\zeta - z_0| > 0$, then

$$f^{(n)}(\zeta) = \frac{n!}{2\pi i} \int_{\partial \mathbb{D}_r(z_0)} \frac{f(z)}{(z-\zeta)^{n+1}} dz$$

for all $n \in \mathbb{N}_0$.

Corollary (Cauchy's Estimates). Let $z_0 \in \mathbb{C}$ and $R > 0$. If $f : \mathbb{D}_R(z_0) \to \mathbb{C}$ is holomorphic and $0 < r < R$, then

$$\left| f^{(n)}(z_0) \right| \leq \frac{n!}{r^n} \sup_{z \in \partial \mathbb{D}_r(z_0)} |f(z)|$$

for all $n \in \mathbb{N}_0$.

General Cauchy's Integral Formula and Cauchy-Goursat Theorem.

As mentioned, both Cauchy's Integral Formula and the Cauchy-Goursat Theorem can be generalized further; the proof of the integral formula is fairly technical and is therefore omitted, but the most general form is provided:

Cauchy's Integral Formula. Let $U \subseteq \mathbb{C}$ be an open set, and let Γ be a collection of closed contours in U. If $f : U \to \mathbb{C}$ is holomorphic and $W_\Gamma(z) = 0$ for all $z \in \mathbb{C} \setminus U$, then:

$$W_\Gamma(\zeta) f(\zeta) = \frac{1}{2\pi i} \int_\Gamma \frac{f(z)}{z - \zeta} dz$$

The Cauchy-Goursat theorem is a consequence of this integral formula:

Cauchy-Goursat Theorem. Let $U \subseteq \mathbb{C}$ be an open set, and Γ a collection of closed contours in U. If $f : U \to \mathbb{C}$ is holomorphic and $W_\Gamma(z) = 0$ for all $z \in \mathbb{C} \setminus U$, then

$$\int_\Gamma f(z)dz = 0$$

4.5 Isolated Singularities and Residue Theory

A discussion of singularities of holomorphic functions requires some additional background. The *Laurent series*, which is a generalization of the power series to include all integer indices, is particularly useful for this purpose.

Laurent Series. A **Laurent series** is a series of the form

$$\sum_{k=-\infty}^{+\infty} c_k(z-a)^k = \cdots + \frac{c_{-2}}{(z-a)^2} + \frac{c_{-1}}{z-a} + c_0 + c_1(z-a) + c_2(z-a)^2 + \cdots$$

A series of functions can be extended in this way; for a sequence given by $(f_k)_{k\in\mathbb{Z}} = (\cdots, f_{-2},\ f_{-1},\ f_0,\ f_1,\ f_2,\ \cdots)$, the series is constructed by

$$\sum_k f_k := \left(\sum_{k=-n}^{n} f_k\right)_{n\in\mathbb{N}}$$

If $\sum_k f_k$ converges, its limit is denoted by

$$\sum_{k=-\infty}^{+\infty} f_k(x) := \lim_{n\to+\infty} \sum_{k=-n}^{n} f_k(x)$$

The terms of a Laurent series with negative index are called the *principal part*; the terms with nonnegative index are called the *regular part*. Note that power series are a special case of the Laurent series, with principal part equal to zero (i.e. $c_{-1} = c_{-2} = \cdots = 0$.

The *convergence of a Laurent series* can be established by applying Cauchy-Hadamard to the principal and regular parts separately. Define the *annulus* of inner radius $r \geq 0$ and outer radius $R > r$ centered at a to be the (open) set given by:

$$\mathbb{A}_{r,R}(a) := \{z \in \mathbb{C} \mid r < |z-a| < R\}$$

Now, the following theorem characterizes convergence of a Laurent series.

Theorem (Convergence of Laurent Series). Let $\sum_k c_k(z-a)^k$ be a Laurent series. Define:

$$r := \limsup_{n\to+\infty} |c_{-n}|^{1/n}, \qquad R := \limsup_{n\to+\infty} \frac{1}{\limsup\limits_{n\to+\infty} |c_n|^{1/n}}$$

Then, the series of functions $\sum_k c_k(z-a)^k$ converges locally uniformly on $\mathbb{A}_{r,R}(a)$ and absolutely uniformly on a compact set $C \subseteq \mathbb{A}_{r,R}(a)$, and $\sum_k c_k(z-a)^k$ diverges for $|z-a| < r$ and $|z-a| > R$.

Proof. Consider the restriction $k \in \mathbb{N}_0$. This indexes the regular part of $\sum_k c_k(z-a)^k$, which is a power series with radius of convergence R (by Cauchy Hadamard).

Consider now the restriction $k \in \mathbb{N}$ to the series $\sum_k c_{-k}(z-a)^{-k}$. Treating this as a power series, Cauchy Hadamard gives a radius of convergence given by:

$$r = \frac{1}{\limsup\limits_{k\to+\infty} |c_{-k}|^{-k}} = \frac{1}{\left(\limsup\limits_{k\to+\infty} |c_{-k}|^{k}\right)^{-1}} = \limsup_{k\to+\infty} |c_{-k}|^k$$

■

The Laurent series will be useful for discussing *isolated singularities* of holomorphic functions, which are (loosely speaking) points which cause undefined behavior, such as division by zero. Naturally, singularities are closely related to zeroes, or points at which a function "vanishes," which will be discussed first.

Let $U \subseteq \mathbb{C}$ be open. Suppose $f : U \to \mathbb{C}$ is analytic and $\zeta \in U$ is a zero of f, i.e. $f(\zeta) = 0$. Define:

$$V := \{n \in \mathbb{N} \mid f^{(n)}(\zeta) \neq 0\}$$

and define $m := \begin{cases} \min V, & V \neq \varnothing \\ +\infty, & V = \varnothing \end{cases}$. m is called the **vanishing order** of f. An equivalent characterization of the vanishing order is as follows:

(i) The vanishing order of f at ζ is $+\infty$ if and only if $f \equiv 0$ on every open subset of U containing ζ.

(ii) The vanishing order m of f at ζ is finite if and only if there exists an analytic function $g : U \to \mathbb{C}$ such that $g(\zeta) \neq 0$ and

$$f(z) = (z - \zeta)^m g(z)$$

As previously mentioned, singularities are points which cause undefined behavior (or more broadly, "poor" behavior, such as discontinuity or non-differentiability). A singularity z_0 of a holomorphic function f is *isolated* if there exists $\varepsilon > 0$ such that f is holomorphic on $\mathbb{A}_{0,\varepsilon}(z_0)$. For the following discussion, let f be a holomorphic function on $\mathbb{A}_{0,\varepsilon}(z_0)$, with Laurent expansion $f(z) = \sum_{n=0}^{+\infty} c_n (z - z_0)^n$. All isolated singularities can be classified as *removable* singularities, *poles*, or *essential singularities*. These definitions, as well as useful equivalent characterizations, are listed:

Removable Singularities. z_0 is a *removable singularity* of f if $c_{-n} = 0$ for all $n \in \mathbb{N}_0$.

- Equivalently, z_0 is a removable singularity if $\lim\limits_{z \to z_0} f(z)$ exists. Note that this is also equivalent to the condition that $\lim\limits_{z \to z_0} (z - z_0) f(z) = 0$. In essence, removable singularities can be *removed* if there is a function g which "extends" f, i.e. $g \equiv f$ on $\mathbb{A}_{0,\varepsilon}(z_0)$ with $g(z_0) = \lim\limits_{z \to z_0} f(z)$.

Poles. z_0 is a *pole* if there exists $k \in \mathbb{N}$ such that $c_{-m} \neq 0$. If $c_{-n} = 0$ for all $m > n$, then f has an m-order pole at z_0.

- Equivalently, z_0 is a pole of order m if $\lim\limits_{z \to z_0} (z - z_0)^m f(z) \neq 0$ exists, which is equivalent to the statement that $f(z) = \frac{g(z)}{(z - z_0)^m}$ for some holomorphic g with $g(z_0) \neq 0$. Yet another characterization is that z_0 is a pole of f if and only if $\lim\limits_{z \to z_0} f(z) = \infty$, where $\infty \in \hat{\mathbb{C}} := \mathbb{C} \cup \{\infty\}$ is the complex infinity. The general intuition for a pole is that f acts like a function which is well-behaved, besides its denominator $(z - z_0)^m$, which exclusively causes f to explode in magnitude.

Essential Singularities. z_0 is an *essential singularity* of f if $c_{-n} \neq 0$ for infinitely many $n \in \mathbb{N}$.

- Equivalently, z_0 is an *essential singularity* if $\lim\limits_{z \to z_0} f(z)$ does not exist in $\hat{\mathbb{C}}$.

Behavior of Essential Singularities. The behavior of functions near an essential singularity can be quite surprising. In an effort to develop some intuition for this type of isolated singularity, two theorems are stated and an example is given. For this discussion, let f be holomorphic on $\mathbb{A}_{0,\varepsilon}(z_0)$, and let z_0 be an essential singularity of f.

Casorati-Weierstrass Theorem. If z_0 is an essential singularity of f and $w \in \hat{\mathbb{C}}$. Then, there exists a sequence $(z_n)_{n \in \mathbb{N}}$ in $\mathbb{A}_{0,\varepsilon}(z_0)$ such that $\lim\limits_{z \to z_0} f(z_n) = w$.

This result can be shown quite quickly from the definition, but the true behavior near an essential singularity is even more unexpected than this (already quite surprising) result. However, to prove this fact requires more advanced discussion than is present in these notes:

Picard's Theorem. If z_0 is an essential singularity of f, then there exists $w_0 \in \mathbb{C}$ such that for all $\varepsilon_0 \in (0, \varepsilon)$ and $w \in \mathbb{C} \backslash \{w\}$, the set $\{z \in \mathbb{A}_{0,\varepsilon_0}(z_0) \mid f(z) = w\}$ is infinite. In particular, on every annulus about z_0, f attains every value, with potentially the exception of w_0, infinitely many times.

Example. Consider the function $f(z) = e^{1/z}$. We can find the coefficients by substituting $1/z$ into the power series expansion for exp:

$$f(1/z) = e^{1/z} = \sum_{n=0}^{+\infty} \frac{z^{-n}}{n!} = \sum_{n=-\infty}^{0} \frac{z^n}{(-n)!} \implies c_n = \begin{cases} \frac{1}{(-n)!}, & n \leq 0 \\ 0, & n > 0 \end{cases}$$

Therefore, f has an essential singularity at $z_0 = 0$.

The **residues** of a function are closely related to the concept of isolated singularities. Let f be a complex function with isolated singularity $z_0 \in \mathbb{C}$, and let $f(z) = \sum_{n=-\infty}^{+\infty} c_n (z - z_0)^n$ be the Laurent expansion of f about z_0. Then, the number

$$\operatorname*{Res}_{z=z_0} f(z) := c_{-1}$$

is called the **residue** of f at z_0. The *residue theorem* relates these residues to the evaluation of integrals.

Residue Theorem. Let $U \subseteq \mathbb{C}$ be open, $S \subseteq U$ be finite, and Γ be a collection of contours in $U \setminus S$. If $W_\Gamma(z) = 0$ for all $z \in \mathbb{C} \setminus U$ and $f : U \setminus S \to \mathbb{C}$ is holomorphic (with isolated singularities at the points in S), then:

$$\int_\Gamma f(z)dz = 2\pi i \sum_{z_0 \in S} W_\Gamma(z_0) \operatorname*{Res}_{z=z_0} f(z) \tag{Residue Theorem}$$

In other words, when integrating around a contour which does not wind around points outside of U, it suffices to count how many times the contour winds around the singularities and to evaluate the residues of f at these singularities. The residue theorem relates these numbers to the value of the desired contour integral. The proof can be carried out by applying the generalized Cauchy-Goursat theorem.

There is a useful formula for the residue provided by an encircled pole, which makes the evaluation of residues simple for many functions. The proof follows by examining the coefficients of the Taylor expansion of f.

Formula for Residues at Poles. Suppose z_0 is an n-order pole of f. Then,

$$\operatorname*{Res}_{z=z_0} f(z) = \frac{1}{(n-1)!} \lim_{z \to z_0} \frac{d^{n-1}}{dz^{n-1}} \left[(z - z_0)^n f(z) \right] \tag{Residue at Poles}$$

A **branch point** is a distinct type of singularity which applies to *multifunctions*.

Definition. Let $S \subseteq \mathbb{C}$ be a set. A complex **multifunction** is a function $F : S \to 2^{\mathbb{C}}$, i.e. an assignment of a set, $F(z) \subseteq \mathbb{C}$, to each $z \in S$. A **branch** of the multifunction F is a single-valued function $f : S_0 \to \mathbb{C}$, with $S_0 \subseteq S$ and $f(z) \in F(z)$ for all $z \in S_0$.

Branches of common multifunctions. Some discussion will now be carried out regarding examples of multifunctions, and common branches.

Example 1. $F(z) = \arg z$ is a multifunction. The two common branches for arg are $\operatorname{Arg} z \in (-\pi, \pi]$, which is discontinuous along the negative real axis, and $\operatorname{HArg} z \in [0, 2\pi)$, which is discontinuous along the positive real axis.

Example 2. The nth root of z, $F(z) = \sqrt[n]{z}$ satisfying $F(z)^n = z$, is a multifunction. In particular, F maps values in $\mathbb{C} \setminus \{0\}$ to the set:

$$\sqrt[n]{z} = \left\{ \sqrt[n]{|z|} \exp\left(i \cdot \frac{\theta + 2\pi k}{n} \right) \,\middle|\, k = 0, 1, \ldots, n-1 \right\}$$

Therefore, there are n solutions to the nth root of a complex number.

Example 3. The square root of z, $F(z) = \sqrt{z}$ satisfying $F(z)^2 = z$, is clearly a multifunction (by Example 2). The **principal branch** f of $\sqrt{z}$, which is the "standard" choice of branch, is found using the above formula (with $n = 2$) with the convention that f is discontinuous along the negative real axis:

$$f(z) = \sqrt{|z|} e^{\frac{i}{2} \operatorname{Arg} z}$$

where the discontinuity comes from the argument. Another branch is found by applying the formula from Example 2 with $k = 1$, in which case the discontinuity is along the positive real axis instead.

Example 4. The logarithm of z, $F(z) = \log z$ satisfying $e^{F(z)} = z$, is a multifunction. In particular, F maps values in $z \in \mathbb{C} \setminus \{0\}$ to the set:

$$\log z = \log |z| + i \arg z = \{\log |z| + i\theta \mid \theta \in \arg z\}$$

where $\log |z|$ is the (single-valued) real logarithm. The **principal branch**, commonly denoted $L(z)$, is given by:

$$L(z) = \log |z| + i \operatorname{Arg} z$$

where L is defined on $\mathbb{C} \setminus (-\infty, 0]$ with discontinuity along the negative real axis.

A **branch point** is a point which, when encircled, produces distinct values which disagree when the path returns to its original point. ∞ is called a branch point of F is 0 is a branch point of $F(1/z)$. More precisely, for a multifunction $F : S \to 2^{\mathbb{C}}$, z_0 is a branch point of F if for every $\varepsilon > 0$ such that $\mathbb{A}_{0,\varepsilon}(z_0) \subseteq S$, there exists a nonempty domain $U \subseteq \mathbb{A}_{0,\varepsilon}(z_0)$ and an analytic branch $f : U \to \mathbb{C}$ of F for which there is no analytic continuation onto all of $\mathbb{A}_{0,\varepsilon}(z_0)$.

Example. As an example, the branch points of $\sqrt{z}$ are precisely $\{0, \infty\}$. We can see this as follows:

Consider the point $-\varepsilon/2 \in \mathbb{A}_{0,\varepsilon}(z_0)$. Taking values $z \in \mathbb{A}_{0,\varepsilon}(z_0) \cap \underbrace{\{z \in \mathbb{C} \mid \operatorname{Im} z > 0\}}_{:=\mathbb{H}}$, note that:

$$\sqrt{z} = \sqrt{|z|}e^{\frac{i}{2}\operatorname{Arg} z} \xrightarrow[z\in\mathbb{A}_{0,\varepsilon}(z_0)\cap\mathbb{H}]{z\to\varepsilon/2} \frac{\varepsilon}{2}e^{\frac{i}{2}\pi} = i\sqrt{\frac{\varepsilon}{2}}$$

However, if we take $w \in \mathbb{A}_{0,\varepsilon}(z_0) \cap \underbrace{\{w \in \mathbb{C} \mid \operatorname{Im} w < 0\}}_{:=\mathbb{H}_-}$, note that:

$$\sqrt{w} = \sqrt{|w|}e^{\frac{i}{2}\operatorname{Arg} z} \xrightarrow[w\in\mathbb{A}_{0,\varepsilon}(z_0)\cap\mathbb{H}_-]{w\to\varepsilon/2} \frac{\varepsilon}{2}e^{\frac{i}{2}(-\pi)} = -i\sqrt{\frac{\varepsilon}{2}}$$

That is, approaching the point $-\varepsilon/2$ by circling 0 produces a different result when approaching from above vs from below. Now, consider $\sqrt{1/z}$. Substitute $w = 1/z$. Clearly, $w = 0$ is a branch point of $\sqrt{w}$ from the previous discussion. Therefore, ∞ is a branch point of $\sqrt{z}$. It can be shown that no other point is a branch point, although this process is tedious.

A common process to produce an analytic branch of a multifunction involves restricting the domain with a **branch cut**. A branch cut is obtained by subtracting a non-self-intersecting curve which connects the branch points of a multifunction from the domain of the multifunction. Doing so prevents encircling the branch points, and therefore allows the construction of an analytic branch on this restricted domain. There are many choices of branch cuts, but simple cuts are generally preferable.

Example. $\sqrt{z}$ has branch points at 0 and ∞. Restricting the domain of $\sqrt{z}$ to the set $U := \mathbb{C} \setminus (-\infty, 0]$ causes the principal branch of $\sqrt{z}$ to be well-defined. It can be shown that the branch points of $\log z$ are also 0 and ∞, so the same branch cut guarantees the ability to construct an analytic branch of log, and in fact the principal branch of log is well-defined on U. These are not the only valid branch cuts; any branch cut connecting 0 to the point "at infinity" would work. For example, removing all points $\gamma(t) = t(i+1),\ t \geq 0$ would remove a line from the first quadrant to connect 0 and ∞, but the principal branch of both $\sqrt{z}$ and log are not valid analytic branches on this domain.

Real Integrals with Complex Methods. The study of singularities allows many real-valued integrals to be evaluated analytically by taking advantage of the properties of singularities for complex functions. Two examples will be provided to illustrate this method: one utilizing residue theory, and one utilizing properties of multifunctions.

Example 1. Evaluate the following integral using residue theory:

$$I_a := \int_{-\infty}^{+\infty} \frac{1}{(x^2+a^2)^2}dx = \lim_{R\to+\infty}\int_{-R}^{R} \frac{1}{(x^2+a^2)^2}dx,\ a > 0$$

The following procedure may be used generally for similar integrals:

1. Express the integral in terms of a contour integral in the complex plane.
2. Complete the original contour to a closed contour.
3. Use the residue theorem to evaluate the integral over the closed contour.
4. Estimate remainder terms (i.e. by the ML estimate), and show that these vanish in the limit.

Now, applying these steps to I_a:

Step 1. Note that the straight-line contour $\sigma_{-R\to R}$ corresponds to a segment of the real axis, and note that z is real-valued on the real axis. Therefore,

$$I_a = \lim_{R\to+\infty}\int_{\sigma_{-R\to R}} \frac{1}{(z^2+a^2)^2}dz$$

Step 2. Let η_R be the semi-circular contour in the upper-half plane, i.e. η_R traces out $\mathbb{H} \cap \partial\mathbb{D}_R$ in the counter-clockwise direction. Define the concatenation $\gamma := \sigma_{-R\to R} \cdot \eta_R$. Note that:

$$\int_\gamma \frac{1}{(z^2+a^2)^2}dz = \int_{\sigma_{-R\to R}} \frac{1}{(z^2+a^2)^2}dz + \int_{\eta_R} \frac{1}{(z^2+a^2)^2}dz$$

Step 3. By the residue theorem,

$$\begin{aligned}\int_\gamma \frac{1}{(z^2+a^2)^2}dz &= 2\pi i \operatorname*{Res}_{z=ai} \frac{1}{(z^2+a^2)^2} \\ &= 2\pi i \frac{d}{dz}\frac{1}{(z+ia)^2} \\ &= \frac{\pi}{2a^3}\end{aligned}$$

Step 4. By the ML estimate and the reverse triangle inequality,

$$\begin{aligned}\int_{\eta_R} \frac{1}{(z^2+a^2)^2}dz &\leq \pi R \cdot \sup_{z\in\partial\mathbb{D}_R} \left|\frac{1}{(z^2+a^2)^2}\right| \\ &\leq \pi R \cdot \frac{1}{(R^2-a^2)^2} \\ &= \frac{\pi R}{(R^2-a^2)^2} \xrightarrow{R\to+\infty} 0\end{aligned}$$

Putting steps 1 through 4 together,

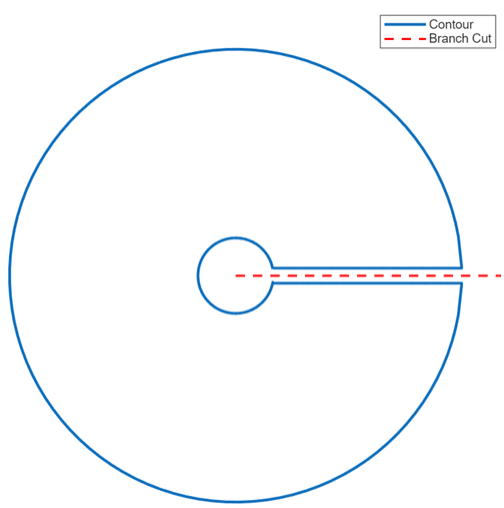
$$\frac{\pi}{2a^3} = \int_\gamma \frac{1}{(z^2+a^2)^2}dz = \lim_{R\to+\infty}\left[\int_{\sigma_{-R\to R}} \frac{1}{(z^2+a^2)^2}dz + \cancelto{0}{\int_{\eta_R} \frac{1}{(z^2+a^2)^2}dz}\right] = I_a$$

Example 2. Use a branch cut of log to evaluate

$$I := \int_0^{+\infty} \frac{1}{(x+1)^2+4}dx = \lim_{(r,R)\to(0,+\infty)} \underbrace{\int_r^R \frac{1}{(x+1)^2+4}dx}_{I_{r,R}}$$

Define $f(z) := \frac{L(z)}{(z+1)^4+4}$, where $L(z)$ is the branch defined by $L(z) := \log|z| + i\,\mathrm{HArg}\,z$. Note that f has singularities at $z = -1 \pm 2i$, as well as at $z = 0$ where $\log|z|$ has a singularity; this is the reason why, even though I is not an improper integral at its lower bound, we still consider the limit.

Let $0 < r < \sqrt{5} < R$ and consider the "keyhole" contour $\gamma_{r,R}$ as shown below (moving clockwise along the inner radius and counterclockwise along the outer radius):



In the limit, the two lines lie "along the branch cut" of L. This property can be exploited to effectively evaluate this integral. Consider the constituent parts of $\int_{\gamma_{r,R}} f(z)dz$. The keyhole contour has four parts: an inner contour of radius r, which contributes $-\int_{\partial\mathbb{D}_r}$, an outer contour of radius R which contributes $\int_{\partial\mathbb{D}_R} f(z)dz$, and two lines along the real axis, which contribute:

$$\begin{aligned}
\int_r^R \frac{L(z)}{(z+1)^2+4}dz - \int_r^R \frac{L(z)}{(z+1)^2+4}dz \xrightarrow[\mathbb{H}_- \ni z \to t]{\mathbb{H} \ni z \to t} \quad & \int_r^R \frac{\log t + 0i}{(t+1)^2+4}dt - \int_r^R \frac{\log t + 2\pi i}{(t+1)^4+4}dt \\
&= -2\pi i \int_r^R \frac{1}{(t+1)^4+4}dt \\
&= -2\pi i I_{r,R}
\end{aligned}$$

Therefore,

$$\int_{\gamma_{r,R}} f(z)dz = -2\pi i I_{r,R} - \int_{\partial\mathbb{D}_r} f(z)dz + \int_{\partial\mathbb{D}_R} f(z)dz$$

Note that $\gamma_{r,R}$ encircles two singularities of f, namely $-1 \pm 2i$. By the residue theorem,

$$\begin{aligned}
\int_{\gamma_{r,R}} f(z)dz &= 2\pi i \operatorname*{Res}_{z=-1+2i} f(z) + 2\pi i \operatorname*{Res}_{z=-1-2i} f(z) \\
&= 2\pi i \left[\operatorname*{Res}_{-1+2i} \frac{L(z)}{(z+1)^2+4} + \operatorname*{Res}_{-1-2i} \frac{L(z)}{(z+1)^2+4} \right] \\
\text{(by pole formula)} \quad &= 2\pi i \left[\frac{L(-1+2i)}{(-1+2i)-(-1-2i)} + \frac{L(-1-2i)}{(-1-2i)-(-1+2i)} \right] \\
&= 2\pi i \frac{L(-1+2i) - L(-1-2i)}{4i} \\
&= \frac{\pi}{2} \left[\log\sqrt{5} + i\,\mathrm{HArg}(-1+2i) - \log\sqrt{5} - i\,\mathrm{HArg}(-1-2i) \right] \\
&= \frac{i\pi}{2} \left([\pi - \arctan(2)] - [\pi + \arctan(2)] \right) \\
&= -\pi i \arctan(2)
\end{aligned}$$

Now, the ML estimate will show that the remainder terms are zero. If $\rho > 0$ and $\rho \neq 1$, we have:

$$\left| \int_{\partial\mathbb{D}_\rho} f(z)dz \right| \leq 2\pi\rho \cdot \sup_{z \in \partial\mathbb{D}_\rho} \left| \frac{\log|z| + i\,\mathrm{HArg}\, z}{(z+1)^2+4} \right| \leq 2\pi\rho(\log|\rho| + 2\pi) \sup_{z \in \partial\mathbb{D}_\rho} \left| \frac{1}{(z+1)^2+4} \right|$$

If $\rho < 1$, then:

$$2\pi\rho(\log|\rho| + 2\pi) \sup_{z \in \partial\mathbb{D}_\rho} \left| \frac{1}{(z+1)^2+4} \right| \leq \frac{2\pi\rho(\log|\rho| + 2\pi)}{(\rho+1)^2+4} = 2\pi \underbrace{\frac{\log|\rho^\rho| + 2\pi\rho}{(\rho+1)^2+4}}_{\rho<1 \implies 1/e \leq |\rho^\rho| < 1} < 2\pi \frac{1+2\pi\rho}{(\rho+1)^2+4} \xrightarrow{\rho \to 0} 0$$

If $\rho > 1$, then:

$$2\pi\rho(\log|\rho| + 2\pi) \sup_{z \in \partial\mathbb{D}_\rho} \left| \frac{1}{(z+1)^2+4} \right| \leq \underbrace{\frac{2\pi\rho \cdot (\rho^{1/2} + 2\pi)}{(\rho+1)^2+4}}_{\log\rho < \sqrt{\rho} \text{ asymptotically}} \xrightarrow{\rho \to +\infty} 0$$

Therefore, in the limit, both $\int_{\partial\mathbb{D}_r} f(z)dz \xrightarrow{r \to 0} 0$ and $\int_{\partial\mathbb{D}_R} f(z)dz \xrightarrow{R \to +\infty} 0$. Therefore, we have that:

$$I = \lim_{(r,R) \to (0,+\infty)} I_{r,R} = -\frac{1}{2\pi i} \int_{\gamma_{r,R}} f(z)dz = \frac{\pi i \arctan 2}{2\pi i} = \boxed{\frac{\arctan 2}{2}}$$

This technique is often referred to as the "log trick," and it can be used to solve many similar integrals.

A final result from the study of residues is presented for its utility. Its proof is not presented, as the material required is

somewhat detached from the discussion of singularities.

Rouche's Theorem. Let $U \subseteq \mathbb{C}$ be an open set, and let γ be a Jordan contour in U such that the inside of γ is fully contained in U. If $f, g : U \to \mathbb{C}$ are holomorphic and

$$|g(z) - f(z)| < f(z)$$

for all $z \in \gamma^*$ (i.e. *along* the path of the contour), then f and g have the same number of zeros (counted with order) inside of γ.

Practically speaking, Rouche's theorem is a tool which allows one to count how many zeros occur within a given domain, which is generally not an obvious, by considering a more obvious function. Rouche's theorem is often utilized by selecting a function f such that $\left|\frac{g(z)-f(z)}{f(z)}\right| < 1$, where f is chosen so that the number of zeros of f within the desired domain is immediately apparent.

Example. How many zeroes does the polynomial

$$g(z) := z^6 + 9z^4 + z^3 + 2z + 4$$

have on $\mathbb{D}$?

Solution. It is obvious that g has 6 zeroes on the complex plane, but it is far from obvious how many lie within the unit disk in particular. However, by taking $f(z) := 9z^4$, and considering the following quantity:

$$\left|\frac{g(z) - f(z)}{f(z)}\right| = \left|\frac{z^6 + z^3 + 2z + 4}{9z^2}\right| \leq \left|\frac{z^4}{9}\right| + \left|\frac{z}{9}\right| + \left|\frac{2}{9z}\right| + \left|\frac{4}{9z^2}\right| \overset{z\in\partial\mathbb{D}}{=} \frac{1}{9} + \frac{1}{9} + \frac{2}{9} + \frac{4}{9} = \frac{8}{9} < 1$$

we see that Rouche's theorem applies, and g has the same number of zeroes on $\mathbb{D}$ as f. Clearly, $9z^4$ has four zeroes (counted with order) at $z = 0$, so g has four zeroes on $\mathbb{D}$.

In principle, this result can be applied to *any* function, not just polynomials.

4.6 The Laplace Equation with Dirichlet Boundary Conditions

Complex analysis can be applied to solve the Laplace Equation with Dirichlet boundary conditions. This particular partial differential equation comes up in several other areas. Notably, the Laplace equation is the heat equation $u_t = \alpha\nabla^2 u$ with $u_t \equiv 0$, i.e. at steady-state. Other examples emerge in fluid mechanics and electrodynamics. First, some definitions and properties are presented for a general class of functions called *harmonic functions*.

Definition. Let $U \subseteq \mathbb{C}$ be open. Let $u : U \to \mathbb{C}$ be twice-differentiable on U. U is called **harmonic** if

$$\Delta u \equiv \nabla^2 u := \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \equiv 0$$

Note that for any holomorphic $f : U \to \mathbb{C}$ where $f(z) = u(z) + iv(z)$, the Cauchy-Riemann equations give us that:

$$\Delta u = \partial_1^2 u + \partial_2^2 u = \partial_1\partial_2 v - \partial_2\partial_1 v = \underbrace{\partial_1\partial_2 v - \partial_1\partial_2 v}_{\text{interchangeable order}} = 0$$

Therefore, the real part u of f is harmonic. Similarly, the imaginary part v of f is also harmonic. The functions u and v are called **harmonic conjugates** in U. This constitutes one simple way to check whether a function u is harmonic: it suffices to show that u is the real part (or the imaginary part) of a holomorphic function.

Another method to find harmonic conjugates is by partial integration. In particular, suppose u is a harmonic function. Then, the function v must satisfy:

$$v(x, y) = \begin{cases} \displaystyle\int \frac{\partial v(x,y)}{\partial y} dy = \int \frac{\partial u(x,y)}{\partial x} dy \\ \displaystyle\int \frac{\partial v(x,y)}{\partial x} dx = -\int \frac{\partial u(x,y)}{\partial y} dx \end{cases}$$

where this relationship comes directly from integrating the Cauchy-Riemann equations.

Another property of harmonic functions is given, which also serves as an alternative characterization of harmonic functions.

Mean value property for harmonic functions. Let $U \subseteq \mathbb{C}$ be an open set, $z_0 \in U$, and $R > 0$ such that $\mathbb{D}_R(z_0) \subseteq U$. A function $u : U \to \mathbb{R}$ is harmonic if an only if it has the mean value property:

$$u(z_0) = \frac{1}{2\pi r} \int_{\partial \mathbb{D}_r(z_0)} u(z)|dz| \quad \forall r \in (0, R) \tag{Mean Value Property}$$

This kind of integral is called an **arclength integral**, defined for any contour $\gamma : [a, b] \to \mathbb{C}$ and any function $g : \gamma^* \to \mathbb{C}$ to be the following complex number:

$$\int_\gamma g(z)|dz| := \int_a^b g(\gamma(t))|\gamma'(t)|dt$$

Some additional properties will be outlined, which can be shown to follow quickly from each other.

Identity Theorem for Harmonic Functions. suppose $U \subseteq \mathbb{C}$ is a domain, and $u : U \to \mathbb{R}$ is harmonic. If there exists $z_0 \in U$ and $r > 0$ such that $\mathbb{D}_r(z_0) \subseteq U$ and u is constant on $\mathbb{D}_r(z_0)$, then u is constant on all of U.

Note that this result is much weaker than the identity theorem for analytic functions.

Maximum and Minimum Principles with Boundary for Harmonic Functions. Let $U \subseteq \mathbb{C}$ be a non-empty, bounded domain. If $u : \overline{U} \to \mathbb{R}$ is harmonic in U and continuous in $\overline{U}$, then there exists a $z_0 \in \partial U$ such that $u(z_0) = \max\{u(z) \mid z \in \overline{U}\}$ and a $z_1 \in \partial U$ such that $u(z_1) \min\{u(z) \mid z \in \overline{U}\}$.

Statement of The Laplace Equation with Dirichlet Boundary Conditions. Let $U \subseteq \mathbb{C}$ be an open set such that $\partial U \neq \varnothing$. Let $h : \partial U \to \mathbb{R}$. A function $u : U \to \mathbb{R}$ is a solution to the Laplace Equation with Dirichlet boundary condition h if u satisfies:

$$\Delta u \equiv 0 \quad \text{on } U \tag{Laplace's Equation}$$

$$\lim_{U \ni z \to \zeta \in \partial U} u(z) = h(\zeta) \tag{Dirichlet BC}$$

Note that the Laplace equation is precisely the statement that u is harmonic, and the Dirichlet boundary condition is the statement that u agrees with prescribed boundary data h.

Theorem (Uniqueness of Solutions). If U is bounded and $h : \partial U \to \mathbb{R}$ is continuous, then there exists at most one solution, u, to the Laplace equation on U with Dirichlet boundary condition h.

Proof. Suppose $u_1, u_2 : \overline{U} \to \mathbb{R}$ are two solutions to the Laplace equation with Dirichlet boundary data h. Since u_1 and u_2 are complex differentiable, u_1 and u_2 are continuous. Also note that $u_1 = h = u_2$ on ∂U. We show a corollary.

Corollary. Suppose $U \subseteq \mathbb{C}$ is a nonempty bounded domain. Suppose $u_1, u_2 : \overline{U} \to \mathbb{R}$ are harmonic in U and continuous in $\overline{U}$. If $u_1 = u_2$ on ∂U, then $u_1 = u_2$ on all of $\overline{U}$.

Proof of the Corollary. Define $u := u_1 - u_2$. Note that u is harmonic in U and continuous in $\overline{U}$. By the maximum and minimum principle with boundary, there exists $z_0, z_1 \in U$ such that

$$\max_{z \in \overline{U}} u(z) = u(z_0) = 0 = u(z_1) = \min_{z \in \overline{U}} u(z_1)$$

which implies that u is constant.

With the corollary, $u_1 = u_2$ on $\overline{U}$. ■

A discussion of existence is much more complicated, but one set for which a solution always exists is on closed disks.

Theorem (Existence of Solutions on Disks). Define the *Poisson kernel* for all $z, \zeta \in \mathbb{C}$, $z \neq \zeta$ as follows:

$$P(z, \zeta) := \frac{|\zeta|^2 - |z|^2}{|z - \zeta|^2}$$

Let $R > 0$, and let $h : \partial\mathbb{D}_R$ be continuous. The function $u : \overline{\mathbb{D}}_R \to \mathbb{R}$ defined by:

$$u := \frac{1}{2\pi R} \int_{\partial\mathbb{D}_R} h(z) P(z, \zeta) |d\zeta| \qquad \text{(Poisson's Integral Formula)}$$

is a solution to the Laplace Equation on $\mathbb{D}_R$ with Dirichlet boundary data h.

The arclength integral which gives u is called *Poisson's integral formula*. The proof that u is harmonic is relatively straightforward, with one only needing to show that u is the real part of a differentiable function. It is much more challenging to show that $u(z) \to h(\zeta)$ as $\mathbb{D}_R \ni z \to \zeta \in \partial\mathbb{D}_R$, but it is true.

The disk is not the only domain on which the Laplace equation can be solved subject to Dirichlet boundary conditions. Solutions can be transformed between topologically equivalent sets by constructing a suitably well-behaved map (i.e. a homeomorphism). Let $S \subseteq \mathbb{C}$. Define $\mathrm{cl}_\infty(S)$ to be the closure of S in $\hat{\mathbb{C}}$, and let $\partial_\infty S$ be the boundary of S in $\hat{\mathbb{C}}$; namely, if S is bounded, then $\mathrm{cl}_\infty(S) = \overline{S}$ and $\partial_\infty S = \partial S$, and if S is unbounded this set includes ∞.

Definition. Let $U, V \subseteq \mathbb{C}$ be domains. A **Riemann map** from U to V is a homeomorphism $\Phi : \mathrm{cl}_\infty(U) \to \mathrm{cl}_\infty(V)$ such that $\Phi(\partial_\infty U) = \partial_\infty V$.

Definition. A bounded subset $\Omega \subseteq \mathbb{C}$ is called a **Jordan domain** if there exists a Jordan curve γ such that Ω is the set of points enclosed by γ.

Theorem (Riemann Mapping Theorem). Suppose $\Omega \subseteq \mathbb{C}$ is a Jordan domain. Then, there exists a Riemann map from Ω to $\mathbb{D}$. In addition, if $z_1, z_2, z_3 \in \partial\Omega$ are distinct points (oriented counterclockwise), and $w_1, w_2, w_3 \in \partial\mathbb{D}$ are distinct points (oriented counterclockwise), then there exists a unique Riemann map $\Phi : \overline{\Omega} \to \overline{\mathbb{D}}$ such that $\Phi(z_i) = w_i$.

The proof of the Riemann mapping theorem is outside the scope of this material, but it can be used to guarantee solutions exist on Jordan domains. Since a Riemann map is guaranteed to exist,

Theorem (Existence of Solutions on Jordan Domains). Suppose $\Omega \subseteq \mathbb{C}$ is a Jordan domain, and let $h : \partial\Omega \to \mathbb{R}$ be continuous Dirichlet boundary data. Choose a Riemann map Φ from Ω to $\mathbb{D}$ (which exists by the Riemann mapping theorem), and define

$$h_{\mathbb{D}}(z) := h(\Phi^{-1}(z)) \quad \forall z \in \partial\mathbb{D}$$

If $u_{\mathbb{D}}$ is a solution to the Laplace equation on $\mathbb{D}$ with Dirichlet boundary data $h_{\mathbb{D}}$ (given by Poisson's integral formula), then the function

$$u := u_{\mathbb{D}} \circ \Phi$$

is a solution to the Laplace equation on Ω with Dirichlet boundary condition h.

The only question, then, is how to construct Riemann maps. This can be done by studying a well-behaved class of conformal maps called **fractional linear transformations** (FLTs). For the following discussion, let A be a matrix of the form $A := \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where $a, b, c, d \in \mathbb{C}$ and $\det A \neq 0$, i.e. A is invertible, and its inverse is given by $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

The fractional linear transformation associated with A is the map $T_A : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ defined by:

$$T_A(z) := \frac{az + b}{cz + d}$$

with the convention that $\frac{1}{\infty} = 0$, $\frac{1}{0} = \infty$, and $\frac{\infty}{\infty} = 1$. Some nice properties follow, which are immediately obvious (outside of the mentioned edge-cases, in which case the results can be shown by applying the conventions): T_A is a continuous map, $T_A \circ T_B = T_{AB}$, and $T_A = T_B \iff A = cB$ for some $c \in \mathbb{C} \setminus \{0\}$.

Perhaps the most important property, which motivates the study of FLTs at all, is that T_A is a homeomorphism with $T_A^{-1} = T_{A^{-1}}$. Namely, T_A is simple to invert, and T_A acts as a conformal map between two subsets of $\hat{\mathbb{C}}$. Another key result which is much less obvious is that FLTs which share three points are identical.

Uniqueness Property of FLTs. Let A, B be matrices as above. Let $z_1, z_2, z_3 \in \hat{\mathbb{C}}$ such that $T_A(z_i) = T_B(z_i)$ for all $i \in \{1, 2, 3\}$. Then, $T_A = T_B$, i.e. $A = cB$ for some $c \neq 0$. That is, if two FLTs agree at three points, they are the same FLT.

This fact allows the construction of FLTs by assigning a mapping to only three points.

Construction of FLTs. Let (z_1, z_2, z_3) be distinct and (w_1, w_2, w_3) be distinct. The following discussion will provide a guide for constructing an FLT from (z_1, z_2, z_3) to (w_1, w_2, w_3), and therefore for constructing an FLT between any two spaces.

Definition. Let $z \in \hat{\mathbb{C}}$, and consider four points (z_0, z_1, z_2, z_3). The **cross-ratio** is defined as the quantity

$$[z_0, z_1, z_2, z_3] := \frac{z_0 - z_1}{z_0 - z_3} \frac{z_2 - z_3}{z_2 - z_1}$$

If (z_1, z_2, z_3) is a collection of distinct points, this is an FLT, i.e. the map

$$z \mapsto [z, z_1, z_2, z_3] = \frac{z - z_1}{z - z_3} \frac{z_2 - z_3}{z_2 - z_1}$$

is an FLT (note that the quantity $\frac{z_2 - z_3}{z_2 - z_1}$ is a constant). Note in particular that $F(z_1) = 0$, $F(z_2) = 1$, and $F(z_3) = \infty$, i.e. F maps the triplet (z_1, z_2, z_3) to the triplet $(0, 1, \infty)$. This very specific FLT can be used to construct *any* FLT between triplets of points, as the following discussion shows.

Consider the FLTs defined by $F(z) := (z, z_1, z_2, z_3)$, and $G(z) := [w, w_1, w_2, w_3]$. Then, note that:

$$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} \xrightarrow{F} \begin{bmatrix} 0 \\ 1 \\ \infty \end{bmatrix} \xleftarrow{G} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$$

Therefore,

$$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} \xrightarrow{F} \begin{bmatrix} 0 \\ 1 \\ \infty \end{bmatrix} \xrightarrow{G^{-1}} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \implies \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} \xrightarrow{F \circ G^{-1}} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$$

This shows that an FLT can be constructed in general between any two triplets of distinct points. Specific steps are as follows:

1. Define $T_A := [z, z_1, z_2, z_3]$, i.e. an FLT from (z_1, z_2, z_3) to $(0, 1, \infty)$.
2. Define $T_B := [w, w_1, w_2, w_3]$, i.e. an FLT from (w_1, w_2, w_3) to $(0, 1, \infty)$.
3. Compute B^{-1}, which is simple for a 2×2 matrix.
4. Compute $T_A \circ T_B^{-1} = T_{AB^{-1}}$, i.e. multiply AB^{-1} and construct the FLT from the resulting transformation.

An important fact allows one to construct FLTs which map desired sets under certain conditions, rather than just three points.

Definition. A set $C \subseteq \hat{\mathbb{C}}$ is a **generalized circle** if C is a circle (i.e. the boundary of some disk in $\mathbb{C}$) or if $C = L \cup \{\infty\}$ for some line $L \subseteq \mathbb{C}$.

This definition makes sense if one considers the limiting case of a circle, where the radius is infinitely large and becomes, in the limit, a perfect line dividing the plane in two.

Fact. A generalized circle has two sides: an inside, and an outside. If C is a true circle, the inside of C is the inside of the circle, and the outside is the rest of $\hat{\mathbb{C}}$; if C is a line, the inside is one half of the plane (not including ∞), and the outside is the rest of $\hat{\mathbb{C}}$, i.e. including ∞. In either case, the sides of C are nonempty, disjoint, connected sets $S_1, S_2 \subseteq \hat{\mathbb{C}}$ such that $\hat{\mathbb{C}} \setminus C = S_1 \cup S_2$, i.e. C separates the extended complex plane into two pieces.

Theorem (Action of FLTs on Generalized Circles). Suppose $F : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ is an FLT and $C \subseteq \hat{\mathbb{C}}$ is a generalized circle with sides S_1 and S_2. Then, the image $F(C) \subseteq \hat{\mathbb{C}}$ is a generalized circle with sides $F(S_1)$ and $F(S_2)$.

This theorem is somewhat painful to prove, but not particularly surprising; note that FLTs are constructed in such a way as to be homeomorphisms, i.e. mappings which preserve topology. It is therefore natural that this fundamental aspect of the topology of generalized circles, i.e. that they have a connected inside and a connected outside, is preserved. Regardless, this fact allows one to determine how an FLT maps sets by checking (or assigning, through an FLT) only three points on the boundary and one point on the inside or outside.

Example. Construct an FLT from $\mathbb{D}$ to the upper-half plane $\mathbb{H}$.

Solution. We will map $\partial\mathbb{D}$ to $\mathbb{R} = \partial\mathbb{H}$, then use the theorem about the action of FLTs on generalized circles to deduce which side of the circle gets mapped to where, and apply a rotation if necessary. Let us try mapping $(1, i, -1) \to (0, 1, \infty)$. Define:

$$T_A = [z, 1, i, -1] = \frac{z-1}{z+1}\frac{i+1}{i-1} = -i\frac{z-1}{z+1} \implies A = -i\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

Then, define:

$$T_B = [z, 0, 1, \infty] = \frac{z-0}{z-\infty}\frac{1-\infty}{1-0} = \frac{z}{1} \implies \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Clearly, $B = I_2$ is its own inverse, and it is the identity, which means that:

$$F := T_{AB^{-1}} = T_A = -i\frac{z-1}{z+1}$$

is an FLT mapping $\partial\mathbb{D}$ to $\mathbb{R}$. Now, we use the fact that $\mathbb{R}$ and $\partial\mathbb{D}$ are generalized circles and investigate which side of $\partial\mathbb{D}$ maps to which side of $\mathbb{R}$. Since $0 \in \partial\mathbb{D}$, i.e. 0 is on the inside of $\partial\mathbb{D}$, it suffices to check this point. Note that:

$$F(0) = -i\frac{0-1}{0+1} = i \in \mathbb{H}$$

As an aside, the fact that there is a Riemann map from $\overline{\mathbb{D}}$ to $\overline{\mathbb{H}}$, i.e. the mapping F which was just found, implies that the Laplace Equation can be solved subject to Dirichlet boundary conditions on $\mathbb{H}$. The exact form of the solution can be found through some (fairly painful) algebra.

Theorem (Existence of Solutions in the Upper-Half Plane). Suppose $h : \mathbb{R} \to \mathbb{R}$ is a continuous function such that

$$h_\infty := \lim_{\mathbb{R}\ni x\to+\infty} h(x) = \lim_{\mathbb{R}\ni x\to-\infty} h(x) \in \mathbb{R}$$

Then, there exists a solution u to the Laplace equation on $\mathbb{H}$ subject to boundary data h, given by

$$u(x, y) = \frac{y}{\pi}\int_{-\infty}^{+\infty} \frac{h(t)}{(t-x)^2 + y^2}dt \qquad \text{where } (x, y) \in \mathbb{H}$$

This follows from transforming the formula which solves the Laplace equation on $\mathbb{D}$ via the above FLT.